# RELATIVE BREUIL MODULES AND SEMISTABLE $\mathbb{Z}_P$-LOCAL SYSTEMS

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Abstract. Let $p \geq 3$ be a prime. We extend the integral classification of lattices in semistable representations ([Liu08]) to the setting of $p$-adic local systems on affinoid rigid analytic varieties admitting semistable formal models. For a small semistable base ring $R$, we define a category of relative Breuil modules of weight $h$ over the associated Breuil ring $S_R$, and prove that the objects with underlying projective modules are equivalent to $G_R$-stable $\mathbb{Z}_p$-lattices in prismatic semistable representations with Hodge–Tate weights in $[0, h]$ when $eh < p - 1$. In particular, there is some restricted ramification. The proof combines Moon's classification of strongly divisible lattices over the completed Shilov points ([Moo24]) with the purity theorem for semistability of [DLMS24b].

PRELIMINARY DRAFT

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## 1. Introduction

Fix a prime number $p$. Let $k$ be a perfect field of characteristic $p > 2$, $W(k)$ its ring of Witt vectors, $K_0 = W(k)[1/p]$, $K/K_0$ a finite totally ramified extension, and $e = e(K/K_0)$ the absolute ramification index. In other words, $K$ is a complete discrete valued field of mixed characteristic $(0, p)$ with perfect residue field. Let $\mathcal{O}_K$ denote the ring of integers of $K$, $\pi$ a uniformizer, and $E(u) \in W(k)[u]$ the minimal polynomial of $\pi$.

The earliest results on the classification of integral lattices in $p$-adic Hodge theory are due to Fontaine and Laffaille [FL82], who worked over an unramified base ring and with small Hodge–Tate weights ($0 \leq h < p-1$, the so-called *Fontaine–Laffaille range*). Later, Breuil introduced what he called *strongly divisible modules*, now known as *Breuil modules*, and conjectured that these should classify lattices in semistable representations in the Fontaine–Laffaille range (see [Bre02] and its references), with unrestricted ramification. Breuil's conjecture was proved by Liu [Liu08]. Kisin recast much of the integral theory in terms of finite-height Frobenius modules, now known as *Breuil–Kisin modules* [Kis06] which play a central role in many modern approaches and this paper. We establish a relative version of Liu's classification for semistable $\mathbb{Z}_p$-local systems over a semistable base ring, albeit with restricted ramification.

The prismatic theory [BS22, BS21] has dramatically improved our understanding of integral $p$-adic Hodge theory. If $X$ is a $p$-adic formal scheme, the absolute prismatic site $X_{\mathbb{\Delta}}$ consists of diagrams $\operatorname{Spf} A \hookleftarrow \operatorname{Spf} A/I \to X$ where $(A, I)$ is a bounded prism. There is also a logarithmic variant, established by Koshikawa [Kos22]. For a bounded $p$-adic log formal scheme $(X, M_X)$, he associates the *absolute log prismatic site* $(X, M_X)_{\mathbb{\Delta}}$ consisting of the same diagrams as before but where $(A, I, M_{\operatorname{Spf} A})$ is a *log prism* (cf. *loc. cit.* for the precise definition). In both the "classical" and logarithmic cases, these sites come equipped with a structure sheaf of commutative rings $\mathcal{O}_{\mathbb{\Delta}}$ with a Frobenius endomorphism $\varphi$, and a distinguished ideal sheaf $\mathcal{I}_{\mathbb{\Delta}} \subset \mathcal{O}_{\mathbb{\Delta}}$ (each obtained by evaluating at the corresponding structures of a (log) prism $(A, I)$ or $(A, I, M_{\operatorname{Spf} A})$). The cohomology of the structure sheaf $\mathcal{O}_{\mathbb{\Delta}}$ interpolates between classical $p$-adic cohomology theories, such as the étale cohomology of the generic fiber and crystalline cohomology of the special fiber ([BS22, Thrm. 1.8] or [KY25, Thrm. 2] in the logarithmic case).

The sheaves on the prismatic site also interpolate between the coefficient objects of étale and crystalline cohomology, namely $\mathbb{Z}_p$-local systems and filtered isocrystals. The category of *prismatic $F$-crystals* $\operatorname{Vect}^{\varphi}(X_{\mathbb{\Delta}})$ is the category of pairs $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}})$ consisting of a vector bundle $\mathcal{E}_{\mathbb{\Delta}}$ on $X_{\mathbb{\Delta}}$ and a $\varphi$-semilinear map

$$\varphi_{\mathcal{E}_{\mathbb{\Delta}}} : \mathcal{E}_{\mathbb{\Delta}} \longrightarrow \mathcal{E}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]$$

whose linearization

$$\Phi_{\mathcal{E}_{\mathbb{\Delta}}} : \varphi^* \mathcal{E}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}] \xrightarrow{\sim} \mathcal{E}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]$$

is an isomorphism. This is the same datum as the usual definition in terms of the linearized Frobenius isogeny. There is a fully faithful *étale realization functor*

$$T_{\mathrm{et}} : \mathrm{Vect}^{\varphi}(X_{\mathbb{\Delta}}) \to \mathrm{Loc}_{\mathbb{Z}_p}(X_\eta).$$

The main theorem [BS21, Thrm. 1.2] of Bhatt and Scholze states that when $X = \operatorname{Spf} \mathcal{O}_K$, the essential image of the étale realization consists of crystalline Galois representations of $G_K$ (identifying $\mathrm{Loc}_{\mathbb{Z}_p}((\operatorname{Spf} \mathcal{O}_K)_\eta) \simeq \mathrm{Rep}_{\mathbb{Z}_p}(G_K)$). The category of prismatic $F$-crystals and the étale and crystalline realization functors also make sense in the log prismatic setup. In this context, [DL26] show that the étale realization establishes an equivalence between log prismatic $F$-crystals and semistable integral $G_K$-representations.

The geometric character of the prismatic theory allows one to consider the theory above in the context of general $p$-adic formal schemes, not just $\operatorname{Spf} \mathcal{O}_K$. In work of Guo and Reinecke [GR24], the authors define a category $\mathrm{Vect}^{\mathrm{an},\varphi}(X_{\mathbb{\Delta}})$ of *analytic prismatic $F$-crystals* generalizing prismatic $F$-crystals to compatible systems of vector bundles on $\operatorname{Spec} A \setminus V(p, I)$ for each prism $(A, I)$. They prove the following.

**Theorem 1.1** ([GR24, DLMS24a])**.** *Let $X$ be a smooth $p$-adic formal scheme over $\mathcal{O}_K$. There is an equivalence of categories*

$$T_{\mathrm{et}} : \mathrm{Vect}^{\mathrm{an},\varphi}(X_{\mathbb{\Delta}}) \xrightarrow{\sim} \mathrm{Loc}^{\mathrm{cris}}_{\mathbb{Z}_p}(X_\eta)$$

The notion of crystallinity for a $\mathbb{Z}_p$-local system is due to Brinon [Bri08], who constructs relative crystalline period rings for suitable $p$-adic base rings and defines crystallinity by the corresponding admissibility condition. An analogous notion of semistability is not available for general local systems. Instead, for a semistable $p$-adic formal log scheme $(X, M_X)$, [DLMS24b, Def. 3.26] calls a $\mathbb{Z}_p$-local system $\mathbb{L}$ on $X_\eta$ *prismatic $X$-semistable* if it lies in the essential image of the étale realization functor

$$T_{\mathrm{et}} : \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \longrightarrow \mathrm{Vect}((X, M_X)_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]^{\wedge}_p)^{\varphi=1} \xrightarrow{\sim} \mathrm{Loc}_{\mathbb{Z}_p}(X_\eta).$$

The first arrow is scalar extension along $\mathcal{O}_{\mathbb{\Delta}} \to \mathcal{O}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]$, and the isomorphism is [DLMS24b, Thrm. 3.14] ([BS21, Cor. 3.8] when $M_X$ is trivial).

The main result of [DLMS24b] is the *prismatic purity theorem*, which is crucial to our paper. Let $(X, M_X)$ be a semistable $p$-adic formal log scheme over $\mathcal{O}_K$. Its irreducible components determine completed traits $\Delta_j$ with natural morphisms $\iota_j : \Delta_j \to X$, and the associated rank-one points of $X_\eta$ are the Shilov points. We use the precise notation fixed in Definition 2.7 below.

**Theorem 1.2** ([DLMS24b, Thrm. 1.6])**.** *A $\mathbb{Z}_p$-local system $\mathbb{L}$ on $X_\eta$ is prismatic $X$-semistable if and only if $(\iota_j)^*\mathbb{L}$ is prismatic $\Delta_j$-semistable on $(\Delta_j)_\eta$ for each $1 \le j \le m$.*

Furthermore, prismatic semistability at the Shilov points can be detected via $\mathcal{O}\mathbb{B}_{\mathrm{st}}$-admissibility [DLMS24b, Thrm. 4.1]; in particular, the corresponding lattices are classified by strongly divisible modules [Moo24].

There is yet another equivalent notion of semistability. Fix a $\mathbb{Z}_p$-local system $\mathbb{L}$ on $X_{\eta,\mathrm{pro\acute{e}t}}$, and $(\mathcal{E}_{\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathbb{Q}}})$ an $F$-isocrystal on the special fiber $(X_1, M_{X_1}) := (X, M_X) \otimes_{\mathbb{Z}_p} \mathbb{Z}/p$. Following Faltings [Fal89, p. 67], [DLMS24b] define these objects to be *associated* if there is a Frobenius-compatible isomorphism of sheaves of $\mathbb{B}_{\mathrm{cris}}$-modules on $X_{\eta,\mathrm{pro\acute{e}t}}$ $\mathcal{E}_{\mathbb{Q}}(\mathbb{B}_{\mathrm{cris}}) \simeq \mathbb{L} \otimes_{\mathbb{Z}_p} \mathbb{B}_{\mathrm{cris}}$.

**Theorem 1.3** ([DLMS24b, Cor. 5.2])**.** *A $\mathbb{Z}_p$-local system on $X_\eta$ is prismatic semistable if and only if it is associated to an $F$-isocrystal on $(X_1, M_{X_1})_{\mathrm{CRIS}}$.*

Furthermore, there is an *isocrystalline realization functor*

$$D_{\mathrm{isoc}} : \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \to \mathrm{Vect}^{\varphi}_{\mathbb{Q}}((X_1, M_{X_1})_{\mathrm{CRIS}}).$$

The interpolation of étale and crystalline coefficient objects is made precise as follows:

**Proposition 1.4** ([DLMS24b, Prop. 3.42]). *For an analytic prismatic $F$-crystal $\mathcal{E}_{\mathbb{\Delta}}$ over $(X, M_X)$, its étale realization $T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$ and its isocrystalline realization $D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}})$ are associated.*

Key to our main theorem is the Breuil realization, an integral refinement of the isocrystalline realization in the Fontaine–Laffaille range which we eventually show to be an equivalence of categories. We make this precise as follows: set $X = \operatorname{Spf} R$ with

$$R := \mathcal{O}_K \left\langle T_1, \dots, T_m, T_{m+1}^{\pm}, \dots, T_d^{\pm} \right\rangle / (T_1 \cdots T_m - \pi).$$

We call such a ring a *small base ring*. Let $M_X$ be the log structure on $X$ induced by $\mathbb{N}^d \to R, e_i \mapsto T_i$ for $i = 1, \dots, d$. Then set

$$\mathfrak{S} := W(k) \left\langle T_1, \dots, T_m, T_{m+1}^{\pm}, \dots, T_d^{\pm} \right\rangle [[u]] / (T_1 \cdots T_m - u)$$

with Frobenius $\varphi(T_i) = T_i^p$. A semistable $p$-adic formal scheme over $\mathcal{O}_K$ refers to a $p$-adic formal scheme which is étale locally of the form $(\operatorname{Spf} R, M_{\operatorname{Spf} R})$ for a small base ring $R$ as above. The tuple $(\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$ is an object of $(X, M_X)_{\mathbb{\Delta}}$ via the map $\mathfrak{S}/(E(u)) \xrightarrow{\sim} R : u \mapsto \pi$, called the *Breuil–Kisin log prism*. This construction extends to *small affine* formal schemes $X$, namely affine formal schemes admitting an étale map to $\operatorname{Spf} R$. We also set $S$ to be the $p$-adically completed divided power envelope of $\mathfrak{S} \to R/p : u \mapsto \pi$, and equip it with the log structure $M_{\operatorname{Spf} S}$ induced from $(\mathfrak{S}, M_{\operatorname{Spf} \mathfrak{S}})$ via the composition $\mathfrak{S} \xrightarrow{\varphi} \mathfrak{S} \to S$ where the second arrow is the inclusion. As such, we obtain a map of log prisms $\varphi : (\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}}) \to (S, (p), M_{\operatorname{Spf} S})$. The ring $S$ also comes with the usual $p$-adically completed divided power filtration $\operatorname{Fil}^\bullet S$, where $\operatorname{Fil}^i S \subset S$ is the $p$-adic closure of the ideal generated by $\gamma_j(E(u))$ for $j \geq i$.

Choose an integer $h \geq 0$ so that $eh < p - 1$. Over $\mathfrak{S}$, there is the category of *projective Breuil–Kisin modules of height at most $h$* $\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S})$, consisting of pairs $(\mathfrak{M}, \varphi_{\mathfrak{M}})$ where $\mathfrak{M}$ is a projective $\mathfrak{S}$-module and $\varphi_{\mathfrak{M}}$ is a $\varphi$-semilinear endomorphism whose linearization $\Phi_{\mathfrak{M}}$ has cokernel killed by $E(u)^h$. Over $S$, there is the category of *projective Breuil modules of weight $h$* $\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(S)$, consisting of triples $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h)$ where $\mathscr{M}$ is a projective $S$-module, $\operatorname{Fil}^h \mathscr{M}$ is a certain submodule of $\mathscr{M}$ (which we obtain as the highest-weight part of a certain Nygaard filtration), and $\varphi_h : \operatorname{Fil}^h \mathscr{M} \to \mathscr{M}$ is a $\varphi$-semilinear map whose image generates $\mathscr{M}$, satisfying certain compatibilities with the Frobenius on $S$. We require that $\mathscr{M} / \operatorname{Fil}^h \mathscr{M}$ admit a resolution of length at most one by finite projective $\mathfrak{S}$-modules. There is a natural functor

$$\underline{\mathscr{M}} : \operatorname{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}) \to \operatorname{Mod}^{\varphi,h,\mathrm{proj}}(S)$$

Proposition 4.12, a modest generalization of [Kim15, Prop. 6.3], shows that this is an equivalence of categories. We further enhance the target category to $\operatorname{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S)$, consisting of tuples $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h, \nabla)$ where the underlying triple $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h)$ is an object of $\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(S)$ and

$$\nabla : \mathscr{M} \to \mathscr{M} \otimes_S \omega_S^1$$

is a quasi-nilpotent integrable log connection, which is Griffiths transversal to $\operatorname{Fil}^h \mathscr{M} \subset \mathscr{M}$ and parallel to the Frobenius $\varphi_h$ (Definition 5.1). This gives the target category for the Breuil realization functor. Let $\operatorname{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}})$ denote the full subcategory of $\operatorname{Vect}^{\varphi}((X, M_X)_{\mathbb{\Delta}})$ whose objects $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}})$ are effective and have the property that $\operatorname{coker}(\Phi_{\mathcal{E}_{\mathbb{\Delta}}})$ is killed by $\mathcal{I}_{\mathbb{\Delta}}^h$.

**Proposition 1.5** (§ 5.2). *There is a natural functor compatible with base change to the Shilov points*

$$\operatorname{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}}) \to \operatorname{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S).$$

Conversely, given an object $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h, \nabla)$, we associate to it a $\mathbb{Z}_p[G_R]$-module

$$T_{\mathrm{cris}}(\mathscr{M}) := \left( \mathbb{A}_{\mathrm{cris}}(\overline{R}) \widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M} \right)^{\varphi_h = 1}$$

with $G_R$-action defined via a Taylor formula involving the components of $\nabla$ (see (6.10)). A key point is that this $T_{\mathrm{cris}}(\mathscr{M})$ is in fact a finite free $\mathbb{Z}_p$-module. This follows from Theorems 6.8 and A.7, which give, respectively, the comparison of $T_{\mathfrak{S}}(\mathfrak{M})$ with $T_{\mathrm{cris}}(\mathscr{M})$, as well as the required theory of étale $\varphi$-modules over semistable base rings. Furthermore, by base-change compatibility of the $T_{\mathrm{cris}}$-functor, the prismatic purity theorem of [DLMS24b], and the comparison between the de Rham and Nygaard-filtered realizations (Proposition 3.18), we deduce that

**Proposition 1.6** (Proposition 6.15)**.** *$T_{\mathrm{cris}}(\mathscr{M})$ is a prismatic semistable $G_R$-representation with Hodge–Tate weights in $[0,h]$.*

Finally, we prove that $T_{\mathrm{cris}}$ and $\mathbb{M}_{\mathrm{Breuil}}$ are quasi-inverse up to a Tate/Breuil-Kisin twist. Specifically, if $(\mathcal{E}_{\mathbb{\Delta}},\varphi_{\mathcal{E}_{\mathbb{\Delta}}})$ is the analytic prismatic $F$-crystal whose étale realization gives $T_{\mathrm{cris}}(\mathscr{M})$, we show that

$$\mathscr{M} \simeq \mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\})$$

where $\{-h\}$ denotes a Breuil–Kisin twist, and

$$T_{\mathrm{cris}}(\mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\})) \simeq T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}}).$$

This proves our main theorem.

**Theorem 1.7.** *The functor*

$$T_{\mathrm{cris}} : \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S) \to \mathrm{Rep}^{\mathrm{st}}_{\mathbb{Z}_p,[0,h]}(G_R)$$

*is an equivalence of categories.*

**Outline.** Section 2 establishes preliminary results and notation on the log prisms and the absolute prismatic site, in particular definitions and compatibilities between the Breuil–Kisin and Breuil prisms, as well as the $\mathbb{A}_{\mathrm{inf}}$ and $\mathbb{A}_{\mathrm{cris}}$-period prisms. Section 3 recalls the theory of log prismatic $F$-crystals, prismatic semistability, and the relationships among the crystalline, isocrystalline, connection, and filtered realizations.

Section 4 defines our various categories of Breuil–Kisin and Breuil modules, including torsion variants. Kim's equivalence of categories [Kim15, Prop. 6.4.1] is also generalized in Section 4.5 (Prop. 4.12). Section 5 defines our category of relative Breuil modules with log connection and the Breuil realization functor.

Section 6 attaches a continuous $G_{R_\infty}$-representation $T_{\mathfrak{S}}(\mathfrak{M})$ to a Breuil–Kisin module and a continuous $G_{R_\infty}$-representation $T_{\mathrm{cris}}(\mathscr{M})$ to the associated Breuil module, and proves the canonical comparison

$$T_{\mathrm{cris}}(\mathscr{M}) \simeq T_{\mathfrak{S}}(\mathfrak{M})(h).$$

Section 6.5 extends the $G_{R_\infty}$-action on $T_{\mathrm{cris}}(\mathscr{M})$ to $G_R$, and proves that this representation is prismatic semistable with Hodge–Tate weights in $[0,h]$. Finally, Section 7 establishes the desired equivalence between semistable local systems with Hodge–Tate weights in $[0,h]$ and relative Breuil modules.

**Notation.** For a formal scheme $X$, we use $X_\eta$ to denote its adic generic fiber. For any locally Noetherian adic space $\mathcal{X}$, we denote by $\mathcal{X}_{\mathrm{pro\acute{e}t}}$ its proétale site (cf. [Sch13, §3]). Write $\mathbb{Z}_p$ for the sheaf $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ on $\mathcal{X}_{\mathrm{pro\acute{e}t}}$, and $\mathbb{Z}_p(1) := \varprojlim \mu_{p^n}$, and further set

$$\mathbb{Z}_p(n) := \begin{cases} \mathbb{Z}_p(1)^{\otimes n} & n \geq 0 \\ \mathbb{Z}_p(-n)^\vee & n < 0. \end{cases}$$

A *local system* on $\mathcal{X}$ will mean a locally free sheaf $\mathbb{L}$ of $\mathbb{Z}_p$-modules on $\mathcal{X}_{\mathrm{pro\acute{e}t}}$, and for such $\mathbb{L}$, we write $\mathbb{L}(n) := \mathbb{L} \otimes_{\mathbb{Z}_p} \mathbb{Z}_p(n)$ for the *Tate twists* of $\mathbb{L}$.

Fix $1 \leq m \leq d$, and set $R_0 = \mathcal{O}_K \langle T_1, \dots, T_m, T_{m+1}^{\pm}, \dots, T_d^{\pm} \rangle / (T_1 \cdots T_m - \pi)$. We equip $R_0$ with the prelog structure $\mathbb{N}^d \to R_0$ sending $e_i \mapsto T_i$ for $1 \leq i \leq d$. Note that when $m = d = 1$, we have $T_1 = \pi$ and we are in the situation of [Yao23]. We call a connected $\mathcal{O}_K$-algebra $R$ with a $p$-completely étale map $\square : R_0 \to R$ a *semistable base ring*. A $p$-adic log formal scheme $(X, M_X)$ is called *small affine with framing* $\square$ if it is of the form $(X, M_X) = (\operatorname{Spf} R, \mathbb{N}^d \to R_0 \xrightarrow{\square} R)^a$ where the $a$-superscript denotes the associated log structure (see [DLMS24b] for notation). A $p$-adic log formal scheme $(X, M_X)$ is *semistable* if it is étale locally isomorphic to a small affine.

To study $p$-adic local systems, we use the log prismatic sites of [Kos22, KY25]. We write a prelog ring as $(A, M_A)$, where the structure map $\alpha_A : M_A \to A$ is suppressed from the notation, and we write $(A, M_{\operatorname{Spf} A})$ for its associated log ring. Accordingly, a prelog prism is denoted by $(A, I, M_A)$ and its associated log prism by

$$(A, I, M_{\operatorname{Spf} A}) := (A, I, M_A)^a.$$

Morphisms displayed between log prisms are always understood to include the corresponding morphisms of log structures.

We frequently take $p$-adic completions. If $A$ is a ring and $M$ is an $A$-module, we write $A^\wedge$ and $M^\wedge$ to denote their $p$-adic completions, unless otherwise specified. If $A \to B$ is a morphism of $p$-complete rings, we write

$$B \widehat{\otimes}_A M := (B \otimes_A M)^\wedge_p$$

for completed scalar extension.

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## 2. Log-Prismatic and Crystalline Preliminaries

In this section, we recall some results from [DLMS24b] that we will use throughout the paper. Our basic setup follows [DLMS24b], which in turn depends on the log prismatic theory introduced in [Kos22, KY25]. We refer to these papers for the basic definitions ($\delta_{\log}$-structures and triples, (bounded) prelog prisms, associated log prisms, etc.). Throughout this section, $(X, M_X)$ denotes a bounded $p$-adic fine and saturated log formal scheme with semistable reduction, where $M_X$ is the canonical divisorial log structure associated with the special fiber. Here bounded means that, affine-locally on $X$, the $p^\infty$-torsion in $\mathcal{O}_X$ is killed by a fixed power of $p$. In the semistable cases considered below, $\mathcal{O}_X$ is $p$-torsion-free, so boundedness is automatic.

Let $(A, I, M_A)$ be a prelog prism. We write

$$(A, I, M_{\operatorname{Spf} A}) := (A, I, M_A)^a$$

for its associated log prism (cf. Remark 2.2 below).

**Definition 2.1** (cf. [KY25, Rem. 4.6])**.** Let $(X, M_X)_{\mathbb{\Delta}}$ denote the *absolute prismatic site* of $(X, M_X)$: an object of this site consists of a log prism $(A, I, M_{\operatorname{Spf} A}) = (A, I, M_A)^a$ together with a diagram of log formal schemes

$$(\operatorname{Spf} A, M_{\operatorname{Spf} A}) \hookleftarrow (\operatorname{Spf} A/I, M_{\operatorname{Spf} A/I}) \to (X, M_X).$$

Fix objects $(A, I, M_{\operatorname{Spf} A})$, $(A', I', M_{\operatorname{Spf} A'})$ with diagrams as above. Maps between such objects are specified by commutative diagrams

$$\begin{array}{ccccc} (\operatorname{Spf} A', M_{\operatorname{Spf} A'}) & \longleftarrow & (\operatorname{Spf}(A'/I'), M_{\operatorname{Spf}(A'/I')}) & \longrightarrow & (X, M_X) \\ \downarrow f & & \downarrow \bar{f} & & \| \\ (\operatorname{Spf} A, M_{\operatorname{Spf} A}) & \longleftarrow & (\operatorname{Spf}(A/I), M_{\operatorname{Spf}(A/I)}) & \longrightarrow & (X, M_X). \end{array}$$

We equip this category with the strict flat topology: a morphism $f : (A', I', M_{\operatorname{Spf} A'}) \to (A, I, M_{\operatorname{Spf} A})$ (with diagrams suppressed) is a cover if $\operatorname{Spf} A' \to \operatorname{Spf} A$ is $(p, I)$-completely faithfully flat map of formal schemes and the induced map $f^* M_{\operatorname{Spf} A} \to M_{\operatorname{Spf} A'}$ is an isomorphism (this is so-called *strictness* of a morphism of log (formal) schemes).

We promote this to a ringed site as follows: the *structure sheaf* $\mathcal{O}_{\mathbb{\Delta}}$ is the sheaf associated to the presheaf $(A, I, M_{\operatorname{Spf} A}) \mapsto A$. We also have the ideal sheaf $\mathcal{I}_{\mathbb{\Delta}} : (A, I, M_{\operatorname{Spf} A}) \mapsto I$.

*Remark* 2.2. We will usually work commutative-algebraically, so our maps between prisms will be in the category of prisms, rather than the opposite category underlying the absolute prismatic site. As such, we denote our prisms as $(A, I, M_{\operatorname{Spf} A})$ rather than $(\operatorname{Spf} A, I, M_{\operatorname{Spf} A})$ as in [DLMS24b].

*Remark* 2.3. Many results in [DLMS24b, §3] are first proved for the *strict* absolute prismatic site $(X, M_X)^{\mathrm{str}}_{\mathbb{\Delta}}$. Fortunately, [DLMS24b, Rem. 3.6, 3.15] establish equivalences between the Laurent $F$-crystals in the strict and non-strict sites, making the distinction between these sites immaterial for this paper, so we choose to work in the non-strict setting.

2.1. **Breuil–Kisin log prisms.** Fix $1 \le m \le d$, and set

$$R_0 = \mathcal{O}_K \left\langle T_1, \dots, T_m, T_{m+1}^{\pm}, \dots, T_d^{\pm} \right\rangle / (T_1 \cdots T_m - \pi). \tag{2.1}$$

We equip $R_0$ with the prelog structure $\mathbb{N}^d \to R_0$ sending $e_i \mapsto T_i$ for $1 \le i \le d$. We call a connected $\mathcal{O}_K$-algebra $R$ with a $p$-completely étale map $\square : R_0 \to R$ a *semistable base ring*. A $p$-adic log formal scheme $(X, M_X)$ is called *small affine with framing* $\square$ if it is of the form $(X, M_X) = (\operatorname{Spf} R, \mathbb{N}^d \to R_0 \xrightarrow{\square} R)^a$. We will often refer to the pair $(X, M_X)$ simply as a *small affine*.

Following [DLMS24b], we let $\mathfrak{S}_{R_0} = W \left\langle T_1, \dots, T_m, T_{m+1}^{\pm}, \dots, T_d^{\pm} \right\rangle [[u]]/(T_1 \cdots T_m - u)$, a $(p, E(u))$-adically complete $W$-algebra such that $\mathfrak{S}_{R_0}/(E(u)) \simeq R_0$. We equip $\mathfrak{S}_{R_0}$ with the prelog structure $\mathbb{N}^d \to \mathfrak{S}_{R_0} : e_i \mapsto T_i$, lifting the one on $R_0$ via $\mathfrak{S}_{R_0} \to R_0$. Since $\square$ is $p$-completely étale, it lifts uniquely to a $(p, E(u))$-completely étale map

$$\square_{\mathfrak{S}} : \mathfrak{S}_{R_0} \to \mathfrak{S}_{R,\square}.$$

When there is no risk of confusion, we write $\mathfrak{S}_R$ or even $\mathfrak{S}$ for $\mathfrak{S}_{R,\square}$, called the *Breuil–Kisin ring* associated to $R$. We further enhance the prelog ring $(\mathfrak{S}_R, \mathbb{N}^d)$ to a log prism, first by setting $\delta_{\log}(e_i) = 0$ and $\delta(T_j) = 0$ on $\mathfrak{S}_{R_0}$, so that $(\mathfrak{S}_{R_0}, \delta, (E(u)), \mathbb{N}^d, \delta_{\log})$ is a prelog prism; we abbreviate it to $(\mathfrak{S}_{R_0}, (E(u)), \mathbb{N}^d)$. By [Kos22, Lem. 2.13], $(\mathfrak{S}_R, (E(u)), \mathbb{N}^d \to \mathfrak{S}_{R_0} \to \mathfrak{S}_R)$ admits a bounded prelog prism structure.

**Definition 2.4** ([DLMS24b, Def. 2.6])**.** The *Breuil–Kisin log prism* $(\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$ (with respect to $\pi$ and $\square$) is the log prism associated to $(\mathfrak{S}, (E(u)), \mathbb{N}^d)$. It is an object of $((\operatorname{Spf} R, \mathbb{N}^d)^a)^{\mathrm{str}}_{\mathbb{\Delta}}$ via $R \simeq \mathfrak{S}/(E(u))$.

**Proposition 2.5** ([DLMS24b, Lem. 2.7–2.9] )**.** *Let $(\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$ denote the Breuil–Kisin log prism in the site $(X, M_X)_{\mathbb{\Delta}}$. Then*

(a) *The Frobenius $\varphi_{\mathfrak{S}} : \mathfrak{S} \to \mathfrak{S}$ is faithfully flat.*
(b) *The object $(\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$ covers the final object of the topos associated to $(X, M_X)_{\mathbb{\Delta}}$ (resp. $(X, M_X)^{str}_{\mathbb{\Delta}}$).*
(c) *Let $(A, I, M_{\operatorname{Spf} A})$ be a log prism in the opposite category $(X, M_X)^{str,op}_{\mathbb{\Delta}}$. Then the coproduct*

$$(A, I, M_{\operatorname{Spf} A}) \coprod (\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$$

*of* $(A, I, M_{\operatorname{Spf} A})$ *and* $(\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$ *exists in* $(X, M_X)^{str,op}_{\mathbb{\Delta}}$*, and the structure map*

$$(A, I, M_{\operatorname{Spf} A}) \to (A, I, M_{\operatorname{Spf} A}) \coprod (\mathfrak{S}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}})$$

*is a cover.*

Let $\mathcal{O}_{L_0}$ be a Cohen ring–a mixed characteristic complete discrete valuation ring (CDVR) with maximal ideal $(p)$–whose residue field has a finite $p$-basis given by the mod $p$ reduction of $\{X_1, \dots, X_b\} \subset \mathcal{O}_{L_0}$. Write $L_0 = \mathcal{O}_{L_0}[p^{-1}]$, and equip $\mathcal{O}_{L_0}$ with the Frobenius $\varphi : \mathcal{O}_{L_0} \to \mathcal{O}_{L_0}$ given by $\varphi(X_i) = X_i^p$ (see [BM90, Cor. 1.2.7(iii)]). Let $L/L_0$ be a finite totally ramified extension, $\mathcal{O}_L$ the ring of integers of $L$, $\pi_L \in \mathcal{O}_L$ a uniformizer, and $E_L(u) \in \mathcal{O}_{L_0}[u]$ be the minimal polynomial of $\pi_L$ over $L_0$. We equip $\mathcal{O}_L$ with the prelog structure $\mathbb{N} \to \mathcal{O}_L : n \mapsto \pi_L^n$. Setting $Y = \operatorname{Spf} \mathcal{O}_L$, we let $(Y, M_Y)_{\mathbb{\Delta}}$ denote the absolute prismatic site of the log formal scheme $(Y, M_Y)$ associated to $(\mathcal{O}_L, \mathbb{N} \to \mathcal{O}_L)$. For the examples relevant to this paper, $L_0$ will contain $K_0$, and $L = L_0 \otimes_{K_0} K$, and $E_L(u)$ will be simply $E(u)$. This will be necessary for results dependent on [Moo24].

Set $\mathfrak{S}_L = \mathcal{O}_{L_0}[[u]]$ (the *Breuil–Kisin ring of* $L$) with Frobenius lift $\varphi(u) = u^p$ compatibly with the $\varphi$-structure on $\mathcal{O}_{L_0}$, and prelog structure $\mathbb{N} \to \mathfrak{S}_L : n \mapsto u^n$. We may thus form the tuple $(\mathfrak{S}_L, (E_L(u)), M_{\operatorname{Spf} \mathfrak{S}_L})$ where $M_{\operatorname{Spf} \mathfrak{S}_L}$ denotes the associated log structure on $\operatorname{Spf} \mathfrak{S}_L$. This is the *Breuil–Kisin log prism of* $L$.

*Remark* 2.6 ([DLMS24b, Rem. 2.16]). The analogue of Proposition 2.5 also holds for the Breuil–Kisin log prism of a CDVR.

Let $R_0$ be the ring from (2.1), with the log structure on $\operatorname{Spf} R_0$ described above, and let $X = \operatorname{Spf} R$ be a small semistable formal scheme with framing $\square : R_0 \to R$. We further require (after passing to an étale localization if necessary) that the framing $\square$ induce a bijection on the sets of irreducible components of $\operatorname{Spf} R_0$ and $\operatorname{Spf} R$.

**Definition 2.7.** Let $\xi_j \in \operatorname{Spec}(R/\pi)$ be the generic point corresponding under this bijection to $(T_j)$, for $1 \le j \le m$. Put $\mathcal{O}_{L_j} := R^\wedge_{\xi_j}$, and set $\Delta_j := \operatorname{Spf} \mathcal{O}_{L_j}$. Equip $\Delta_j$ with the pullback log structure $M_{\Delta_j}$, and write

$$\iota_j : (\Delta_j, M_{\Delta_j}) \longrightarrow (X, M_X)$$

for the natural morphism, which we call is the *Shilov point* associated to the $j$th component. When no confusion can arise, we refer to the collection $(\xi_j, \Delta_j, \iota_j)$ as the *Shilov data*.

Thus $\xi_j = (T_j)$ and $(\pi, T_j) = (T_j)$. The ring $\mathcal{O}_{L_j}$ is a CDVR with uniformizer $\pi$, and $M_{\Delta_j}$ is associated to the prelog structure $\mathbb{N} \to \mathcal{O}_{L_j}$ sending 1 to $\pi$. Choose a subring $\mathcal{O}_{L_0,j} \subset \mathcal{O}_{L_j}$ and a ring endomorphism $\varphi$ of $\mathcal{O}_{L_0,j}$ so that $\mathcal{O}_{L_0,j}$ is a Cohen ring of $\mathcal{O}_{L_j}/(\pi) = \operatorname{Frac}(R/(\pi, T_j))$ with $T_i \in \mathcal{O}_{L_0,j}$ (for $1 \le i \le d$, $i \ne j$), and such that $\varphi$ is a lift of Frobenius on $\mathcal{O}_{L_0,j}/(p)$ with $\varphi(T_i) = T_i^p$. We choose $\mathcal{O}_{L_0,j}$ compatibly with the induced inclusion $k \to \mathcal{O}_{L_j}/(\pi)$, so it carries the unique Frobenius-compatible $W(k)$-algebra structure lifting this inclusion. Since $\pi$ is a uniformizer of $\mathcal{O}_{L_j}$ and $E(u)$ remains its minimal polynomial over $\mathcal{O}_{L_0,j}$, the natural map

$$\mathcal{O}_{L_0,j} \otimes_{W(k)} \mathcal{O}_K \xrightarrow{\sim} \mathcal{O}_{L_j}$$

is an isomorphism. Write $\mathcal{O}_{\mathcal{E}_R} = \mathfrak{S}_R[E(u)^{-1}]^\wedge_p$, and for each $j = 1, \dots, m$, set $\mathfrak{S}_{L_j} := \mathcal{O}_{L_0,j}[[u]]$ and $\mathcal{O}_{\mathcal{E}_{L_j}} := \mathfrak{S}_{L_j}[E(u)^{-1}]^\wedge_p$. There is a map

$$\mathfrak{S}_{R_0} \to \mathfrak{S}_{L_j} : \begin{cases} T_j \mapsto u(T_1 \cdots T_{j-1} T_{j+1} \cdots T_m)^{-1}, & \\ u \mapsto u, & \\ T_i \mapsto T_i, & i \ne j \end{cases}$$

which lifts uniquely to $\mathfrak{S}_R \to \mathfrak{S}_{L_j}$, which is injective, flat, and compatible with $\varphi$. The log structure on $\mathfrak{S}$ induces one on each $\mathfrak{S}_{L_j}$, and as in [DLMS24b, §2.3, 2.4], we obtain a log prism $(\mathfrak{S}_{L_j}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}_{L_j}})$.

**Definition 2.8.** The log prism $(\mathfrak{S}_{L_j}, (E(u)), M_{\operatorname{Spf} \mathfrak{S}_{L_j}})$ is the *Breuil–Kisin log prism associated to the Shilov point* $(T_j)$.

Given a Breuil–Kisin log prism $(\mathfrak{S}, (E(u)), M_{\operatorname{Spf}\mathfrak{S}})$ (for a small affine or CDVR), write

$$(\mathfrak{S}^{(i)}, (E(u)), M_{\operatorname{Spf}\mathfrak{S}^{(i)}})$$

for its $(i+1)$-st self-product in the absolute strict log prismatic site $(X, M_X)^{\mathrm{str}}_{\mathbb{\Delta}}$ (where $X$ is a small affine or $\operatorname{Spf}\mathcal{O}_L$ with the appropriate log structure as defined above). Thus $\mathfrak{S}^{(i)}$ denotes the underlying ring of this log prism. From now until the discussion of the period prisms, we suppress the argument of $E(u)$ (and of $E_L(u)$) when it is clear from context, writing simply $E$ (and $E_L$).

Finally, we recall a minor variant of the *ring-theoretic purity* results from [DLMS24b, §2.4]. The embeddings $\mathfrak{S}_R \to \mathfrak{S}_{L_j}$ extend to $\mathcal{O}_{\mathcal{E}_R} \to \mathcal{O}_{\mathcal{E}_{L_j}}$ for each $1 \le j \le m$. Putting these together, we obtain "diagonal" embeddings

$$\mathfrak{S}_R \to \prod_{j=1}^m \mathfrak{S}_{L_j}, \quad \text{and} \quad \mathcal{O}_{\mathcal{E}_R} \to \prod_{j=1}^m \mathcal{O}_{\mathcal{E}_{L_j}}.$$

**Lemma 2.9.** *We have that*

$$\mathfrak{S}_R = \mathcal{O}_{\mathcal{E}_R} \cap \prod_{j=1}^m \mathfrak{S}_{L_j} \tag{2.2}$$

*viewed as subrings of* $\prod_{j=1}^m \mathcal{O}_{\mathcal{E}_{L_j}}$. *In particular, if* $P$ *is a finite projective* $\mathfrak{S}_R$*-module, then there is an equality*

$$P = (P \otimes_{\mathfrak{S}_R} \mathcal{O}_{\mathcal{E}_R}) \cap \prod_{j=1}^m \left(P \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j}\right).$$

*inside* $\prod_{j=1}^m P \otimes_{\mathfrak{S}_R} \mathcal{O}_{\mathcal{E}_{L_j}}$.

*Proof.* The first assertion follows from [DLMS24b, Lem. 2.24]. The second assertion follows at once by [DLMS24a, Lem. 3.1]. □

Let $\overline{R}$ denote the union of finite $R$-subalgebras $R'$ inside a fixed algebraic closure of $\operatorname{Frac}(R)$ such that $R'[p^{-1}]$ is étale over $R[p^{-1}]$. Let $\mathcal{P}$ denote the set of minimal prime ideals of $\overline{R}$ containing $p$. For each $\mathfrak{P} \in \mathcal{P}$, write $\mathfrak{p} := R \cap \mathfrak{P}$. Then $\mathcal{O}_L(\mathfrak{p}) := (R_{\mathfrak{p}})^\wedge$ is a CDVR whose residue field has a $p$-basis given by $\{T_1, \dots, T_{j-1}, T_{j+1}, \dots, T_d\}$, where the omitted $T_j$ corresponds to $\mathfrak{p}$. Note that $\pi$ is a uniformizer of $\mathcal{O}_L(\mathfrak{p})$. Write $L(\mathfrak{p}) := \mathcal{O}_L(\mathfrak{p})[p^{-1}]$.

For each $\mathfrak{P} \in \mathcal{P}$, fix a continuous ring map $(\overline{R}_{\mathfrak{P}})^\wedge \to (\mathcal{O}_{\overline{L(\mathfrak{p})}})^\wedge$ which extends the map $R_{\mathfrak{p}} \to \mathcal{O}_{\overline{L(\mathfrak{p})}}$. Taking the product over $\mathfrak{P}$'s induces injective maps

$$\overline{R}^\wedge \to \prod_{\mathfrak{P}\in\mathcal{P}} (\overline{R}_{\mathfrak{P}})^\wedge \to \prod_{\mathfrak{P}\in\mathcal{P}} (\mathcal{O}_{\overline{L(\mathfrak{p})}})^\wedge, \quad \text{and} \quad \overline{R}^\flat \to \prod_{\mathfrak{P}\in\mathcal{P}} \mathcal{O}^\flat_{\overline{L(\mathfrak{p})}},$$

where $A^\flat$ denotes the tilt $\varprojlim_{x\mapsto x^p} A/p$.

**Lemma 2.10** ([DLMS24b, Lem. 2.29])**.** *As subrings of* $\prod_{\mathfrak{P}\in\mathcal{P}} W(\overline{L(\mathfrak{p})}^\flat)$,

$$W(\overline{R}^\flat[(\pi^\flat)^{-1}]) \cap \prod_{\mathfrak{P}\in\mathcal{P}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}}) = W(\overline{R}^\flat).$$

We will use the following notation below:

$$A := W(\overline{R}^\flat), \quad B := W\big(\overline{R}^\flat[(\pi^\flat)^{-1}]\big), \quad C := \prod_{\mathfrak{P}\in\mathcal{P}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}}), \quad \text{and} \quad D := \prod_{\mathfrak{P}\in\mathcal{P}} W(\overline{L(\mathfrak{p})}^\flat).$$

Thus Lemma 2.10 states that $A = B \cap C$ inside $D$.

**Lemma 2.11.** *Let* $P$ *be a finite* $\mathfrak{S}_R$*-module satisfying one of the following conditions:*

(a) $P$ *is finite projective over* $\mathfrak{S}_R$*;*
(b) $P$ *admits a finite filtration whose successive quotients are killed by* $p$ *and finite projective over* $\mathfrak{S}_R/(p)$*.*

*Then, inside $P \otimes_{\mathfrak{S}_R} D$, there is an equality*

$$P \otimes_{\mathfrak{S}_R} A = (P \otimes_{\mathfrak{S}_R} B) \cap (P \otimes_{\mathfrak{S}_R} C)\,. \tag{2.3}$$

*Proof.* In case (a), this follows from Lemma 2.10 and the module-intersection formula for finite projective modules [DLMS24a, Lem. 3.1]. We prove case (b) by dévissage. First suppose that $pP = 0$ and that $P$ is finite projective over $\mathfrak{S}_R/(p)$. The characteristic-$p$ equality underlying Lemma 2.10 is $A/(p) = B/(p) \cap C/(p)$ inside $D/(p)$. Applying the finite-projective module-intersection formula over $\mathfrak{S}_R/(p)$ gives (2.3) for $P$. In both the projective case and this characteristic-$p$ case, the maps from the three modules on the left of (2.3) to $P \otimes_{\mathfrak{S}_R} D$ are injective. The same remains true under extensions, by the exact base-change sequences used below, so the indicated intersection is well-defined throughout the dévissage.

We now show that the assertion is preserved under extensions. Suppose

$$0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0$$

is exact and that the formula holds for $P'$ and $P''$. Since these modules have finite filtrations with $\mathfrak{S}_R/(p)$-projective successive quotients, tensoring this sequence with any of the $p$-torsion-free $A, B, C, D$ preserves exactness.

Let $x_B \in P \otimes_{\mathfrak{S}_R} B$ and $x_C \in P \otimes_{\mathfrak{S}_R} C$ have the same image in $P \otimes_{\mathfrak{S}_R} D$. Then their images in $P'' \otimes_{\mathfrak{S}_R} B$ and $P'' \otimes_{\mathfrak{S}_R} C$ come from an element $x''_A \in P'' \otimes_{\mathfrak{S}_R} A$. Choose a lift $x_A \in P \otimes_{\mathfrak{S}_R} A$, and let $\widetilde{x_B}$ (resp. $\widetilde{x_C}$) denote its images in $P \otimes_{\mathfrak{S}_R} B$ (resp. $P \otimes_{\mathfrak{S}_R} C$). Then $x_B - \widetilde{x_B}$ and $x_C - \widetilde{x_C}$ vanish in $P'' \otimes_{\mathfrak{S}_R} B$ (resp. $P'' \otimes_{\mathfrak{S}_R} C$), and in particular have the same image in $P' \otimes_{\mathfrak{S}_R} D$. Since the result is known for $P'$, these differences come from some $x'_A \in P' \otimes_{\mathfrak{S}_R} A$. Therefore $x'_A + x_A \in P \otimes_{\mathfrak{S}_R} A$ maps to both $x_B$ and $x_C$. This proves the reverse inclusion in (2.3). The forward inclusion is tautological. Induction on the length of the filtration proves case (b). □

*Remark* 2.12. Under the hypothesis of part (b) above, we also have

$$P = (P \otimes_{\mathfrak{S}_R} \mathcal{O}_{\mathcal{E}}) \cap \left( P \otimes_{\mathfrak{S}_R} \prod_{j=1}^{m} \mathfrak{S}_{L_j} \right)$$

by essentially the same proof.

2.2. **Breuil prisms and the log crystalline site.** Let $(X, M_X)$ be a semistable log formal scheme, or the log formal spectrum of a CDVR. Set $(X_1, M_{X_1}) = (X, M_X) \otimes_{\mathbb{Z}_p} \mathbb{Z}_p/p$, and let $(X_1, M_{X_1})_{\mathrm{CRIS}}$ denote the absolute log crystalline site of $(X_1, M_{X_1})$ [DLMS24b, Def. B.6]. Suppose $(X, M_X) = (\operatorname{Spf} R, \mathbb{N}^d)^a$ is small affine with framing $\square : R_0 \to R$, and let $(\mathfrak{S}, (E), M_{\operatorname{Spf} \mathfrak{S}}) \in (X, M_X)_{\mathbb{\Delta}}$ be the associated Breuil–Kisin log prism.

Define $S_{R,\square}$ to be the $p$-adic completion of the log PD-envelope of $\mathfrak{S}_{R,\square}$ with respect to the kernel of the exact surjection $\mathfrak{S}_{R,\square} \to R/p$. By [BS22, Cor. 2.39], $S_{R,\square} \simeq \mathfrak{S}_{R,\square}\{\varphi(E)/p\}^{\wedge}_{\delta}$ where $\{-\}_\delta$ denotes adjoining elements as a $\delta$-ring. As before, we will abbreviate $S_{R,\square}$ to $S_R$ or simply $S$ when there is no risk of confusion. We let

$$\operatorname{Fil}^i S = \langle \gamma_j(E) \,:\, j \geq i \rangle^{\wedge}$$

denote the $p$-completed divided power filtration on $S$, and for $i < p$, we let

$$\varphi_i = \varphi/p^i : \operatorname{Fil}^i S \to S \tag{2.4}$$

denote the divided Frobenius. By [Kim15, Rem. 3.2.2], we have that $\operatorname{Fil}^1 S_R + pS_R$ is contained in the Jacobson radical of $S_R$.

There is a natural map $\varphi : \mathfrak{S} \to S$ given by post-composing the inclusion map with the Frobenius on $\mathfrak{S}$. Since $c_1 := \varphi(E)/p \in S$ is a unit, we obtain a well-defined morphism of prisms $(\mathfrak{S}, (E)) \to (S, (p))$. We denote the prelog structure on $S$ induced via the map $\varphi : \mathfrak{S} \to S$ by $\varphi^* \mathbb{N}^d$, and set $(S, (p), M_{\operatorname{Spf} S}) := (S, (p), \varphi^* \mathbb{N}^d)^a$. This is the *Breuil log prism*. The map $\varphi : \mathfrak{S} \to S$ induces a morphism of log prisms

$$\varphi : (\mathfrak{S}, (E), M_{\operatorname{Spf} \mathfrak{S}}) \to (S, (p), M_{\operatorname{Spf} S}).$$

Set $D_X = \operatorname{Spf} S$, and write $M_{D_X}$ for the log structure on $D_X$ induced from the prelog structure $\mathbb{N}^d$ with $e_i \mapsto T_i$. Then the exact closed immersion $X_1 \hookrightarrow D_X$ is a log PD-thickening, and $(X_1, D_X, M_{D_X})$ defines an ind-object $\varinjlim_n (X_1, \operatorname{Spec}(S_{R,\square}/p^n))$ of $(X_1, M_{X_1})^{\mathrm{aff}}_{\mathrm{CRIS}}$.

**Lemma 2.13** ([DLMS24b, Lem. 3.29])**.** *$(X_1, D_X, M_{D_X})$ is a weakly final object.*

For each $i \geq 0$, let $(X_1, \operatorname{Spf} S^{(i)}, M_{\operatorname{Spf} S^{(i)}})$ be the $(i+1)$-st self-product of the object $(X_1, D_X, M_{D_X})$ in $(X_1, M_{X_1})_{\mathrm{CRIS}}$. By taking coproducts of the map (2.4), we obtain morphisms

$$\varphi : (\mathfrak{S}^{(i)}, (E), M_{\operatorname{Spf} \mathfrak{S}^{(i)}}) \to (S^{(i)}, (p), M_{\operatorname{Spf} S^{(i)}})$$

for any $i \geq 0$.

**Lemma 2.14** ([DLMS24b, Lem. 3.32])**.** *The underlying rings of the self-products of Breuil log prisms in the prismatic and strict prismatic sites agree. Moreover, they agree with the underlying rings of the self-products of the log PD-thickening $(X_1 \to D_X, M_{D_X})$ in the log crystalline site.*

Just as for Breuil–Kisin log rings and prisms, we may also define Breuil log rings and prisms in the CDVR case. Let $\mathcal{O}_L$ be a CDVR, so that we have rings $\mathcal{O}_{L_0}$, $L_0$, $L$, and $\mathfrak{S}_L$, and set $(Y, M_Y) = (\operatorname{Spf} \mathcal{O}_L, M_{\operatorname{Spf} \mathcal{O}_L})$ (with notation as in 2.1). As in the small affine situation, we define $S_L$ to be the $p$-adic completion of the log PD-envelope of $\mathfrak{S}_L$ with respect to the kernel of the exact surjection $\mathfrak{S}_L \to \mathcal{O}_L/(p)$. For the same reason, we have an isomorphism $S_L \simeq \mathfrak{S}_L\{\varphi(E_L)/p\}^\wedge_\delta$, and we give $S_L$ the log structure induced from the one on $\mathfrak{S}_L$ via the map $\varphi : \mathfrak{S}_L \to S_L$. Thus $(S_L, (p), M_{\operatorname{Spf} S_L})$ is the Breuil log prism of $L$. The analogues of Lemmas 2.13 and 2.14 hold for Breuil log prisms over a CDVR by the same argument.

**Lemma 2.15.** *For every Shilov point, the natural map*

$$\left(\mathfrak{S}_{L_j} \otimes_{\mathfrak{S}_R} S_R\right)^\wedge_p \xrightarrow{\sim} S_{L_j}$$

*is an isomorphism, and $S_R \to S_{L_j}$ is $p$-completely flat. For every $i \geq 0$,*

$$S_{L_j} \widehat{\otimes}_{S_R} \operatorname{Fil}^i S_R \xrightarrow{\sim} \operatorname{Fil}^i S_{L_j}.$$

*Proof.* The first two claims follow by applying the base change compatibility statement in [BS22, Lem. 2.38] and taking $p$-completions. For the filtration assertion, tensor along the uncompleted PD-envelope $S^{\mathrm{nc}}_R$ along the map $\mathfrak{S}_R \to \mathfrak{S}_{L_j}$, as well as the ideals generated by $\gamma_n(E)$ for $n \geq i$. Since $S^{\mathrm{nc}}_{L_j}$ is also the PD-envelope generated by the image of $E(u)$, we have an isomorphism $\operatorname{Fil}^i S^{\mathrm{nc}}_{L_j} \simeq \operatorname{Fil}^i S^{\mathrm{nc}}_R \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j}$. Reducing modulo powers of $p$ and taking inverse limits gives the asserted isomorphism. □

**Lemma 2.16.** *The Shilov-localization maps induce an injective ring homomorphism*

$$S_R \longrightarrow \prod_{j=1}^m S_{L_j}.$$

*Proof.* We first show that, for every $a \geq 1$, the natural map

$$R/p^a \longrightarrow \prod_{j=1}^m \mathcal{O}_{L_j}/p^a$$

is injective. When $a = 1$, this holds as $R/p$ is Cohen–Macaulay and its associated primes are precisely the generic points $(T_1), \ldots, (T_m)$ of its irreducible components.[1] The assertion for general $a$ follows by a straightforward induction. Taking inverse limits, we find that

$$R \longrightarrow \prod_{j=1}^m \mathcal{O}_{L_j}$$

[1] This statement is remarked in the proof of [DLMS24b, Lem. 2.24], but not proven.

is injective. Now equip $S_R$ and each $S_{L_j}$ with their divided-power filtrations. The standard description of these rings as $p$-adically completed divided-power envelopes gives $\mathrm{gr}^n S_R \simeq R \cdot \gamma_n(E)$ and $\mathrm{gr}^n S_{L_j} \simeq \mathcal{O}_{L_j} \cdot \gamma_n(E)$ for every $n \geq 0$. These identifications are compatible with the Shilov-localization maps. Consequently, the map on associated graded pieces is identified with

$$R \cdot \gamma_n(E) \longrightarrow \prod_{j=1}^{m} \mathcal{O}_{L_j} \cdot \gamma_n(E),$$

and is therefore injective.

Finally, the divided-power filtration on $S_R$ is separated. Indeed, if a nonzero element $x \in S_R$ mapped to zero in every $S_{L_j}$, there would be an $n \geq 0$ such that $x \in \mathrm{Fil}^n S_R$ but $x \notin \mathrm{Fil}^{n+1} S_R$. Its nonzero initial term in $\mathrm{gr}^n S_R$ would map to zero in every $\mathrm{gr}^n S_{L_j}$, contradicting the injectivity shown above. Thus $x = 0$. □

**Lemma 2.17.** *For each $j$, regard $S_R \cap \mathfrak{S}_{L_j}$ as a subring of $S_R$, where the intersection is taken inside $S_{L_j}$. Then*

$$\mathfrak{S}_R = \bigcap_{j=1}^{m} (S_R \cap \mathfrak{S}_{L_j}).$$

*Proof.* The inclusion from left to right is clear. For the reverse direction, fix $f \in S_R$ whose image in $S_{L_j}$ belongs to $\mathfrak{S}_{L_j}$ for every $j$. We show that the class of $f$ in $S_R/\mathfrak{S}_R$ vanishes, for which we establish the following criterion: if $a \geq 1$ and an element of $R/p^a$ vanishes after base change to $\mathcal{O}_{L_j}/p^a$ for every $j$, then it is zero. For $a = 1$, this follows directly from the construction of the Shilov points: $R/p$ is Cohen–Macaulay, its associated primes are precisely its generic points $(T_1), \ldots, (T_m)$, and completion of $R_{(T_j)}$ is faithfully flat. The assertion for general $a$ follows by induction, using the $p$-torsion-freeness of $R$ and of the $\mathcal{O}_{L_j}$.

Filter $\mathfrak{S}_R$ by $E^\bullet \mathfrak{S}_R$ and $S_R$ by $\mathrm{Fil}^\bullet S_R$. The inclusion $\mathfrak{S}_R \subset S_R$ is strict for these filtrations, and the induced filtration on the quotient $S_R/\mathfrak{S}_R$ is given by

$$\mathrm{Fil}^i(S_R/\mathfrak{S}_R) := (\mathrm{Fil}^i S_R + \mathfrak{S}_R)/\mathfrak{S}_R \simeq \mathrm{Fil}^i S_R/E^i\mathfrak{S}_R.$$

We claim that this filtration is separated. Let the class of $f \in S_R$ belong to $\mathrm{Fil}^i(S_R/\mathfrak{S}_R)$ for every $i \geq 1$. For each $i$, choose $a_i \in \mathfrak{S}_R$ such that $f - a_i \in \mathrm{Fil}^i S_R$. Then

$$a_{i+1} - a_i \in \mathfrak{S}_R \cap \mathrm{Fil}^i S_R = E^i\mathfrak{S}_R,$$

where the equality is the strictness of $\mathfrak{S}_R \hookrightarrow S_R$. Since $E$ is topologically nilpotent and $\mathfrak{S}_R$ is complete, the sequence $(a_i)$ converges to some $a \in \mathfrak{S}_R$. The ideals $E^i\mathfrak{S}_R$ are closed, so $a - a_i \in E^i\mathfrak{S}_R \subseteq \mathrm{Fil}^i S_R$. Hence $f - a \in \mathrm{Fil}^i S_R$ for every $i$. The divided-power filtration on $S_R$ is separated, and therefore $f = a \in \mathfrak{S}_R$. Thus the class of $f$ in $S_R/\mathfrak{S}_R$ is zero.

We also find that the associated graded is

$$\begin{aligned}
\mathrm{gr}^n(S_R/\mathfrak{S}_R) &\simeq (\mathrm{Fil}^n S_R/E^n\mathfrak{S}_R)/(\mathrm{Fil}^{n+1} S_R/E^{n+1}\mathfrak{S}_R) \\
&\simeq (\mathrm{Fil}^n S_R/\mathrm{Fil}^{n+1} S_R)/(E^n\mathfrak{S}_R/E^{n+1}\mathfrak{S}_R) \\
&\simeq (\gamma_n(E)R)/(E^n R) \\
&\simeq R/n!R \qquad (2.5)
\end{aligned}$$

for any $n \geq 1$. The same calculation holds in the CDVR case, i.e.

$$\mathrm{gr}^n(S_{L_j}/\mathfrak{S}_{L_j}) \simeq \mathcal{O}_{L_j}/n!\mathcal{O}_{L_j}$$

for any $n \geq 1$.

Suppose that the class of $f$ in $S_R/\mathfrak{S}_R$ were nonzero. By separatedness, there is a smallest $n_0 \geq 0$ where $f$ has nonzero image $f_n \in \mathrm{gr}^{n_0}(S_R/\mathfrak{S}_R)$, which by (2.5) is represented by a nonzero element of $R/n_0!R$. Since the image of $f$ belongs to $\mathfrak{S}_{L_j}$ for every $j$, the image of $f_{n_0}$ in $\mathcal{O}_{L_j}/n_0!\mathcal{O}_{L_j}$ vanishes for each $j$. As $n_0!$ is a unit times a power of $p$, our criterion above implies that the class of $f$ vanishes, a contradiction, and thus $f \in \mathfrak{S}_R$ as desired. □

2.3. **The Period Prisms $\mathbb{A}_{\mathrm{inf}}$ and $\mathbb{A}_{\mathrm{cris}}$.** We use the notation of 2.1. Let $R$ (resp. $\mathcal{O}_L$) be a semistable base ring (resp. CDVR). Define $\mathbb{A}_{\mathrm{inf}}(\overline{R}) := W(\overline{R}^\flat)$ and $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}}) := W(\mathcal{O}_{\overline{L}}^\flat)$. Fix compatible systems of $p$-power roots $\pi^\flat := (\pi^{1/p^n})_{n\geq 0}$ and $T_i^\flat := (T_i^{1/p^n})_{n\geq 0}$ for $i = 1,\dots,d$ in $\overline{R}$, chosen so that $\prod_{i=1}^m T_i^\flat = \pi^\flat$ in $\overline{R}^\flat$. In the CDVR case, fix compatible systems $\pi_L^\flat := (\pi_L^{1/p^n})_{n\geq 0}$ and $X_i^\flat := (X_i^{1/p^n})_{n\geq 0}$ for $i = 1,\dots,b$ in $\mathcal{O}_{\overline{L}}$. Together with the chosen Frobenius lifts on the coefficient rings, these choices define Frobenius-compatible maps

$$\mathfrak{S}_R \to \mathbb{A}_{\mathrm{inf}}(\overline{R}) : u \mapsto [\pi^\flat],\ T_i \mapsto [T_i^\flat],$$
$$\mathfrak{S}_L \to \mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}}) : u \mapsto [\pi_L^\flat],\ X_i \mapsto [X_i^\flat].$$

The prelog structures on $\mathfrak{S}_R$ and $\mathfrak{S}_L$ induce prelog (and $\delta_{\log}$) structures on $\mathbb{A}_{\mathrm{inf}}(\overline{R})$ and $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}})$ respectively, and we may thus form the associated log prisms

$$(\mathbb{A}_{\mathrm{inf}}(\overline{R}), (E([\pi^\flat])), \mathbb{N}^d)^a \quad \text{and} \quad (\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}}), (E_L([\pi_L^\flat])), \mathbb{N})^a.$$

We write their associated log structures as $M_{\operatorname{Spf}\mathbb{A}_{\mathrm{inf}}(\overline{R})}$ and $M_{\operatorname{Spf}\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}})}$, respectively. By [DLMS24b, Rem. 2.5], these induced log structures are independent of the choices of $p$-power roots, and thus $G_R$ (resp. $G_L$) acts on $(\mathbb{A}_{\mathrm{inf}}(\overline{R}), (E([\pi^\flat])), \mathbb{N}^d)^a$ (resp. $(\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}}), (E_L([\pi_L^\flat])), \mathbb{N})^a$). The rings $\widehat{\overline{R}}$ and $(\mathcal{O}_{\overline{L}})_p^\wedge$ are integral perfectoid ([BMS18, Def. 3.5, Lem. 3.10]), so Fontaine's $\theta$ maps are surjective. Their kernels are generated by $E([\pi^\flat])$ and $E_L([\pi_L^\flat])$, respectively.

Let $\mathbb{A}_{\mathrm{cris}}(\overline{R})$ denote the $p$-complete log PD-envelope of $\mathbb{A}_{\mathrm{inf}}(\overline{R})$ with respect to $\ker\theta$; we use $\mathrm{Fil}^\bullet \mathbb{A}_{\mathrm{cris}}$ to denote the PD filtration (see Section 2.2). Thus the map $\mathfrak{S}_R \to \mathbb{A}_{\mathrm{inf}}(\overline{R})$ uniquely extends to $S_R \to \mathbb{A}_{\mathrm{cris}}(\overline{R})$, inducing a log structure $M_{\operatorname{Spf}\mathbb{A}_{\mathrm{cris}}(\overline{R})}$ on $\mathbb{A}_{\mathrm{cris}}(\overline{R})$ (independent of the choices of $p$-power roots above). In particular, $(\mathbb{A}_{\mathrm{cris}}(\overline{R}), (p), M_{\operatorname{Spf}\mathbb{A}_{\mathrm{cris}}(\overline{R})})$ is the associated $\mathbb{A}_{\mathrm{cris}}$ log prism. We can place these maps into a commutative diagram

$$\begin{array}{ccc} (\mathfrak{S}_R, (E(u)), \mathbb{N}^d)^a & \longrightarrow & (\mathbb{A}_{\mathrm{inf}}(\overline{R}), (E([\pi^\flat])), \mathbb{N}^d)^a \\ \downarrow{\scriptstyle\varphi} & & \downarrow{\scriptstyle\varphi} \\ (S_R, (p), \varphi^*\mathbb{N}^d)^a & \longrightarrow & (\mathbb{A}_{\mathrm{cris}}(\overline{R}), (p), \varphi^*\mathbb{N}^d)^a. \end{array}$$

**Lemma 2.18.** *The diagram above is commutative, and the untwisted structural maps on the underlying rings induce isomorphisms*

$$\mathbb{A}_{\mathrm{cris}}(\overline{R}) \simeq \mathbb{A}_{\mathrm{inf}}(\overline{R})\widehat{\otimes}_{\mathfrak{S}_R} S_R$$

*and equalities* $(\mathrm{Fil}^i S_R)\mathbb{A}_{\mathrm{cris}}(\overline{R}) = \mathrm{Fil}^i \mathbb{A}_{\mathrm{cris}}(\overline{R})$ *for every* $i \geq 0$. *In particular, there is a natural isomorphism*

$$\mathbb{A}_{\mathrm{cris}}(\overline{R})\widehat{\otimes}_{S_R} N \simeq \mathbb{A}_{\mathrm{inf}}(\overline{R})\widehat{\otimes}_{\mathfrak{S}_R} N$$

*for any $p$-complete $S_R$-module $N$.*

*Proof.* This follows from the same argument as Lemma 2.15 since $\mathbb{A}_{\mathrm{cris}}(\overline{R})$ is the PD envelope of the image of $E(u)$ in $\mathbb{A}_{\mathrm{inf}}(\overline{R})$ under the map $\mathfrak{S}_R \to \mathbb{A}_{\mathrm{inf}}(\overline{R})$. □

The same discussion applies to the CDVR case. Henceforth, we suppress the arguments of $E(u)$, $E([\pi^\flat])$, $E_L(u)$, and $E_L([\pi_L^\flat])$ whenever they are clear from context, writing simply $E$ or $E_L$ accordingly.

2.4. **Log crystals and connections.** We recall the setup from [DLMS24b, §B.2]. Let $(X, M_X)$ be either (1) a small affine or (2) the log formal spectrum of a CDVR (with notation as in 2.1). Let $(X_1, M_{X_1}) := (X, M_X) \otimes_{\mathbb{Z}_p} \mathbb{Z}/p$, and let $(X_1, M_{X_1})_{\mathrm{CRIS}}$ denote the big log crystalline site, as in [DLMS24b, Def. B.5]. We generally drop subscripts from Breuil–Kisin and Breuil rings and prisms to treat the small affine and CDVR cases simultaneously. As in [DLMS24b, Ex. B.22], set $D = \operatorname{Spf} S$ with log structure $M_D$. The pair $(X_1, D)$ is weakly final in $(X_1, M_{X_1})^{\mathrm{aff}}_{\mathrm{CRIS}}$ by [DLMS24b, Lem. 3.29]. Let $(D^{(i)}, M_{D^{(i)}}) = (\operatorname{Spf} S^{(i)}, M_{\operatorname{Spf} S^{(i)}})$ denote its $(i+1)$-st self-product in

the log crystalline site. We have the module of *logarithmic differentials* $\omega^1_{(\mathfrak{S},M_{\mathfrak{S}})/\mathbb{Z}_p}$ ([Kat89, Def. 1.7]). Since we always endow $\mathfrak{S}$ with the log structures of 2.1, we write $\omega^1_{\mathfrak{S}/\mathbb{Z}_p}$ instead of $\omega^1_{(\mathfrak{S},M_{\mathfrak{S}})/\mathbb{Z}_p}$ for simplicity. We have

$$\omega^1_{\mathfrak{S}_R/\mathbb{Z}_p} = \bigoplus_{i=1}^{d} \mathfrak{S}_R \, d\log T_i \quad \text{with } d\log u = \sum_{i=1}^{m} d\log T_i$$

in the small affine case, and

$$\omega^1_{\mathfrak{S}_L/\mathbb{Z}_p} = \mathfrak{S}_L \, d\log u \oplus \bigoplus_{i=1}^{b} \mathfrak{S}_L \, d\log(X_i)$$

where $\{X_i\}_{i=1}^b \subset \mathcal{O}_{L_0}$ is a lift of a $p$-basis of the residue field of $\mathcal{O}_L$ in the CDVR case. Set $\omega^1_S := S \otimes_{\mathfrak{S}} \omega^1_{\mathfrak{S}/\mathbb{Z}_p}$. The universal log derivation on $\mathfrak{S}$ extends to a derivation $d : S \to \omega^1_S$. A *log connection* on an $S$-module $\mathscr{M}$ is an additive map

$$\nabla : \mathscr{M} \to \mathscr{M} \otimes_S \omega^1_S$$

such that $\nabla(fm) = f\nabla(m) + m \otimes df$ for $f \in S$ and $m \in \mathscr{M}$. We extend the connection to higher differentials by maps

$$\nabla^i : \mathscr{M} \otimes_S \omega^i_S \to \mathscr{M} \otimes_S \omega^{i+1}_S$$

defined by $\nabla^i(m \otimes \omega) = \nabla(m) \wedge \omega + m \otimes d\omega$. We say $\nabla$ is *integrable* if $\nabla^1 \circ \nabla = 0$ (equivalently, $\nabla^{i+1} \circ \nabla^i = 0$ for every $i \geq 0$). In the small affine case, let $N_{T_i} : \mathscr{M} \to \mathscr{M}$ be the component of $\nabla$ along $d\log T_i$ for $i = 1, \ldots, d$. In the CDVR case, define $N_u$ and $N_{X_i}$ analogously using the displayed basis. Integrability is equivalent to pairwise commutativity of the corresponding operators [BO15, Thm. 4.8]. We say that $\nabla$ is *quasi-nilpotent* if the pullback $(\mathscr{M}, \nabla)$ to $S/p^n$ is quasi-nilpotent in the sense of [Kat89, Thm. 6.2(iii)] and [Ogu94, p. 19] for every $n \geq 1$. Now we recall the following definition.

**Definition 2.19** (cf. [DLMS24b, Def. B.23])**.** Let $\mathscr{M}$ be an $S$-module. An *HPD-stratification* on $\mathscr{M}$ is an isomorphism

$$\eta : S^{(1)} \otimes_{p_0,S} \mathscr{M} \to S^{(1)} \otimes_{p_1,S} \mathscr{M}$$

on $S^{(1)}$ satisfying the cocycle condition on $S^{(2)}$ (cf. [DLMS24b, Rem. B.24]). In particular, the base change of $\eta$ along the "multiplication" map $S^{(1)} \to S$ is the identity on $\mathscr{M}$.

The following result is an immediate application of the cited proposition.

**Proposition 2.20** ([DLMS24b, Prop. B.25])**.** *The following categories are equivalent:*

(a) $\mathrm{CR}((X_1, M_{X_1})_{\mathrm{CRIS}})$ *of coherent crystals on* $(X_1, M_{X_1})_{\mathrm{CRIS}}$*;*
(b) *the category of finitely generated quasi-coherent $S$-modules together with HPD-stratifications;*
(c) *the category of finite-type $S$-modules $\mathscr{M}$ together with a quasi-nilpotent integrable log connection*

$$\nabla : \mathscr{M} \to \mathscr{M} \otimes_S \omega^1_S.$$

We briefly describe the equivalence. Let $s_i$, $i = 0, 1$ denote the coface maps $s_i : S \to S^{(1)}$ and $s_{ij}$, $i = 0, 1$, $j = 1, 2$ denote the coface maps $s_{ij} : S^{(1)} \to S^{(2)}$. Evaluating a coherent crystal $\mathcal{E}$ on the weakly final object $(X_1, D)$ and its self-products along the diagram

$$S \underset{s_1}{\overset{s_0}{\rightrightarrows}} S^{(1)} \underset{s_{12}}{\overset{s_{01}}{\overset{s_{02}}{\rightrightarrows}}} S^{(2)}$$

gives (1) a finite-type $S$-module and (2) an HPD-stratification: the crystal condition gives an isomorphism after pullback along $s_0$ and $s_1$, and (3) the cocycle condition follows by checking that the relevant diagram commutes after publlback along the various projections to $S^{(2)}$. The weak finality of $(X_1, D)$ and crystalline descent show that the HPD-stratification uniquely determines the coherent crystal, establishing the equivalence of (a) and (b). Given a pair $(\mathscr{M}, \eta)$ consisting of an

$S$-module with HPD-stratification $\eta : s_0^*\mathscr{M} \xrightarrow{\sim} s_1^*\mathscr{M}$, put $\ell_t := \log\left(\frac{p_0^*t}{p_1^*t}\right)$. The logarithmic Taylor formula in the small affine case is

$$\eta(1 \otimes m) = \sum_{j_1,\dots,j_d \geq 0} \prod_{i=1}^{d} \gamma_{j_i}(\ell_{T_i}) \otimes N_{T_1}^{j_1} \cdots N_{T_d}^{j_d}(m), \tag{2.6}$$

where the right-hand side lies in $p_1^*\mathscr{M}$. In the CDVR case the analogous formula is

$$\eta(1 \otimes m) = \sum_{j_0,\dots,j_b \geq 0} \gamma_{j_0}(\ell_u) \prod_{i=1}^{b} \gamma_{j_i}(\ell_{X_i}) \otimes N_u^{j_0} N_{X_1}^{j_1} \cdots N_{X_b}^{j_b}(m).$$

The corresponding connection in the small affine case is

$$\nabla(m) = \sum_{i=1}^{d} N_{T_i}(m) \otimes d\log T_i,$$

with the analogous formula $\nabla(m) = N_u(m) \otimes d\log u + \sum_{i=1}^{b} N_{X_i}(m) \otimes d\log X_i$ in the CDVR case. The cocycle condition guarantees that $\nabla$ is an integrable connection (cf. [Kat89, Thm. 6.2] in the logarithmic case, following [BM90, Thm. 4.2.11] in the classical case); hence the corresponding differential operators commute, and the convergence condition is precisely quasi-nilpotence. Moreover, this recipe is reversible, thereby establishing the equivalence of (b) and (c).

## 3. Prismatic $F$-Crystals, Semistability, and Realizations

We recall basic definitions and results on prismatic $F$-crystals, Laurent $F$-crystals, and various realization functors in the small affine and log CDVR settings. Most material is from [DLMS24b, §3], and a critical result we restate for our purposes is [Ino25, Prop. 6.23] comparing Hodge and Nygaard filtrations on the filtered $F$-isocrystal attached to an analytic prismatic $F$-crystal (Proposition 3.18). We also show that, under our ramification restriction, analytic prismatic $F$-crystals of height at most $h$ are genuine prismatic $F$-crystals (Proposition 3.6).

3.1. **Log prismatic $F$-crystals.** In this section, we take for $(X, M_X)$ either (1) a small affine or (2) the formal log spectrum of a CDVR. For a log prism $(A, I, M_{\operatorname{Spf} A}) \in (X, M_X)_{\mathbb{\Delta}}$, the $\delta$-structure on $A$ induces a Frobenius $\varphi_A$ on $A$, and $\varphi_A$ acts on $\operatorname{Spec} A - V(p, I)$.

**Definition 3.1** (cf. [DLMS24b, Def. 3.3])**.** Throughout, $\varphi$ denotes the semilinear Frobenius and $\Phi$ its linearization.

1. Write $\operatorname{Vect}^{\varphi}(A, I)$ for the category of pairs $(\mathfrak{M}_A, \varphi_{\mathfrak{M}_A})$, where $\mathfrak{M}_A$ is a finite projective $A$-module and
$$\varphi_{\mathfrak{M}_A} : \mathfrak{M}_A \longrightarrow \mathfrak{M}_A[I^{-1}]$$
is $\varphi_A$-semilinear, such that its $A$-linearization
$$\Phi_{\mathfrak{M}_A} : \mathfrak{M}_A^*[I^{-1}] := \big(A \otimes_{\varphi_A, A} \mathfrak{M}_A\big)[I^{-1}] \xrightarrow{\sim} \mathfrak{M}_A[I^{-1}], \quad a \otimes m \longmapsto a\varphi_{\mathfrak{M}_A}(m),$$
is an isomorphism. We call the objects of this category *projective Kisin modules over $A$*.

1'. Write $\operatorname{Vect}^{\mathrm{an},\varphi}(A, I)$ for the category of pairs $(\mathcal{E}_A, \varphi_{\mathcal{E}_A})$, where $\mathcal{E}_A$ is a vector bundle on $\operatorname{Spec} A - V(p, I)$ and
$$\varphi_{\mathcal{E}_A} : \mathcal{E}_A \longrightarrow \mathcal{E}_A[I^{-1}]$$
is $\varphi_A$-semilinear, such that its linearization
$$\Phi_{\mathcal{E}_A} : \varphi_A^*\mathcal{E}_A[I^{-1}] \xrightarrow{\sim} \mathcal{E}_A[I^{-1}]$$
is an isomorphism. We call the objects of this category *analytic Kisin modules over $A$*.

2. A *prismatic $F$-crystal* $(\mathcal{E}, \varphi_{\mathcal{E}})$ on the logarithmic prismatic site $(X, M_X)_{\mathbb{\Delta}}$ is a vector bundle $\mathcal{E}$ on $((X, M_X)_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}})$ together with a $\varphi$-semilinear map

$$\varphi_{\mathcal{E}} : \mathcal{E} \longrightarrow \mathcal{E}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]$$

whose linearization is an isomorphism

$$\Phi_{\mathcal{E}} : \varphi^*\mathcal{E}[\mathcal{I}_{\mathbb{\Delta}}^{-1}] \xrightarrow{\sim} \mathcal{E}[\mathcal{I}_{\mathbb{\Delta}}^{-1}].$$

Thus $\varphi_{\mathcal{E}}(m) = \Phi_{\mathcal{E}}(1 \otimes m)$, and conversely $\Phi_{\mathcal{E}}(a \otimes m) = a\varphi_{\mathcal{E}}(m)$; in particular, this is the same datum as the convention of [DLMS24b, Def. 3.3]. Write $\mathrm{Vect}^{\varphi}((X, M_X)_{\mathbb{\Delta}})$ for the category of prismatic $F$-crystals on $(X, M_X)$. By [DLMS24b, Prop. 3.2], there is a natural equivalence of categories

$$\mathrm{Vect}^{\varphi}((X, M_X)_{\mathbb{\Delta}}) \simeq \lim_{(A,I,M_{\mathrm{Spf}\,A}) \in (X,M_X)_{\mathbb{\Delta}}} \mathrm{Vect}^{\varphi}(A, I).$$

2'. The category $\mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$ of *analytic prismatic $F$-crystals* over $(X, M_X)_{\mathbb{\Delta}}$ is defined as

$$\mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) := \lim_{(A,I,M_{\mathrm{Spf}\,A}) \in (X,M_X)_{\mathbb{\Delta}}} \mathrm{Vect}^{\mathrm{an},\varphi}(A, I).$$

A prismatic $F$-crystal $(\mathcal{E}, \varphi_{\mathcal{E}})$ is *effective* if $\varphi_{\mathcal{E}}$ takes values in $\mathcal{E}$, or equivalently if $\Phi_{\mathcal{E}}$ restricts to a map $\varphi^*\mathcal{E} \to \mathcal{E}$, which we again denote by $\Phi_{\mathcal{E}}$. We use the analogous definition for (analytic) Kisin modules and analytic prismatic $F$-crystals. We write $\mathrm{Vect}^{\varphi,\mathrm{eff}}((X, M_X)_{\mathbb{\Delta}})$ for the full subcategory of $\mathrm{Vect}^{\varphi}((X, M_X)_{\mathbb{\Delta}})$ consisting of effective prismatic $F$-crystals.

An effective prismatic $F$-crystal $(\mathcal{E}, \varphi_{\mathcal{E}})$ has *height at most $h$* if

$$\mathcal{I}_{\mathbb{\Delta}}^{h}\mathcal{E} \subseteq \Phi_{\mathcal{E}}(\varphi^*\mathcal{E}) \subseteq \mathcal{E},$$

or equivalently if $\mathrm{coker}(\Phi_{\mathcal{E}})$ is killed by $\mathcal{I}_{\mathbb{\Delta}}^{h}$. For an effective (analytic) Kisin module $(\mathfrak{M}_A, \varphi_{\mathfrak{M}_A})$ over $(A, I)$, the analogous condition is

$$I^h\mathfrak{M}_A \subseteq \Phi_{\mathfrak{M}_A}(\mathfrak{M}_A^*) \subseteq \mathfrak{M}_A.$$

The least such $h$ is called the *height*. We denote the category of effective (analytic) Kisin modules of height at most $h$ by $\mathrm{Vect}^{(\mathrm{an}),\varphi,h}(A, I)$ and effective prismatic $F$-crystals of height at most $h$ by $\mathrm{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}})$. We may also then form the category of *effective analytic prismatic $F$-crystals of height at most $h$*

$$\mathrm{Vect}^{\mathrm{an},\varphi,h}((X, M_X)_{\mathbb{\Delta}}) := \varprojlim_{(A,I,M_{\mathrm{Spf}\,A}) \in (X,M_X)_{\mathbb{\Delta}}} \mathrm{Vect}^{\mathrm{an},\varphi,h}(A, I).$$

We will drop the subscript $A$ from $(\mathfrak{M}_A, \varphi_{\mathfrak{M}_A})$ when it is clear from context.

*Remark* 3.2 (cf. [DLMS24b, §3.1]). For a prism $(A, I)$, its *Breuil–Kisin twist* $A\{1\}$ is defined in [BL22, §2.3, 2.5]. This yields an object $\mathcal{O}_{\mathbb{\Delta}}\{1\} \in \mathrm{Vect}^{\varphi}((X, M_X)_{\mathbb{\Delta}})$ whose underlying $\mathcal{O}_{\mathbb{\Delta}}$-module is invertible. For each $n \in \mathbb{Z}$, write $\mathcal{O}\{n\} := (\mathcal{O}_{\mathbb{\Delta}}\{1\})^{\otimes n}$. We have a canonical isomorphism $\varphi^*\mathcal{O}_{\mathbb{\Delta}}\{n\} \simeq \mathcal{I}_{\mathbb{\Delta}}^{-n}\mathcal{O}_{\mathbb{\Delta}}\{n\}$ by *loc. cit.*. For any $\mathcal{E}_{\mathbb{\Delta}} \in \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$, we define $\mathcal{E}_{\mathbb{\Delta}}\{n\} := \mathcal{E}_{\mathbb{\Delta}} \otimes_{\mathcal{O}_{\mathbb{\Delta}}} \mathcal{O}_{\mathbb{\Delta}}\{n\}$. If $\mathcal{E}_{\mathbb{\Delta}}$ is genuine, then $\mathcal{E}_{\mathbb{\Delta}}\{n\}$ is effective for $n \ll 0$ (cf. [DLMS24b, Rem. 3.10(1)]).

**Lemma 3.3.** *Let $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$, and let $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon) \in \mathrm{DD}_{\mathfrak{S}}$ be its associated analytic Breuil–Kisin module with descent datum. For every $n \in \mathbb{Z}$, evaluation on the Breuil–Kisin prism identifies the analytic Breuil–Kisin module with descent datum associated to $\mathcal{E}_{\mathbb{\Delta}}\{n\}$ with*

$$(\mathfrak{M}\{n\}, \varphi_{\mathfrak{M}\{n\}}, \varepsilon\{n\}) := (\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon) \otimes \mathcal{O}_{\mathbb{\Delta}}\{n\}(\mathfrak{S}).$$

*Moreover, the étale realization is compatible with these twists: there is a natural isomorphism*

$$T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}}\{n\}) \xrightarrow{\sim} T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})(n).$$

*Proof.* The first assertion follows from the compatibility of evaluation with tensor products and from the definition of the descent datum by evaluation on the first self-product of the Breuil–Kisin prism. The second assertion is the compatibility of the symmetric monoidal étale realization with Breuil–Kisin and Tate twists; see [BS21, Construction 4.8, Example 4.9]. □

Let $(\mathfrak{S}, (E), M_{\mathrm{Spf}\,\mathfrak{S}})$ be the Breuil–Kisin log prism associated to $(X, M_X)$, and let

$$p_0, p_1 : (\mathfrak{S}, (E), M_{\mathrm{Spf}\,\mathfrak{S}}) \rightrightarrows (\mathfrak{S}^{(1)}, (E), M_{\mathrm{Spf}\,\mathfrak{S}^{(1)}})$$

denote the two projection morphisms to its first self-product in the log prismatic site (Proposition 2.5).

**Definition 3.4** ([DLMS24b, Def. 3.7])**.** Let $\mathrm{DD}_{\mathfrak{S}} \coloneqq \mathrm{DD}_{\mathfrak{S}_R,\square}$ denote the category of *(integral) Kisin descent data*: objects are triples $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon)$ where

- $\mathfrak{M}$ is a torsion free finite $\mathfrak{S}_R$-module such that $\mathfrak{M}[p^{-1}]$ is $\mathfrak{S}_R[p^{-1}]$-projective[2], $\mathfrak{M}[E^{-1}]$ is $\mathfrak{S}[E^{-1}]$-projective, and $\mathfrak{M} = \mathfrak{M}[p^{-1}] \cap \mathfrak{M}[E^{-1}]$;
- $\varphi_{\mathfrak{M}} : \mathfrak{M} \to \mathfrak{M}[E^{-1}]$ is $\varphi_{\mathfrak{S}}$-semilinear, and its linearization
$$\Phi_{\mathfrak{M}} : \mathfrak{M}^*[E^{-1}] := \big(\mathfrak{S} \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}\big)[E^{-1}] \xrightarrow{\sim} \mathfrak{M}[E^{-1}]$$
is an isomorphism of $\mathfrak{S}[E^{-1}]$-modules;
- $\varepsilon\colon \mathfrak{S}^{(1)} \otimes_{p_0,\mathfrak{S}} \mathfrak{M} \xrightarrow{\cong} \mathfrak{S}^{(1)} \otimes_{p_1,\mathfrak{S}} \mathfrak{M}$ is an isomorphism of $\mathfrak{S}^{(1)}$-modules that satisfies the cocycle condition over $\mathfrak{S}^{(2)}$ and is compatible with $\Phi_{\mathfrak{M}}$ after inverting $E$. Explicitly, if $\varepsilon^*$ denotes the induced isomorphism between the Frobenius pullbacks, then
$$\varepsilon \circ (1 \otimes \Phi_{\mathfrak{M}}) = (1 \otimes \Phi_{\mathfrak{M}}) \circ \varepsilon^*.$$

Morphisms of $\mathrm{DD}_{\mathfrak{S}}$ are $\mathfrak{S}$-linear maps which are compatible with all the structures. An object is *effective* if $\varphi_{\mathfrak{M}}$ takes values in $\mathfrak{M}$, equivalently if $\Phi_{\mathfrak{M}}(\mathfrak{M}^*) \subseteq \mathfrak{M}$, and it has *height at most* $h$ if

$$E^h\mathfrak{M} \subseteq \Phi_{\mathfrak{M}}(\mathfrak{M}^*) \subseteq \mathfrak{M}.$$

We let $\mathrm{DD}^h_{\mathfrak{S}}$ denote the category of objects of height at most $h$, $\mathrm{DD}^{\mathrm{vb}}_{\mathfrak{S}}$ the full subcategory of $\mathrm{DD}_{\mathfrak{S}}$ for which $\mathfrak{M}$ is projective, and $\mathrm{DD}^{\mathrm{vb},h}_{\mathfrak{S}}$ for the intersection of the two.

Evaluation of an analytic prismatic $F$-crystal $(\mathcal{E}, \varphi_{\mathcal{E}})$ along the diagram

$$p_0, p_1 : (\mathfrak{S}, (E), M_{\mathrm{Spf}\,\mathfrak{S}}) \to (\mathfrak{S}^{(1)}, (E), M_{\mathrm{Spf}\,\mathfrak{S}^{(1)}})$$

yields an object $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon)$ of $\mathrm{DD}_{\mathfrak{S}}$, where $\mathfrak{M} = \mathcal{E}((\mathfrak{S}, (E), M_{\mathrm{Spf}\,\mathfrak{S}}))$ and

$$\varphi_{\mathfrak{M}} = \varphi_{\mathcal{E}}((\mathfrak{S}, (E), M_{\mathrm{Spf}\,\mathfrak{S}})) : \mathfrak{M} \longrightarrow \mathfrak{M}[E^{-1}].$$

The descent datum $\varepsilon$ is the identification

$$\varepsilon : \mathfrak{S}^{(1)} \otimes_{p_0,\mathfrak{S}} \mathfrak{M} \simeq \mathcal{E}((\mathfrak{S}^{(1)}, (E), M_{\mathrm{Spf}\,\mathfrak{S}^{(1)}})) \simeq \mathfrak{S}^{(1)} \otimes_{p_1,\mathfrak{S}} \mathfrak{M}$$

which satisfies the cocycle condition $p_{12}^*\varepsilon \circ p_{01}^*\varepsilon = p_{02}^*\varepsilon$ over $\mathfrak{S}^{(2)}$ (where $p_{ij}^*$ denotes $\mathfrak{S}^{(2)} \otimes_{p_{ij},\mathfrak{S}^{(1)}} (-)$).

**Lemma 3.5** (cf. [DLMS24b, Lem. 3.8])**.** *The functor defined above gives an equivalence of categories*

$$\mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \xrightarrow{\sim} \mathrm{DD}_{\mathfrak{S}}$$

*which restricts to an equivalence*

$$\mathrm{Vect}^{\varphi}((X, M_X)_{\mathbb{\Delta}}) \xrightarrow{\sim} \mathrm{DD}^{\mathrm{vb}}_{\mathfrak{S}}.$$

*Moreover, it preserves effectivity and, for every $h \geq 0$, restricts to equivalences*

$$\mathrm{Vect}^{\mathrm{an},\varphi,h}((X, M_X)_{\mathbb{\Delta}}) \xrightarrow{\sim} \mathrm{DD}^h_{\mathfrak{S}}, \quad \textit{and} \quad \mathrm{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}}) \xrightarrow{\sim} \mathrm{DD}^{\mathrm{vb},h}_{\mathfrak{S}}.$$

[2]By [DLMS24b, Rem. 3.10(2), Prop. 5.14], this condition is redundant.

*Proof.* The two underlying equivalences are [DLMS24b, Lem. 3.8], and their compatibility with effectivity is [DLMS24b, Lem. 3.5]. It remains only to check the assertion about height. Evaluation on the Breuil–Kisin log prism carries $\Phi_{\mathcal{E}}$ to $\Phi_{\mathfrak{M}}$ and carries the distinguished ideal $\mathcal{I}_{\mathbb{\Delta}}$ to $(E)$. Consequently, the inclusion

$$\mathcal{I}_{\mathbb{\Delta}}^h \mathcal{E} \subseteq \Phi_{\mathcal{E}}(\varphi^* \mathcal{E}) \subseteq \mathcal{E}$$

evaluates to

$$E^h \mathfrak{M} \subseteq \Phi_{\mathfrak{M}}(\mathfrak{M}^*) \subseteq \mathfrak{M}.$$

Conversely, under the descent equivalence the latter inclusions base change to the corresponding inclusions on the prismatic cover used to reconstruct $\mathcal{E}$. The height condition is equivalently the vanishing of the morphism

$$\mathcal{I}_{\mathbb{\Delta}}^h \mathcal{E} \longrightarrow \operatorname{coker}(\Phi_{\mathcal{E}}),$$

and vanishing of a morphism of finite quasi-coherent modules is faithfully flat local. The condition therefore descends to $\mathcal{E}$. Thus the height-at-most-$h$ conditions are equivalent, and in particular the least possible heights agree. □

This lemma gives the Kisin-module description of prismatic $F$-crystals. Recall that $e$ is the absolute ramification index of $\mathcal{O}_K$.

### 3.2. **Projectivity under small ramification.**

**Proposition 3.6.** *Let $(\mathfrak{M}, \varphi_{\mathfrak{M}})$ be an analytic Kisin module over $\mathfrak{S}_R$ of height at most $h$ and suppose $eh < p-1$. Then $\mathfrak{M}$ is finite projective.*

*Proof.* Put $\mathfrak{S} = \mathfrak{S}_R$. Fix a maximal ideal $\mathfrak{q}$ of $\mathfrak{S}$, denote the $\mathfrak{q}$-adic completion of $\mathfrak{S}$ by $\mathfrak{S}_{\mathfrak{q}}^{\wedge}$. Write $\mathfrak{M}_{\mathfrak{q}}^{\wedge} := \mathfrak{S}_{\mathfrak{q}}^{\wedge} \otimes_{\mathfrak{S}} \mathfrak{M}$, and $(\mathfrak{M}^{\wedge})_{\mathfrak{q}}^* := \mathfrak{S}_{\mathfrak{q}}^{\wedge} \otimes_{\varphi, \mathfrak{S}} \mathfrak{M}$. It suffices to show that $\mathfrak{M}_{\mathfrak{q}}^{\wedge}$ is projective at every such maximal ideal $\mathfrak{q}$ of $\mathfrak{S}$. The Frobenius $\varphi$ on $\mathfrak{S}$ uniquely extends to $\mathfrak{S}_{\mathfrak{q}}^{\wedge}$, and by the proof of [DLMS24a, Lem 2.7], $\varphi$ is faithfully flat on $\mathfrak{S}_{\mathfrak{q}}^{\wedge}$. By the proof of [DLMS24b, Prop. 5.4], we can find some $s_i \in \mathfrak{q}$ so that

$$\mathfrak{S}_{\mathfrak{q}}^{\wedge} = \begin{cases} W(k_{\mathfrak{q}})[[s_1, \dots, s_{d-1}]][[u]] \\ W(k_{\mathfrak{q}})[[s_1, \dots, s_{d-a}]][[T_1, \dots, T_a]] \quad 2 \leq a \leq d, \end{cases} \tag{3.1}$$

for a finite extension $k_{\mathfrak{q}}$ of $k$. The first case occurs when $\mathfrak{q}$ contains exactly one $T_i$, and the second when (up to renaming) $T_1, \dots, T_a \in \mathfrak{q}$ for $a \geq 2$. In the first case, $\mathfrak{q} = (p, u, s_1, \dots, s_{d-1})$, and in the second, $\mathfrak{q} = (p, T_1, \dots, T_a, s_1, \dots, s_{d-a})$. The following argument does not distinguish between these cases, so we simply write

$$\mathfrak{S}_{\mathfrak{q}}^{\wedge} = W(k_{\mathfrak{q}})[[x_1, \dots, x_d]]$$

with $x_1 = u$ and $x_i = s_{i-1}$ for $2 \leq i \leq d$ in the first case, and $x_i = T_i$ for $1 \leq i \leq a$ and $x_i = s_{i-a}$ for $a+1 \leq i \leq d$. We write $u = wv$ with $w = u$ and $v = 1$ in the first case, and $w = x_1 \cdots x_a$ and $v = T_{a+1} \cdots T_m \in \mathfrak{S}_{\mathfrak{q}}^{\times}$ in the second. Let $r = eh$, and note that $E^h \equiv u^{eh} = u^r = w^r v^r \bmod p$. We now suppress most $\mathfrak{q}$-subscripts for simplicity. By flatness, we have that $(\mathfrak{M}^{\wedge})^* := \mathfrak{S}^{\wedge} \otimes_{\varphi, \mathfrak{S}_{\mathfrak{q}}} \mathfrak{M}_{\mathfrak{q}}$ is $E$-torsion-free, and the height-condition supplies maps

$$\Phi : (\mathfrak{M}^{\wedge})^* \to \mathfrak{M}^{\wedge} \quad \text{and} \quad \Psi : \mathfrak{M}^{\wedge} \to (\mathfrak{M}^{\wedge})^*$$

with $\Phi\Psi = E^h$ and $\Psi\Phi = E^h$. Indeed, $\Phi$ is injective as it becomes an isomorphism after inverting $E$. Let $\overline{\mathfrak{S}} = \mathfrak{S}^{\wedge}/p$ and $\overline{\mathfrak{M}} := \mathfrak{M}^{\wedge}/p$. Reducing modulo $p$ and absorbing the unit into $\Psi$, we obtain $\overline{\mathfrak{S}}$-linear maps

$$f : \overline{\mathfrak{M}}^* \to \overline{\mathfrak{M}} \quad \text{and} \quad g : \overline{\mathfrak{M}} \to \overline{\mathfrak{M}}^*$$

satisfying $fg = w^r$ and $gf = w^r$. Moreover, $\overline{\mathfrak{M}}[1/w]$ is projective, since $\mathfrak{M}[1/E]$ is projective.

Fix $i > 0$ and set $H = \operatorname{Ext}^i_{\overline{\mathfrak{S}}}(\overline{\mathfrak{M}}, \overline{\mathfrak{S}})$. These are finite modules, and $H[1/w] = 0$ because $\overline{\mathfrak{M}}_{\mathfrak{q}}[1/w]$ is projective. Flatness of $\varphi$ implies

$$\operatorname{Ext}^i_{\overline{\mathfrak{S}}}(\overline{\mathfrak{M}}^*, \overline{\mathfrak{S}}) \simeq H^* := \overline{\mathfrak{S}} \otimes_{\varphi, \overline{\mathfrak{S}}} H.$$

The maps $f$ and $g$ induce maps $f : H \to H^*$ and $g : H^* \to H$ (arrows reversed since $\mathrm{Ext}^i_{\overline{\mathfrak{S}}}(-, \overline{\mathfrak{S}})$ is contravariant) whose composites $gf$ and $fg$ give multiplication by $w^r$. Since $H[1/w] = 0$ and $H$ is a finite $\overline{\mathfrak{S}}$-module, we have $w^n H = 0$ for some $n \geq 0$. Choose the smallest such $n$; we aim to show $n = 0$, i.e. $H = 0$. Since $w^r$ on $H^*$ factors through $H_i$, we have $w^{n+r}H^* = 0$. Write $n + r = pq + t$ with $0 \leq t < p$. Then for any $z \in H$,

$$0 = w^{n+r}(1 \otimes z) = w^t \otimes w^q z \quad \text{in } H^*.$$

Now note that the monomials $x^\alpha = x_1^{\alpha_1} \cdots x_d^{\alpha_d}$ with $0 \leq \alpha_i < p$ form a basis of $\overline{\mathfrak{S}}$ as a module over itself via the Frobenius, and $w^t = x_1^t \cdots x_a^t$ is one of these basis elements. Hence $H \to H^* : z \mapsto w^t \otimes z$ is injective, hence $w^q z = 0$. Since $z$ was arbitrary, $w^q H = 0$. If $n > 0$, then $q = \lfloor \frac{n+r}{p} \rfloor < n$ since $r < p - 1$, contradicting the minimality of $n$. Hence $n = 0$ and $H = 0$ for any $i > 0$, meaning that the $\overline{\mathfrak{M}}$ is projective. Since $\mathfrak{M}^\wedge$ is finitely-generated, torsion-free, and complete, it is itself projective. Since $\mathfrak{q}$ was arbitrary and completion is faithfully flat, $\mathfrak{M}$ is projective. □

*Remark* 3.7. Under the first case of (3.1), the proof of [LM20, Prop. 4.3] shows that $\mathfrak{M}$ is projective, but this argument does not handle the second case.

*Remark* 3.8. The restriction functor $\mathrm{Vect}^\varphi((X, M_X)_{\mathbb{\Delta}}) \to \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$ realizes prismatic $F$-crystals as a full subcategory of analytic prismatic $F$-crystals. By Proposition 3.6, an analytic prismatic $F$-crystal of height at most $h$ is a genuine prismatic $F$-crystal whenever $eh < p - 1$.

3.3. **Laurent $F$-crystals and the étale realization.** Let $(X, M_X)$ be a bounded fine saturated $p$-adic log formal scheme. We recall the definition of Laurent $F$-crystals on the absolute log prismatic site and their relation to $\mathbb{Z}_p$-local systems on the generic fiber $X_\eta$ of $X$.

**Definition 3.9** ([DLMS24b, Def. 3.12])**.** A *Laurent $F$-crystal* on $(X, M_X)_{\mathbb{\Delta}}$ consists of a pair $(\mathcal{E}, \varphi_\mathcal{E})$ where $\mathcal{E}$ is a crystal in vector bundles on the ringed site $((X, M_X)_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]_p^\wedge)$ and $\varphi_\mathcal{E} : \mathcal{E} \to \mathcal{E}$ is $\varphi$-semilinear, with linearization

$$\Phi_\mathcal{E} : \varphi^*\mathcal{E} \xrightarrow{\sim} \mathcal{E}$$

an isomorphism. Write $\mathrm{Vect}((X, M_X)_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]_p^\wedge)^{\varphi=1}$ for the category of Laurent $F$-crystals on $(X, M_X)_{\mathbb{\Delta}}$.

Let $R$ be a small affine, or let $R = \mathcal{O}_L$ be a CDVR, and let $(X, M_X) = (\mathrm{Spf}\, R, M_{\mathrm{Spf}\, R})$ be the log $p$-adic formal scheme from § 2.1. With the notation from the end of 2.1, let $G_R = \mathrm{Gal}(\overline{R}[p^{-1}]/R[p^{-1}])$ if $R$ is small affine, and let $G_R = \mathrm{Gal}(\overline{L}/L)$ if $R = \mathcal{O}_L$.

**Theorem 3.10** ([DLMS24b, Thrm. 3.14, Rem. 3.15])**.** *Assume that $(X, M_X)$ is semistable or the log formal spectrum of a CDVR. There are natural equivalences of categories*

$$\mathrm{Vect}((X, M_X)_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]_p^\wedge)^{\varphi=1} \xrightarrow{\sim} \mathrm{Loc}_{\mathbb{Z}_p}(X_\eta) \xrightarrow{\sim} \mathrm{Rep}_{\mathbb{Z}_p}(G_R)$$

*mapping a Laurent $F$-crystal $(\mathcal{E}, \varphi_\mathcal{E})$ to $\mathcal{E}(\mathbb{A}_{\mathrm{inf}}(R))^{\varphi=1}$.*

*Remark* 3.11. The theorem above, as stated in [DLMS24b], uses the *strict* prismatic site, but [DLMS24b, Rem. 3.6] states that restriction of prismatic $F$-crystals from the usual prismatic site to the strict one is an equivalence. We work exclusively with the general (nonstrict) prismatic site, as the technical simplifications afforded by strictness do not enter this paper.

3.4. **Prismatic semistability and purity.**

**Definition 3.12** ([DLMS24b, Def. 3.17], [DLMS24b, Def. 3.26])**.** Let $(X, M_X)$ be a semistable $p$-adic formal scheme or the log formal spectrum of a CDVR, and let $R = \Gamma(X, \mathcal{O}_X)$.

(1) The *étale realization functor* is defined to be the composite

$$T_{\mathrm{et}} : \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \to \mathrm{Vect}((X, M_X)_{\mathbb{\Delta}}, \mathcal{O}_{\mathbb{\Delta}}[\mathcal{I}_{\mathbb{\Delta}}^{-1}]_p^\wedge)^{\varphi=1} \xrightarrow[\text{Thrm. 3.10}]{\simeq} \mathrm{Loc}_{\mathbb{Z}_p}(X_\eta).$$

(2) By [DLMS24b, Prop. 3.20], the étale realization functor is fully faithful. A $\mathbb{Z}_p$-local system $\mathbb{L}$ on $X_\eta$ is *prismatic semistable* if it is in the essential image of $T_{\mathrm{et}}$.

(3) Let $R = \Gamma(X, \mathcal{O}_X)$, and let $V$ be a $G_R$-representation. We say that $V$ is *prismatic semistable* if the associated local system $\mathbb{L}$ on $X_\eta$ is prismatic semistable.

*Remark* 3.13. We also write $T_{\mathrm{et}}$ for the composite

$$\mathrm{DD}_{\mathfrak{S}_R} \to \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \xrightarrow{T_{\mathrm{et}}} \mathrm{Loc}_{\mathbb{Z}_p}(X_\eta)$$

where the first arrow is the quasi-inverse to the functor of Lemma 3.5.

Let $(X, M_X)$ be a semistable $p$-adic log formal scheme over $\mathcal{O}_K$, and retain the notation $\xi_j$, $(\Delta_j, M_{\Delta_j})$, and $\iota_j : (\Delta_j, M_{\Delta_j}) \to (X, M_X)$ for its Shilov data from Definition 2.7. The morphisms $\iota_j$ induce maps $(\iota_j)_\eta : (\Delta_j)_\eta \to X_\eta$ on generic fibers. The main theorem of [DLMS24b] is the following purity theorem.

**Theorem 3.14** ([DLMS24b, Thrm. 4.1, Thrm. 5.1, Cor. 5.5])**.** *Let $X$ be a semistable formal scheme, and let $\mathbb{L}$ be a $\mathbb{Z}_p$-local system on $X_\eta$. The following are equivalent:*

(a) *$\mathbb{L}$ is prismatic semistable;*
(b) *$(\iota_j)_\eta^*\mathbb{L}$ is prismatic semistable for each $j = 1, \dots, m$;*
(c) *$(\iota_j)_\eta^*\mathbb{L}$ is $\mathcal{O}\mathbb{B}_{\mathrm{st}}$-admissible for each $j = 1, \dots, m$.*

3.5. **The integral crystalline and isocrystalline realizations.** Let $(X, M_X)$ be a small affine or the log formal spectrum of a CDVR, and set $(X_1, M_{X_1}) := (X, M_X) \otimes_{\mathbb{Z}_p} \mathbb{Z}_p/p$. On this "special fiber," we consider the absolute log crystalline site $(X_1, M_{X_1})_{\mathrm{CRIS}}$ (cf. [DLMS24b, Def. B.1]). Write $\mathrm{Vect}^\varphi((X_1, M_{X_1})_{\mathrm{CRIS}})$ for the category of locally free $F$-crystals on this site. Thus an object is a locally free crystal $\mathcal{E}_{\mathrm{cris}}$ equipped with a semilinear Frobenius

$$\varphi_{\mathcal{E}_{\mathrm{cris}}} : \mathcal{E}_{\mathrm{cris}} \longrightarrow \mathcal{E}_{\mathrm{cris}}$$

whose linearization

$$\Phi_{\mathcal{E}_{\mathrm{cris}}} : \varphi^*\mathcal{E}_{\mathrm{cris}} \longrightarrow \mathcal{E}_{\mathrm{cris}}$$

is an isogeny. We recall the constructions from [DLMS24b, §3.5]. Fix $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^\varphi((X, M_X)_{\mathbb{\Delta}})$, and let $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon) \in \mathrm{DD}_{\mathfrak{S}}$ be the associated Breuil–Kisin module with descent datum. For each $n \geq 1$, set

$$\mathscr{M}_n := S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}/p^n\mathfrak{M}.$$

Taking the base change of $\varepsilon$ gives an isomorphism

$$\eta_n : S^{(1)} \otimes_{p_0,S} \mathscr{M}_n \xrightarrow{\sim} S^{(1)} \otimes_{p_1,S} \mathscr{M}_n$$

satisfying the cocycle condition; such data constitute an *HPD-stratification* ([DLMS24b, Def. B.23]). The object

$$(\mathscr{M}, \eta) := \varprojlim_n (\mathscr{M}_n, \eta_n)$$

determines a locally free crystal $\mathcal{E}_{\mathrm{cris}}$ on $(X_1, M_{X_1})_{\mathrm{CRIS}}$ ([DLMS24b, Prop. B.25]). The HPD-stratification functorially determines a quasi-nilpotent integrable connection

$$\nabla : \mathscr{M} \to \mathscr{M} \otimes_S \omega^1_{(S,M_S)/\mathbb{Z}_p},$$

and the pairs $(\mathscr{M}, \eta)$ and $(\mathscr{M}, \nabla)$ determine one another (cf. *loc. cit.*). The semilinear Frobenius $\varphi_{\mathfrak{M}}$, equivalently its linearization $\Phi_{\mathfrak{M}}$, induces after inverting $p$ a semilinear Frobenius $\varphi_{\mathscr{M}[p^{-1}]}$ on $\mathscr{M}[p^{-1}]$ whose linearization $\Phi_{\mathscr{M}[p^{-1}]}$ is an isomorphism and is compatible with $\eta[p^{-1}]$. We therefore obtain the $F$-isocrystal $(\mathcal{E}_{\mathrm{cris},\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathrm{cris},\mathbb{Q}}}) \in \mathrm{Vect}^\varphi_{\mathbb{Q}}((X_1, M_{X_1})_{\mathrm{CRIS}})$ determined by the triple

$$(\mathscr{M}[p^{-1}], \varphi_{\mathscr{M}[p^{-1}]}, \eta[p^{-1}]),$$

as in [DLMS24b, Lem. 3.34].

Suppose now that $\mathcal{E}_{\mathbb{\Delta}}$ is effective. Then $\varphi_{\mathfrak{M}}$ takes values in $\mathfrak{M}$, and the formula

$$\varphi_{\mathscr{M}}(s \otimes m) := \varphi_S(s) \otimes \varphi_{\mathfrak{M}}(m)$$

defines a semilinear Frobenius on $\mathscr{M} = S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}$. Frobenius compatibility of the descent datum $\varepsilon$ implies that $\varphi_{\mathscr{M}}$ is compatible with the HPD-stratification $\eta$, and hence defines a Frobenius $\varphi_{\mathcal{E}_{\mathrm{cris}}}$

on the corresponding crystal. Since $\Phi_{\mathfrak{M}}$ is an isomorphism after inverting $E$ and $\varphi(E) = pc_1$ with $c_1 \in S^\times$, the linearization $\Phi_{\mathcal{E}_{\mathrm{cris}}}$ is an isogeny. Thus $(\mathcal{E}_{\mathrm{cris}}, \varphi_{\mathcal{E}_{\mathrm{cris}}})$ is a locally free $F$-crystal.

**Definition 3.15.** The *integral crystalline realization* is the functor

$$D_{\mathrm{cris}} : \mathrm{Vect}^{\varphi,\mathrm{eff}}((X, M_X)_{\mathbb{\Delta}}) \longrightarrow \mathrm{Vect}^{\varphi}((X_1, M_{X_1})_{\mathrm{CRIS}}), \quad (\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \longmapsto (\mathcal{E}_{\mathrm{cris}}, \varphi_{\mathcal{E}_{\mathrm{cris}}}).$$

The rational construction above extends canonically to analytic prismatic $F$-crystals and defines the *isocrystalline realization* functor

$$D_{\mathrm{isoc}} : \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \longrightarrow \mathrm{Vect}^{\varphi}_{\mathbb{Q}}((X_1, M_{X_1})_{\mathrm{CRIS}}), \quad \mathcal{E}_{\mathbb{\Delta}} \longmapsto (\mathcal{E}_{\mathrm{cris},\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathrm{cris},\mathbb{Q}}}).$$

If $\mathcal{E}_{\mathbb{\Delta}}$ has height at most $h$, then its integral crystalline realization satisfies the analogous height condition

$$p^h \mathcal{E}_{\mathrm{cris}} \subseteq \Phi_{\mathcal{E}_{\mathrm{cris}}}(\varphi^* \mathcal{E}_{\mathrm{cris}}) \subseteq \mathcal{E}_{\mathrm{cris}}.$$

Indeed, this follows by applying $S \otimes_{\varphi,\mathfrak{S}} (-)$ to the corresponding inclusions for $\Phi_{\mathfrak{M}}$ and using $\varphi(E) = pc_1$ with $c_1 \in S^\times$.

*Remark* 3.16. This notation differs slightly from [Ino25, Def. 6.14], replacing his $T$ with $D$. Moreover, [DLMS24b] refers to the isocrystalline realization as the "crystalline realization". For every effective prismatic $F$-crystal there is a natural identification

$$D_{\mathrm{cris}}(\mathcal{E}_{\mathbb{\Delta}})[1/p] \simeq D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}}).$$

We now use the conventions of [DLMS24b, §3.6] on period rings and period sheaves. Fix a locally free $F$-isocrystal $(\mathcal{E}_{\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathbb{Q}}})$ on $(X_1, M_{X_1})_{\mathrm{CRIS}}$. Let $U \in \mathrm{Perfd}\,/X_{\eta,\mathrm{pro\acute{e}t}}$, and let $\widehat{U} = \mathrm{Spa}(A, A^+)$ be the associated affinoid perfectoid. Set

$$\mathcal{E}_{\mathbb{Q}}(\mathbb{B}^+_{\mathrm{cris}})(U) := \mathcal{E}_{\mathbb{Q}}(\mathbb{A}_{\mathrm{cris}}(A^+)) \otimes_{\mathbb{B}^+_{\mathrm{cris}}(A^+)} \mathbb{B}^+_{\mathrm{cris}}(U).$$

For a $\mathbb{Z}_p$-local system $\mathbb{L}$ on $X_{\eta,\mathrm{pro\acute{e}t}}$, we say that $\mathbb{L}$ and $(\mathcal{E}_{\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathbb{Q}}})$ are *associated* if there is an isomorphism

$$\alpha_{\mathrm{cris},\mathbb{L},\mathcal{E}_{\mathbb{Q}}} : \mathcal{E}_{\mathbb{Q}}(\mathbb{B}_{\mathrm{cris}}) \xrightarrow{\simeq} \mathbb{L} \otimes_{\mathbb{Z}_p} \mathbb{B}_{\mathrm{cris}}$$

of sheaves of $\mathbb{B}_{\mathrm{cris}}$-modules on $X_{\eta,\mathrm{pro\acute{e}t}}$ that is compatible with Frobenius. Let $\mathfrak{M} \in \mathrm{DD}_{\mathfrak{S}}$ denote the Breuil–Kisin module with descent datum associated to $\mathcal{E}_{\mathbb{\Delta}}$, and let $T = \mathcal{E}(\mathbb{A}_{\mathrm{inf}})^{\varphi=1}$ be the $G_R$-representation associated to $\mathcal{E}_{\mathbb{\Delta}}$ (more precisely, $T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$), as above. By [DLMS24b, Prop. 3.42], $\mathbb{L} := T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$ and $D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}})$ are associated. In our setting (the small affine or CDVR case), this means that there are $(\varphi, G_R)$-equivariant isomorphisms

$$\mathfrak{M} \otimes_{\mathfrak{S}} \mathbb{B}_{\mathrm{cris}}(\overline{R}) \simeq \mathscr{M} \otimes_S \mathbb{B}_{\mathrm{cris}}(\overline{R}) \simeq T \otimes_{\mathbb{Z}_p} \mathbb{B}_{\mathrm{cris}}(\overline{R}).$$

3.6. **The de Rham realization.** Let $\mathrm{MIC}(X_\eta)$ denote the category of vector bundles with integrable connection on $X_\eta$. A *filtered $F$-isocrystal* on $(X, M_X)$ is a pair $((\mathcal{E}_{\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathbb{Q}}}), (\mathbb{E}, \nabla_{\mathbb{E}}, \mathrm{Fil}^\bullet \mathbb{E}))$ consisting of an $F$-isocrystal $(\mathcal{E}_{\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathbb{Q}}})$ and a triple $(\mathbb{E}, \nabla_{\mathbb{E}}, \mathrm{Fil}^\bullet \mathbb{E})$, where $(\mathbb{E}, \nabla_{\mathbb{E}}) = \mathrm{MIC}(\mathcal{E}_{\mathbb{Q}}, \varphi_{\mathcal{E}_{\mathbb{Q}}})$ and $\mathrm{Fil}^\bullet$ is a $\mathbb{Z}$-indexed decreasing, separated, and exhaustive filtration by $\mathcal{O}_{X_\eta}$-submodules of $\mathbb{E}$ satisfying *Griffiths transversality* ($\nabla(\mathrm{Fil}^i \mathbb{E}) \subset \mathrm{Fil}^{i-1} \mathbb{E} \otimes_{\mathcal{O}_{X_\eta}} \omega^1_{X_\eta/\mathbb{Z}_p}$). We denote the category of filtered vector bundles with integrable connection by $\mathrm{MIC}^{\mathrm{fil}}(X_\eta)$ and the category of filtered $F$-isocrystals by $\mathrm{FilIsoc}^{\varphi}(X, M_X)$.

Now we use the de Rham period sheaves $\mathbb{B}^+_{\mathrm{dR}}$, $\mathbb{B}_{\mathrm{dR}}$, $\mathcal{O}\mathbb{B}^+_{\mathrm{dR}}$, and $\mathcal{O}\mathbb{B}_{\mathrm{dR}}$ on $X_{\eta,\mathrm{pro\acute{e}t}}$ defined in [Sch13, Def. 6.1, 6.8]. Given $T \in \mathrm{Rep}_{\mathbb{Z}_p}(G_R)$, we set

$$\begin{aligned} D_{\mathrm{dR}}(T) &:= (T \otimes_{\mathbb{Z}_p} \mathcal{O}\mathbb{B}_{\mathrm{dR}}(\overline{R}))^{G_R} \\ \nabla_{D_{\mathrm{dR}}(T)} &:= \mathrm{id}_T \otimes \nabla_{\mathcal{O}\mathbb{B}_{\mathrm{dR}}(\overline{R})} \\ \mathrm{Fil}^\bullet D_{\mathrm{dR}}(T) &:= (T \otimes_{\mathbb{Z}_p} \mathrm{Fil}^\bullet \mathcal{O}\mathbb{B}_{\mathrm{dR}}(\overline{R}))^{G_R} \end{aligned}$$

**Definition 3.17.** The triple $(D_{\mathrm{dR}}(T), \nabla_{D_{\mathrm{dR}}(T)}, \mathrm{Fil}^\bullet D_{\mathrm{dR}}(T)) \in \mathrm{MIC}^{\mathrm{fil}}(X_\eta)$ is the *de Rham realization* of $T$, and we will often shorten this to $D_{\mathrm{dR}}(T)$ for simplicity. We say the representation $T$ is *de Rham* if $\mathrm{rank}_{\mathbb{Z}_p} T = \mathrm{rank}_{R[p^{-1}]} D_{\mathrm{dR}}(T)$. An integer $i \in \mathbb{Z}$ is a *Hodge–Tate weight* of $T$ if

$$\mathrm{gr}^{-i} D_{\mathrm{dR}}(T) \neq 0.$$

In particular, $\mathbb{Q}_p(1)$ has Hodge–Tate weight 1, and twisting shifts weights by

$$\mathrm{HT}(T(n)) = \mathrm{HT}(T) + n.$$

For integers $a \leq b$, we write $\mathrm{Rep}^{\mathrm{st}}_{\mathbb{Z}_p,[a,b]}(G_R)$ for the category of prismatic semistable $\mathbb{Z}_p$-representations whose Hodge–Tate weights lie in $[a,b]$, and abbreviate $\mathrm{Rep}^{\mathrm{st},h}_{\mathbb{Z}_p}(G_R) := \mathrm{Rep}^{\mathrm{st}}_{\mathbb{Z}_p,[0,h]}(G_R)$. When $R = \mathcal{O}_L$ is a CDVR, prismatic semistability agrees with classical semistability.

If $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$, then the tuple

$$\mathbb{D}_{\mathrm{dR}}(\mathcal{E}_{\mathbb{\Delta}}) := (D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}}), D_{\mathrm{dR}}(T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})))$$

is the *de Rham realization* of $\mathcal{E}_{\mathbb{\Delta}}$.

By [DLMS24b, Prop. 3.45], the de Rham realization $\mathbb{D}_{\mathrm{dR}}(\mathcal{E}_{\mathbb{\Delta}})$ is a filtered $F$-isocrystal.

3.7. **The Nygaard-filtered realization and Hodge–Tate weights.** We define

$$D_{\mathrm{HT}} : \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \xrightarrow{D_{\mathrm{isoc}}} \mathrm{Vect}^{\varphi}_{\mathbb{Q}}((X_1, M_{X_1})_{\mathrm{CRIS}}) \xrightarrow{\mathrm{MIC}} \mathrm{MIC}(X_\eta)$$

where the second arrow is described in [DLMS24b, Rem. B.33]. Now fix $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$, and let $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon) \in \mathrm{DD}(\mathfrak{S})$ be the associated Breuil–Kisin module with descent datum, and $(\mathscr{M}, \varphi_{\mathscr{M}}, \eta)$ denote its base change along $\varphi : \mathfrak{S} \to S$. In the small affine case, its global sections are given by

$$\Gamma(X_\eta, D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}})) \simeq \mathfrak{M}^*[p^{-1}]/E\mathfrak{M}^*[p^{-1}] \simeq \mathscr{M}[p^{-1}] \otimes_{S_R[p^{-1}],q} R[p^{-1}],$$

where $q : S_R[p^{-1}] \to R[p^{-1}]$ maps $\gamma_i(E(u))$ to 0 for all $i \geq 1$. In the CDVR case, the last term reads

$$D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}) \simeq \mathscr{M}[p^{-1}] \otimes_{S_L[p^{-1}],q} L,$$

where $q : S_L[p^{-1}] \to L$ has the same description. In both cases, the connection is obtained from $\nabla : \mathscr{M} \to \mathscr{M} \otimes_S \omega^1_{(S,M_S)/\mathbb{Z}_p}$ by inverting $p$ and then tensoring along $q$. For every $i \in \mathbb{Z}$, define

$$N^i\mathfrak{M}^* := \mathfrak{M}^*[1/p] \cap \Phi_{\mathfrak{M}}^{-1}\big(E^i\mathfrak{M}[1/p]\big) \subseteq \mathfrak{M}^*[1/p, 1/E],$$

where $E^i\mathfrak{M}[1/p]$ is interpreted as a submodule of $\mathfrak{M}[1/p, 1/E]$ when $i < 0$. Define a decreasing filtration on $D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}})$ by

$$\mathrm{Fil}^i_N D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}) := \mathrm{im}\left(N^i\mathfrak{M}^* \longrightarrow \mathfrak{M}^*[1/p]/E\mathfrak{M}^*[1/p]\right). \tag{3.2}$$

This construction is functorial in $\mathcal{E}_{\mathbb{\Delta}}$, since a morphism of analytic prismatic $F$-crystals induces a morphism of Breuil–Kisin modules commuting with the linearized Frobenii and therefore preserving the submodules $N^i\mathfrak{M}^*$. We set

$$D_{\mathrm{Nyg}}(\mathcal{E}_{\mathbb{\Delta}}) := \big(D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}), \nabla, \mathrm{Fil}^\bullet_N\big) \in \mathrm{MIC}^{\mathrm{fil}}(X_\eta),$$

and define the *Nygaard-filtered crystalline realization* by

$$\mathbb{D}_{\mathrm{Nyg}}(\mathcal{E}_{\mathbb{\Delta}}) := \big(D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}}), D_{\mathrm{Nyg}}(\mathcal{E}_{\mathbb{\Delta}})\big).$$

The following proposition shows, in particular, that $\mathrm{Fil}^\bullet_N$ has locally free graded pieces and satisfies Griffiths transversality, so that this construction defines a functor

$$\mathbb{D}_{\mathrm{Nyg}} : \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}}) \longrightarrow \mathrm{FilIsoc}^{\varphi}(X, M_X).$$

**Proposition 3.18.** *Let $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$.*

(a) *There is a natural isomorphism*

$$\mathbb{D}_{\mathrm{Nyg}}(\mathcal{E}_{\mathbb{\Delta}}) \xrightarrow{\sim} \mathbb{D}_{\mathrm{dR}}(\mathcal{E}_{\mathbb{\Delta}})$$

*of filtered $F$-isocrystals.*

(b) *Suppose that there are integers $a \leq b$ such that*

$$E^b\mathfrak{M}[1/p] \subseteq \Phi_{\mathfrak{M}}\big(\mathfrak{M}^*[1/p]\big) \subseteq E^a\mathfrak{M}[1/p] \quad \textit{inside } \mathfrak{M}[1/p, 1/E]. \tag{3.3}$$

*Then*

$$\mathrm{gr}^i D_{\mathrm{dR}}\big(T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})\big) = 0 \quad \textit{unless} \quad a \leq i \leq b.$$

*Equivalently, the Hodge–Tate weights of $T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$ lie in $[-b, -a]$.*

*In particular, if $\mathcal{E}_{\mathbb{\Delta}}$ is effective of height at most $h$, then $T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$ has Hodge–Tate weights in $[-h, 0]$.*

*Proof.* Under the identification

$$D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}) \simeq \mathfrak{M}^*[1/p] \otimes_{\mathfrak{S}_R[1/p]} R[1/p],$$

the filtration (3.2) is precisely the Nygaard filtration of [Ino25, §6.4]. Thus [Ino25, Prop. 6.23] identifies $D_{\mathrm{Nyg}}(\mathcal{E}_{\mathbb{\Delta}})$ with the de Rham realization $D_{\mathrm{dR}}(T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}}))$ as filtered vector bundles with integrable connection. The $F$-isocrystal in both filtered $F$-isocrystals is $D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}})$. This gives the asserted isomorphism of filtered $F$-isocrystals.

We prove the assertion concerning the range of the filtration. Suppose first that $i \leq a$. The right-hand inclusion in (3.3) gives

$$\Phi_{\mathfrak{M}}\big(\mathfrak{M}^*[1/p]\big) \subseteq E^a\mathfrak{M}[1/p] \subseteq E^i\mathfrak{M}[1/p].$$

Hence $N^i\mathfrak{M}^* = \mathfrak{M}^*[1/p]$ and thus $\mathrm{Fil}^i_N D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}) = D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}})$. Now suppose that $i \geq b+1$. The left-hand inclusion in (3.3) implies that

$$\Psi_b : \mathfrak{M}[1/p] \longrightarrow \mathfrak{M}^*[1/p], \quad y \longmapsto \Phi_{\mathfrak{M}}^{-1}(E^b y),$$

is well-defined, and satisfies $\Phi_{\mathfrak{M}} \circ \Psi_b = E^b \,\mathrm{id}_{\mathfrak{M}[1/p]}$. If $x \in N^i\mathfrak{M}^*$, write $\Phi_{\mathfrak{M}}(x) = E^i y$ for some $y \in \mathfrak{M}[1/p]$. Then

$$\Phi_{\mathfrak{M}}(x) = E^{i-b}\Phi_{\mathfrak{M}}\big(\Psi_b(y)\big) = \Phi_{\mathfrak{M}}\big(E^{i-b}\Psi_b(y)\big).$$

Since $\Phi_{\mathfrak{M}}$ is an isomorphism after inverting $E$, it follows that $x = E^{i-b}\Psi_b(y) \in E\mathfrak{M}^*[1/p]$. Thus the image of $N^i\mathfrak{M}^*$ vanishes in $\mathfrak{M}^*[1/p]/E\mathfrak{M}^*[1/p]$, and hence $\mathrm{Fil}^i_N D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}) = 0$ for all $i \geq b+1$. We deduce that

$$\mathrm{gr}^i_{\mathrm{Fil}_N} D_{\mathrm{HT}}(\mathcal{E}_{\mathbb{\Delta}}) = 0 \quad \text{for } i \notin [a, b].$$

By the filtered comparison above, the same holds for $D_{\mathrm{dR}}(T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}}))$. Since our convention is that $j$ is a Hodge–Tate weight precisely when $\mathrm{gr}^{-j} D_{\mathrm{dR}}(T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})) \neq 0$, this is precisely the statement that the Hodge–Tate weights lie in $[-b, -a]$. In particular, that $\mathcal{E}_{\mathbb{\Delta}}$ is effective of height at most $h$ means

$$E^h\mathfrak{M} \subseteq \Phi_{\mathfrak{M}}(\mathfrak{M}^*) \subseteq \mathfrak{M}.$$

Thus (3.3) holds with $a = 0$ and $b = h$, giving Hodge–Tate weights in $[-h, 0]$. □

## 4. Breuil–Kisin and Breuil Modules

We define our purely algebraic categories of Breuil–Kisin and Breuil modules, describe their compatibilities with base change, and extend Kim's equivalence [Kim15, Prop. 6.3] over semistable base rings (Proposition 4.12).

### 4.1. **Regular sequences and torsion-freeness.**

Let $R$ be a small affine with framing $\square : R_0 \to R$, and $\mathfrak{S}_R$ be the associated Breuil–Kisin prism. Let $\mathcal{O}_L$ be a CDVR, and $\mathfrak{S}_L$ be the associated Breuil–Kisin prism.

**Lemma 4.1.** *For any $i \geq 0$, the sequences $(p, E)$ and $(E, p)$ are regular in $\mathfrak{S}_R^{(i)}$, $\mathfrak{S}_L^{(i)}$, $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}})$, and $\mathbb{A}_{\mathrm{inf}}(\overline{R})$.*

*Proof.* This is immediate in the CDVR case, so we first consider the small affine Breuil–Kisin ring. Begin with the basic ring

$$\mathfrak{S}_{R_0} := W\left\langle T_1, \dots, T_m, T_{m+1}^{\pm 1}, \dots, T_d^{\pm 1}\right\rangle [[u]]/(T_1 \cdots T_m - u).$$

Clearly $p$ is a non-zero-divisor, and we have

$$\mathfrak{S}_{R_0}/p = \left(k[T_1, \dots, T_m, T_{m+1}^{\pm 1}, \dots, T_d^{\pm 1}]\right)^{\wedge}_{(T_1 \cdots T_m)},$$

where the displayed ring is the $(T_1 \cdots T_m)$-adic completion; equivalently, it retains the defining presentation with the formal variable $u$ and the relation $u = T_1 \cdots T_m$. This is a flat $k[[u]]$-algebra; hence $u^e = E(u) \bmod p$ is a non-zero-divisor in $\mathfrak{S}_{R_0}/p$. Since $p$ is in the Jacobson radical of $\mathfrak{S}_{R_0}$, it follows that $(E, p)$ is also a regular sequence. Now suppose $\mathfrak{S}_{R_0} \to \mathfrak{S}_R$ is $p$-completely étale. The map $\mathfrak{S}_{R_0}/p \to \mathfrak{S}_R/p$ is étale, and in particular flat, so $\mathfrak{S}_R/p$ is $E$-torsion-free. Note also that $\mathfrak{S}_R$ is classically $p$-complete. That $p$ is a non-zero-divisor in $\mathfrak{S}_R/(E)$ is now a consequence of the following lemma. The same argument applies to the self-products: their structure maps from $\mathfrak{S}_R$ are $(p, E)$-completely flat, so regularity is preserved after reduction modulo either member of the sequence. The statement for the self-products follows as the maps $\mathfrak{S} \to \mathfrak{S}^{(i)}$ are faithfully flat (Proposition 2.5(c)).

Now let $\mathbb{A}_{\mathrm{inf}} = \mathbb{A}_{\mathrm{inf}}(\overline{R})$ or $\mathbb{A}_{\mathrm{inf}}(\mathcal{O}_{\overline{L}})$. The element $p \in \mathbb{A}_{\mathrm{inf}}$ is regular, as it is in the ring of Witt vectors of any perfect ring of characteristic $p > 0$. Moreover, $E$ generates the kernel of Fontaine's $\theta$-map and is therefore a non-zero-divisor by [BMS18, Lem. 3.10]. The quotient $\mathbb{A}_{\mathrm{inf}}/(E)$ is $\widehat{\overline{R}}$ or $\widehat{\mathcal{O}}_{\overline{L}}$, respectively, and both rings are $p$-torsion-free. Hence $(E, p)$ is a regular sequence. Since $\mathbb{A}_{\mathrm{inf}}$ is $(p, E)$-complete, $(p, E)$ is also a regular sequence. □

**Lemma 4.2.** *Let $A \to B$ be a ring map such that (1) $A$ is $f$-torsion-free, (2) $A/f^n \to B/f^n$ is flat for every $n \geq 1$, and (3) $B$ is $f$-adically separated. Then $B$ is $f$-torsion-free.*

*Proof.* Suppose $fb = 0$ for some $b \in B$. Since $A$ is $f$-torsion-free,

$$0 \to A/f^n \xrightarrow{f} A/f^{n+1} \to A/f \to 0$$

is exact for any $n \geq 1$. Since $B/f^{n+1}$ is $A/f^{n+1}$-flat, so is the sequence

$$0 \to B/f^n \xrightarrow{f} B/f^{n+1} \to B/f \to 0.$$

The class of $b$ in $B/f^n$ maps to 0 in $B/f^{n+1}$, hence $b \in f^n B$. Hence $b \in f^n B$ for every $n$. Since $B$ is $f$-adically separated, we conclude that $b = 0$. □

4.2. **Breuil–Kisin modules and torsion variants.** Let $(A, (E), M_{\mathrm{Spf}\, A})$ be a Breuil–Kisin log prism, or one of its self-products, in the small affine or CDVR case. We write $'\mathrm{Mod}^{\varphi}(A)$ for the category of finitely generated $A$-modules $\mathfrak{M}$ equipped with a $\varphi_A$-semilinear endomorphism $\varphi_{\mathfrak{M}}$. Put $\mathfrak{M}^* := A \otimes_{\varphi, A} \mathfrak{M}$ and let $\Phi_{\mathfrak{M}} := 1 \otimes \varphi_{\mathfrak{M}} : \mathfrak{M}^* \longrightarrow \mathfrak{M}$ denote the linearization of $\varphi_{\mathfrak{M}}$. Morphisms in $'\mathrm{Mod}^{\varphi}(A)$ are $A$-linear maps commuting with Frobenius. We write $'\mathrm{Mod}^{\varphi,h}(A)$ for the category of *generalized Breuil–Kisin modules of height at most $h$*: its objects are objects of $'\mathrm{Mod}^{\varphi}(A)$ equipped with an $A$-linear map $\Psi_{\mathfrak{M}} : \mathfrak{M} \longrightarrow \mathfrak{M}^*$ such that $\Psi_{\mathfrak{M}} \circ \Phi_{\mathfrak{M}} = E^h \,\mathrm{id}_{\mathfrak{M}^*}$ and $\Phi_{\mathfrak{M}} \circ \Psi_{\mathfrak{M}} = E^h \,\mathrm{id}_{\mathfrak{M}}$. Morphisms are required to commute with both $\Phi$ and $\Psi$. When $\Phi_{\mathfrak{M}}$ is injective, the map $\Psi_{\mathfrak{M}}$ is uniquely determined by $\Phi_{\mathfrak{M}}$, and the displayed condition is equivalent to $\mathrm{coker}(\Phi_{\mathfrak{M}})$ being killed by $E^h$.

We let $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(A)$ be the full subcategory of $'\mathrm{Mod}^{\varphi,h}(A)$ consisting of objects whose underlying $A$-module is finite projective. Following [Kis09, §1.1.8], we define $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(A)$ to be the smallest full extension-closed subcategory of $'\mathrm{Mod}^{\varphi,h}(A)$ containing every object killed by $p$ whose underlying $A/(p)$-module is finite projective. Equivalently, an object belongs to $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(A)$ if it admits a finite filtration by Breuil–Kisin submodules whose successive quotients are killed by $p$ and finite projective over $A/(p)$. We call the objects of these two subcategories *projective* and *torsion Breuil–Kisin modules*, respectively, and a sequence in either category short exact if it is short exact on the underlying $A$-modules.

When $A = \mathfrak{S}$, an object $\mathfrak{M} \in {}'\mathrm{Mod}^{\varphi,h}(\mathfrak{S})$ is called *étale* if the natural map $\mathfrak{M} \to \mathfrak{M} \otimes_{\mathfrak{S}} \mathcal{O}_{\mathcal{E}_R}$ is injective. It is clear that because $E(u)$ is a unit in $\mathcal{O}_{\mathcal{E}}$, that $\Phi_{\mathfrak{M}}$ (and hence $\varphi_{\mathfrak{M}}$) is injective (cf. [LL23, §6]). We denote the full subcategory of étale objects by $\mathrm{Mod}^{\varphi,h}(\mathfrak{S}) \subset {}'\mathrm{Mod}^{\varphi,h}(\mathfrak{S})$. The projective and torsion categories just defined are full subcategories of $\mathrm{Mod}^{\varphi,h}(\mathfrak{S})$: this is immediate for projective objects, while for torsion objects it follows by dévissage from the regularity of $E$ on $\mathfrak{S}/(p)$ in Lemma 4.1. Thus, throughout the paper, an undecorated Breuil–Kisin module of height at most $h$ means an object of $\mathrm{Mod}^{\varphi,h}(\mathfrak{S})$.

*Remark* 4.3. For a Breuil–Kisin log prism $(\mathfrak{S}, (E), M_{\mathrm{Spf}\,\mathfrak{S}})$, the projective objects in the notation of Section 4.2 are precisely the projective Breuil–Kisin modules of height at most $h$ over its underlying prism from Definition 3.1, i.e.

$$\mathrm{Vect}^{\varphi,h}(\mathfrak{S}, (E)) = \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}).$$

The $\mathrm{Vect}^{\varphi,h}(\mathfrak{S}, (E))$-notation was used in the context of definition of analytic prismatic $F$-crystals. We will exclusively use the $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S})$ notation going forward.

4.3. **From Breuil–Kisin modules to Breuil modules.** Let $(A, (E), M_{\mathrm{Spf}\,A})$ denote a Breuil–Kisin log prism, or one of its self-products, in either the small affine or the CDVR case. Let $(B, (p), M_{\mathrm{Spf}\,B})$ be the corresponding Breuil log prism, and let $\varphi : (A, (E), M_{\mathrm{Spf}\,A}) \longrightarrow (B, (p), M_{\mathrm{Spf}\,B})$ denote the Frobenius morphism of log prisms. Throughout this subsection, let $\mathfrak{M}$ belong to $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(A)$ or $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(A)$.

**Lemma 4.4.** *Let $C$ be an $A$-algebra. Suppose either that $\mathfrak{M}$ is projective and $E$ is a non-zero-divisor on $C$, or that $\mathfrak{M}$ is torsion and $(p, E)$ is a regular sequence on $C$. Then*

$$\mathrm{id}_C \otimes \Phi_{\mathfrak{M}} : C \otimes_A \mathfrak{M}^* \longrightarrow C \otimes_A \mathfrak{M}$$

*is injective. In particular, when $A = \mathfrak{S}_R$, the extension of $\Phi_{\mathfrak{M}}$ along $\mathfrak{S}_R \to \mathbb{A}_{\mathrm{inf}}(\overline{R})$ is injective for both projective and torsion $\mathfrak{M}$.*

*Proof.* After tensoring the identity $\Psi_{\mathfrak{M}} \circ \Phi_{\mathfrak{M}} = E^h \,\mathrm{id}_{\mathfrak{M}^*}$ with $C$, every element in the kernel of $C \otimes_A \Phi_{\mathfrak{M}}$ is killed by $E^h$. If $\mathfrak{M}$ is projective, then $C \otimes_A \mathfrak{M}^*$ is finite projective over $C$ and hence $E$-torsion-free, proving the assertion.

Suppose that $\mathfrak{M}$ is torsion, and let

$$\mathfrak{M}^* = \mathfrak{M}_0^* \supset \mathfrak{M}_1^* \supset \cdots \mathfrak{M}_n^* \supset \mathfrak{M}_{n+1}^* = 0$$

be a filtration with $A/(p)$-projective quotients. Write $\mathfrak{M}_{i,C}^* := \mathfrak{M}_i^* \otimes_A C$. We claim that we still have a filtration

$$\mathfrak{M}_C^* = \mathfrak{M}_{0,C}^* \supset \mathfrak{M}_{1,C}^* \supset \cdots \mathfrak{M}_{n,C}^* \supset \mathfrak{M}_{n+1,C}^* = 0$$

and that the successive quotients are $C/(p)$-projective. Indeed, each sequence

$$0 \to \mathfrak{M}_{i+1}^* \to \mathfrak{M}_i^* \to \mathfrak{M}_i^*/\mathfrak{M}_{i+1}^* \to 0$$

is exact, and $\mathrm{Tor}_1^A(C, \mathfrak{M}_i^*/\mathfrak{M}_{i+1}^*) = 0$ for each $i$ because $C$ is $p$-torsion-free, and the corresponding successive quotients of $C \otimes_A \mathfrak{M}^*$ are finite projective over $C/(p)$ and are therefore $E$-torsion-free (since $E$ is a non-zero-divisor on $C/(p)$). It follows by *dévissage* that $C \otimes_A \mathfrak{M}^*$ is $E$-torsion-free, and the kernel again vanishes. The final assertion follows from Lemma 4.1. □

For $0 \le i \le p-1$, define the *Breuil–Kisin filtration* on $\mathfrak{M}^*$ by

$$\mathrm{Fil}^i \mathfrak{M}^* = \{x \in \mathfrak{M}^* \,:\, \Phi_{\mathfrak{M}}(x) \in E^i \mathfrak{M}\}.$$

The formula $a \otimes m \mapsto \varphi_A(a) \otimes \varphi_{\mathfrak{M}}(m)$ defines a $\varphi_A$-semilinear endomorphism $\varphi_{\mathfrak{M}^*}$ of $\mathfrak{M}^*$. By [LL23, §6.1],

$$\varphi_{\mathfrak{M}^*}(\mathrm{Fil}^i \mathfrak{M}^*) \subset \varphi((a_0^{-1}E)^i)\mathfrak{M}^*,$$

where $a_0 p = E(0)$. Thus we may define

$$\varphi_i : \mathrm{Fil}^i \mathfrak{M}^* \to \mathfrak{M}^*, \quad \varphi_i(x) := \frac{\varphi_{\mathfrak{M}^*}(x)}{\varphi((a_0^{-1}E)^i)}.$$

The maps $\Phi_{\mathfrak{M}}$ and multiplication by $E$ are injective by the preceding lemma applied with $C = A$ (using Lemma 4.1). Thus for $\bullet \in \{\mathrm{proj}, \mathrm{tor}\}$ and an object $\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\bullet}$,

$$\Psi_{\mathfrak{M}} : \mathfrak{M} \xrightarrow{\sim} \mathrm{Fil}^h \mathfrak{M}^*$$

is an isomorphism. We define the divided Frobenius intrinsically by

$$\varphi_h\big(\Psi_{\mathfrak{M}}(m)\big) := \varphi(a_0)^h(1 \otimes m).$$

This agrees with the preceding quotient formula whenever division is unique, since

$$\varphi_{\mathfrak{M}^*}\big(\Psi_{\mathfrak{M}}(m)\big) = \varphi(E)^h(1 \otimes m).$$

We now recall the associated Breuil-module data. For $i \geq 0$, write $\mathrm{Fil}^i B$ for the $p$-completed divided power filtration on $B$. We have that $\varphi(\mathrm{Fil}^i B) \subset p^i B$ for $0 \leq i \leq p-1$, so we define the *divided Frobenii* $\varphi_i : \mathrm{Fil}^i B \to B$ via $\varphi_i := p^{-i}\varphi$. We also set $c_1 = \varphi(E)/p \in B^\times$.

Let ${}'\mathrm{Mod}^{\varphi,h}(B)$ denote the category whose objects are triples $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h)$ consisting of

(1) a $B$-module $\mathscr{M}$,
(2) a $B$-submodule $\mathrm{Fil}^h \mathscr{M} \subset \mathscr{M}$ containing $\mathrm{Fil}^h B \cdot \mathscr{M}$.
(3) a $\varphi$-semilinear map $\varphi_h : \mathrm{Fil}^h \mathscr{M} \to \mathscr{M}$ such that for all $f \in \mathrm{Fil}^h B$ and $m \in \mathscr{M}$, we have
$$\varphi_h(fm) = c_1^{-h}\varphi_h(f)\varphi_h(E^h m).$$
(4) $\varphi_h(\mathrm{Fil}^h \mathscr{M})$ generates $\mathscr{M}$ as a $B$-module.

Morphisms in this category are maps on the underlying $B$-modules preserving the $\mathrm{Fil}^h$-submodules and commuting with $\varphi_h$-maps. We often simplify notation by writing $\mathscr{M}$ to represent the triple $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h)$. We call a sequence $0 \to \mathscr{M}' \to \mathscr{M} \to \mathscr{M}'' \to 0$ short exact in this category if it is short exact as a sequence of $B$-modules and induces a short exact sequence on the $\mathrm{Fil}^h$-submodules as well.

Let $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(B)$ be the full subcategory of ${}'\mathrm{Mod}^{\varphi,h}(B)$ consisting of objects for which $\mathscr{M}$ is finite projective over $B$ and the quotient $Q := \mathscr{M}/\mathrm{Fil}^h \mathscr{M}$, viewed as an $A$-module through the untwisted structural map $A \to B$, is finitely presented and satisfies $\mathrm{pd}_A(Q) \leq 1$. Equivalently, $Q$ admits a resolution of length at most one by finite projective $A$-modules. The ring $A$ paired with $B$ is part of this convention; in particular, for $B = S_R$ (resp. $S_L$) we use $A = \mathfrak{S}_R$ (resp. $\mathfrak{S}_L$). We define $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(B)$ to be the smallest full extension-closed subcategory containing every object killed by $p$ whose underlying $B/(p)$-module is finite projective. Equivalently, its objects admit a finite filtration by Breuil submodules whose successive quotients are killed by $p$ and finite projective over $B/(p)$. We call these the projective and torsion Breuil modules over $B$.

We now define the Breuil-module construction. For $\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\bullet}(A)$, where $\bullet \in \{\mathrm{proj}, \mathrm{tor}\}$, set $\mathscr{M} := B \otimes_{\varphi,A} \mathfrak{M}$,

$$\mathrm{Fil}^h \mathscr{M} := \{x \in \mathscr{M} \ : \ (1 \otimes \Phi_{\mathfrak{M}})(x) \in \mathrm{Fil}^h B \otimes_A \mathfrak{M}\}.$$

and define $\varphi_h : \mathrm{Fil}^h \mathscr{M} \to \mathscr{M}$ as the composite

$$\mathrm{Fil}^h \mathscr{M} \xrightarrow{1\otimes\Phi_{\mathfrak{M}}} \mathrm{Fil}^h B \otimes_A \mathfrak{M} \xrightarrow{\varphi_{B,h}\otimes 1} B \otimes_{\varphi,A} \mathfrak{M} = \mathscr{M}.$$

We call $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h)$ the *Breuil-module data associated to* $\mathfrak{M}$. We will often denote the resulting triple simply by $\mathscr{M}$.

**Lemma 4.5.** *For $\bullet \in \{\mathrm{proj}, \mathrm{tor}\}$, the construction above defines an exact functor*

$$\mathrm{Mod}^{\varphi,h,\bullet}(A) \longrightarrow \mathrm{Mod}^{\varphi,h,\bullet}(B).$$

*In particular, the construction commutes with reduction modulo $p^n$ on projective objects.*

*Proof.* For projective objects, this is [CL09, Prop. 2.1.2], which in turn relies on [Kis09, Prop. 1.1.11]; the required finite-projective resolution of the filtered quotient is verified directly in Lemma 4.8 below. We explain the extension to torsion objects. First suppose that $\mathfrak{M}$ is killed by $p$ and finite projective over $A/(p)$. Zariski locally on $\mathrm{Spec}(A/(p))$, the module $\mathfrak{M}$ is free, and the proof

of [Kis09, Prop. 1.1.11], in its height-at-most-$h$ form [LL23, Prop. 6.12], shows that the associated data is killed by $p$ and finite projective over $B/(p)$.

Let $0 \to \mathfrak{M}_1 \to \mathfrak{M}_2 \to \mathfrak{M}_3 \to 0$ be a short exact sequence in which $\mathfrak{M}_3$ is obtained by successive extensions of such $p$-torsion projective objects. The ring $B$ is $p$-torsion-free, so $\operatorname{Tor}_1^A(B, \mathfrak{P}) = 0$ for every $A/(p)$-projective module $\mathfrak{P}$, where $B$ is regarded as an $A$-module through Frobenius. By dévissage the same vanishing holds with $\mathfrak{P}$ replaced by $\mathfrak{M}_3$, so the associated sequence of underlying $B$-modules is exact. The filtration-lifting argument in the proof of [Kis09, Prop. 1.1.11] then shows that the sequence of $\operatorname{Fil}^h$-submodules is exact as well. Induction on the lengths of the defining filtrations proves exactness on $\operatorname{Mod}^{\varphi,h,\mathrm{tor}}(A)$ and shows that its image lies in $\operatorname{Mod}^{\varphi,h,\mathrm{tor}}(B)$. The final assertion follows immediately from the construction. □

*Remark* 4.6. The divided Frobenii $\varphi_{\mathfrak{M},h}$ and $\varphi_{\mathscr{M},h}$ differ by an $S^\times$ unit on $\operatorname{Fil}^h \mathfrak{M}^*$. Here, we view $\operatorname{Fil}^h \mathfrak{M}^*$ as an $\mathfrak{S}_R$-submodule of $\operatorname{Fil}^h \mathscr{M}$. By definition of the divided Frobenius (see §4.2), we have

$$\varphi_{\mathfrak{M},h}(x) = \varphi(a_0)^h (1 \otimes m)$$

while

$$\varphi_{\mathscr{M},h}(x) = \varphi_h(E^h) \otimes m = c_1^h (1 \otimes m).$$

Setting $c := c_1/\varphi(a_0)$, we deduce that $\varphi_{h,\mathscr{M}}(x) = c^h \varphi_{h,\mathfrak{M}}(x)$.

4.4. **Filtrations, base change, and resolutions of filtered quotients.**

**Lemma 4.7.** *Let $A \to A'$ be a flat, Frobenius-compatible morphism of Breuil–Kisin rings in the small affine or CDVR case, and let $B \to B'$ be the induced morphism of their Breuil rings. Assume that $B' \simeq A' \widehat{\otimes}_A B$ via the untwisted structural maps and that $B \to B'$ is $p$-completely flat, as for the Shilov maps of Lemma 2.15. Let $\mathfrak{M} \in \operatorname{Mod}^{\varphi,h,\bullet}(A)$, where $\bullet \in \{\mathrm{proj}, \mathrm{tor}\}$, put $\mathfrak{M}' := A' \otimes_A \mathfrak{M}$, and let $\mathscr{M}$ and $\mathscr{M}'$ denote the associated Breuil-module data. For every $0 \le i \le p-1$, the following hold:*

(a) $\operatorname{Fil}^i((\mathfrak{M}')^*) = \operatorname{Fil}^i \mathfrak{M}^* \otimes_A A'$;

(b) *Let $\operatorname{Fil}^i \mathscr{M}$ denote the Nygaard filtration, and let $F^i\mathscr{M}$ be the submodule*

$$F^i\mathscr{M} := \operatorname{im}\big(B \otimes_A \operatorname{Fil}^i \mathfrak{M}^* \longrightarrow \mathscr{M}\big) + \operatorname{Fil}^p B \cdot \mathscr{M},$$

*where the displayed map is induced by $\operatorname{Fil}^i \mathfrak{M}^* \hookrightarrow \mathfrak{M}^*$ and the canonical identification $B \otimes_A \mathfrak{M}^* \simeq \mathscr{M}$. Then*

$$F^i\mathscr{M} = \operatorname{Fil}^i \mathscr{M}$$

*and $\operatorname{Fil}^i \mathscr{M}$ is $p$-adically complete. If $\mathfrak{M}$ is projective, it is moreover $p$-saturated in $\mathscr{M}$. Under the canonical identification $\mathscr{M}' = B' \otimes_B \mathscr{M}$, one has*

$$B' \widehat{\otimes}_B \operatorname{Fil}^i \mathscr{M} \xrightarrow{\sim} \operatorname{Fil}^i \mathscr{M}'.$$

*In the torsion case the completed tensor product agrees with the ordinary tensor product.*

*Proof.* (a) The filtration on $\mathfrak{M}$ fits into an exact sequence

$$0 \to \operatorname{Fil}^i \mathfrak{M}^* \to \mathfrak{M}^* \to \mathfrak{M}/E^i\mathfrak{M}.$$

Tensoring along $A \to A'$, we obtain the exact sequence

$$0 \to \operatorname{Fil}^i \mathfrak{M}^* \otimes_A A' \to (\mathfrak{M}')^* \to (\mathfrak{M}/E^i\mathfrak{M}) \otimes_A A'$$

and the last term is $\mathfrak{M}'/E^i\mathfrak{M}'$, again by flatness of $A \to A'$. This argument applies in both the projective and torsion cases.

(b) That $F^i\mathscr{M} \subset \operatorname{Fil}^i \mathscr{M}$ is clear. Now fix $m \in \operatorname{Fil}^i \mathscr{M}$. Using $B = A + \operatorname{Fil}^p B$, choose $m' \in \mathfrak{M}^*$ and $m'' \in \operatorname{Fil}^p B \otimes_A \mathfrak{M}^*$ whose images in $\mathscr{M}$ sum to $m$. By definition, $(1 \otimes \Phi_{\mathfrak{M}})(m)$ lies in the image of $\operatorname{Fil}^i B \otimes_A \mathfrak{M}$, and the same is true of $(1 \otimes \Phi_{\mathfrak{M}})(m'')$ because $i \le p-1$. It follows that the image of $\Phi_{\mathfrak{M}}(m')$ vanishes in

$$(B/\operatorname{Fil}^i B) \otimes_A \mathfrak{M} \simeq (A/E^iA) \otimes_A \mathfrak{M} \simeq \mathfrak{M}/E^i\mathfrak{M}.$$

Thus $m' \in \operatorname{Fil}^i \mathfrak{M}^*$, and hence

$$m \in \operatorname{im}\big(B \otimes_A \operatorname{Fil}^i \mathfrak{M}^* \to \mathscr{M}\big) + \operatorname{Fil}^p B \cdot \mathscr{M} = F^i \mathscr{M}.$$

Put $Q_i = \mathscr{M} / \operatorname{Fil}^i \mathscr{M}$ and set $\mathscr{M}^\varphi := B \otimes_A \mathfrak{M}$. Note that $\operatorname{Fil}^i \mathscr{M}$ is the kernel of

$$1 \otimes \Phi_{\mathfrak{M}} : \mathscr{M} \to \mathscr{M}^\varphi / \operatorname{im}(\operatorname{Fil}^i B \otimes_A \mathfrak{M}),$$

and since $B / \operatorname{Fil}^i B = A/E^i$, the target is isomorphic to $\mathfrak{M}/E^i\mathfrak{M}$. Hence

$$Q_i = \operatorname{im}\big(\mathfrak{M}^* \xrightarrow{\Phi_{\mathfrak{M}}} \mathfrak{M}/E^i\mathfrak{M}\big). \tag{4.1}$$

Suppose first that $\mathfrak{M}$ is projective. Then $Q_i$ is finite over the Noetherian $p$-complete ring $A$, and is therefore $p$-complete. It is also $p$-torsion-free because $\mathfrak{M}/E^i\mathfrak{M}$ is. We deduce that $\operatorname{Fil}^i \mathscr{M}$ is closed in $\mathscr{M}$ and $\operatorname{Fil}^i \mathscr{M} \cap p^n \mathscr{M} = p^n \operatorname{Fil}^i \mathscr{M}$ for every $n$, establishing its completeness and $p$-saturation. Reducing $0 \to \operatorname{Fil}^i \mathscr{M} \to \mathscr{M} \to Q_i \to 0$ modulo $p^n$, tensoring with the flat $B/p^n$-algebra $B'/p^n$, and taking inverse limits gives an exact sequence

$$0 \to B' \widehat{\otimes}_B \operatorname{Fil}^i \mathscr{M} \to \mathscr{M}' \to B' \widehat{\otimes}_B Q_i \to 0$$

because the transition maps on the left are surjective. Lemma 2.15 gives $B' \widehat{\otimes}_B Q_i \simeq A' \otimes_A Q_i$, and flat base change along $A \to A'$ applied to (4.1) identifies $A' \otimes_A Q_i$ with $\mathscr{M}' / \operatorname{Fil}^i \mathscr{M}'$. Thus

$$B' \widehat{\otimes}_B \operatorname{Fil}^i \mathscr{M} \xrightarrow{\sim} \operatorname{Fil}^i \mathscr{M}'.$$

Now suppose that $\mathfrak{M}$ is torsion, and choose $n \geq 1$ such that $p^n \mathfrak{M} = 0$. Since $B \to B'$ is $p$-completely flat, $B'/p^n$ is flat over $B/p^n$. Tensoring the exact sequence

$$0 \to \operatorname{Fil}^i \mathscr{M} \to \mathscr{M} \to Q_i \to 0$$

with $B'/p^n$ therefore gives an exact sequence

$$0 \to B' \otimes_B \operatorname{Fil}^i \mathscr{M} \to \mathscr{M}' \to B' \otimes_B Q_i \to 0.$$

Thus the same argument as above gives

$$B' \widehat{\otimes}_B \operatorname{Fil}^i \mathscr{M} \xrightarrow{\sim} \operatorname{Fil}^i \mathscr{M}',$$

which finishes the proof. □

**Lemma 4.8.** *Let $\mathfrak{M} \in \operatorname{Mod}^{\varphi,h,\mathrm{proj}}(A)$, and let $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h)$ be its associated Breuil-module data over $B$. There is a natural isomorphism*

$$\mathscr{M} / \operatorname{Fil}^h \mathscr{M} \simeq \Phi_{\mathfrak{M}}(\mathfrak{M}^*)/E^h\mathfrak{M} \simeq \operatorname{coker}(\Psi_{\mathfrak{M}} : \mathfrak{M} \to \mathfrak{M}^*).$$

*In particular, $\mathscr{M} / \operatorname{Fil}^h \mathscr{M}$ has a resolution of length at most one by finite projective $A$-modules, and the Breuil module data is an object of $\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(B)$.*

*Proof.* The identification $B / \operatorname{Fil}^h B \simeq A/(E^h)$ and the defining preimage formula for $\operatorname{Fil}^h \mathscr{M}$ identify $Q$ with the image of

$$\mathfrak{M}^*/E^h\mathfrak{M}^* \longrightarrow \mathfrak{M}/E^h\mathfrak{M}.$$

Since $E^h\mathfrak{M} \subseteq \Phi_{\mathfrak{M}}(\mathfrak{M}^*)$, this image is $\Phi_{\mathfrak{M}}(\mathfrak{M}^*)/E^h\mathfrak{M}$. The identity $\Phi_{\mathfrak{M}}\Psi_{\mathfrak{M}} = E^h \operatorname{id}_{\mathfrak{M}}$ now gives the exact sequence

$$0 \longrightarrow \mathfrak{M} \xrightarrow{\Psi_{\mathfrak{M}}} \mathfrak{M}^* \longrightarrow \mathscr{M} / \operatorname{Fil}^h \mathscr{M} \longrightarrow 0.$$

Both terms on the left are finite projective over $A$, proving the assertion. Equivalently, one can use the concrete resolution $0 \to E^h\mathfrak{M} \to \Phi_{\mathfrak{M}}(\mathfrak{M}^*) \to \mathscr{M} / \operatorname{Fil}^h \mathscr{M} \to 0$. □

**Lemma 4.9.** *Let $Q$ be a finitely presented $A$-module annihilated by $E^h$ with $\operatorname{pd}_A(Q) \leq 1$. If $C$ is an $A$-algebra on which $E$ is a non-zero-divisor, then*

$$\operatorname{Tor}_i^A(C, Q) = 0 \quad \textit{for } i > 0.$$

*If, in addition, $C$ is $p$-adically complete and $(p, E)$ is a regular sequence on $C$, then $C \otimes_A Q$ is $p$-torsion-free and $p$-adically complete.*

*Proof.* Choose a finite-projective resolution $0 \to P_1 \xrightarrow{d} P_0 \to Q \to 0$. Since $E^h Q = 0$, there is a map $s : P_0 \to P_1$ with $ds = E^h \operatorname{id}_{P_0}$; injectivity of $d$ also gives $sd = E^h \operatorname{id}_{P_1}$. Consequently, $1 \otimes d$ is injective after any base change on which $E$ is regular. The base-changed resolution proves the Tor vanishing.

Under the additional hypotheses, $E$ is regular on $C/p^n$ for every $n \geq 1$. Thus the resolution remains exact modulo $p^n$. For $n = 1$ this shows that $C \otimes_A Q$ is $p$-torsion-free. Taking inverse limits of the exact sequences modulo $p^n$ recovers the original cokernel: the modules $C \otimes_A P_i$ are $p$-complete, and the transition maps on $(C \otimes_A P_1)/p^n$ are surjective. Hence $C \otimes_A Q$ is $p$-complete as well. □

**Lemma 4.10.** *Let $A \to A'$ and $B \to B'$ be as in Lemma 4.7, and let $\mathscr{M} \in \operatorname{Mod}^{\varphi,h,\mathrm{proj}}(B)$. Put $Q = \mathscr{M}/\operatorname{Fil}^h \mathscr{M}$. Then $\operatorname{Fil}^h \mathscr{M}$ is $p$-adically complete and $p$-saturated in $\mathscr{M}$, and there is a natural exact sequence*

$$0 \to B' \widehat{\otimes}_B \operatorname{Fil}^h \mathscr{M} \to B' \otimes_B \mathscr{M} \to A' \otimes_A Q \to 0.$$

*In particular, the completed tensor product on the left identifies with the $p$-adic closure of the image of $B' \otimes_B \operatorname{Fil}^h \mathscr{M}$ in $B' \otimes_B \mathscr{M}$.*

*Proof.* By Lemma 4.9, applied with $C = A$ and $C = A'$, both $Q$ and $A' \otimes_A Q$ are $p$-torsion-free and $p$-complete. Hence the quotient sequence for $\mathscr{M}$ is exact modulo every $p^n$, and $\operatorname{Fil}^h \mathscr{M}$ is closed and $p$-saturated, therefore $p$-complete. Moreover,

$$B' \widehat{\otimes}_B Q \simeq A' \widehat{\otimes}_A Q = A' \otimes_A Q.$$

Tensoring the quotient sequence modulo $p^n$ with $B'/p^n$ and taking inverse limits now proves the claimed exact sequence, as in Lemma 4.7. Since its quotient is $p$-torsion-free, the topology on the kernel is its intrinsic $p$-adic topology. The ordinary tensor product is dense in its completion, giving the final assertion. □

*Remark* 4.11. Set $Q = \mathscr{M}/\operatorname{Fil}^h \mathscr{M}$. For a projective Breuil module, $E^h Q = 0$ because $E^h \mathscr{M} \subseteq \operatorname{Fil}^h \mathscr{M}$. Lemma 4.9, applied with $C = A$, shows that $Q$ is $p$-torsion-free. If $h = 1$ and both $A$ and $A/(E)$ are regular, the condition $\operatorname{pd}_A(Q) \leq 1$ is equivalent to finite projectivity over $A/(E)$, by the Auslander–Buchsbaum formula applied locally. Thus it recovers the quotient condition of [Kim15, Def. 3.2.3(2)] in weight one.

In the CDVR case, $A = \mathfrak{S}_L$ is a regular local ring of dimension two. For any finite $A/(E^h)$-module $Q$, $p$-torsion-freeness implies $\operatorname{depth}_A(Q) \geq 1$ when $Q \neq 0$, and hence $\operatorname{pd}_A(Q) \leq 1$ by Auslander–Buchsbaum. Therefore, in this case our quotient condition is equivalent to the usual $p$-torsion-freeness condition, for every allowed $h$. Of course when $Q = 0$ there is nothing to say.

4.5. **Equivalence of categories between Breuil–Kisin and Breuil modules.** In this subsection, we compare the projective categories $\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R)$ and $\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(S_R)$. We adapt [Kim15, Prop. 6.3] to semistable base rings and height $h \leq p-2$. Kim discusses the higher-weight extension in [Kim15, Rem. 3.3.3, Rem. 6.4.8]. Some work is needed to accomplish the claim in [Kim15, Rem. 6.4.8], however, as we only assume $\operatorname{pd}_{\mathfrak{S}_R}(\mathscr{M}/\operatorname{Fil}^h \mathscr{M}) \leq 1$, and not any direct projectivity of this quotient. As such, the map $B : \mathscr{M} \to \operatorname{Fil}^1 \mathscr{M}$ of [Kim15, Lem. 6.4.2] cannot be constructed in the same way, as the analogous "Hodge filtration" does not split in the higher-weight case. A simplified variant of [Kim15, Lem. 6.4.2] does hold, and our proof of the equivalence uses a similar successive approximation argument as [Kim15, Lem. 6.4.3].

**Proposition 4.12.** *The construction of Subsection 4.3 defines an equivalence of categories*

$$\operatorname{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R) \xrightarrow{\sim} \operatorname{Mod}^{\varphi,h,\mathrm{proj}}(S_R).$$

**Lemma 4.13** (cf. [Kim15, Lem 6.1.5])**.** *Let $\mathfrak{M}^\bullet := [0 \to \mathfrak{M}_1 \to \mathfrak{M}_2 \to \mathfrak{M}_3 \to 0]$ be a sequence of maps of finite projective $\mathfrak{S}_R$-modules. Then $\mathfrak{M}^\bullet$ is exact if and only if $S_R \otimes_{\varphi,\mathfrak{S}_R} \mathfrak{M}^\bullet$ is exact.*

*Proof.* This is [Kim15, Lem. 6.1.5]. Its proof applies unchanged to the relative rings $\mathfrak{S}_R$ and $S_R$, since it uses only finite projectivity of the terms and the defining properties of the Breuil ring. □

**Lemma 4.14.** *Let*

$$\begin{aligned}
&\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R);\\
&\mathfrak{M}_j = \mathfrak{M} \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_{L_j}),\ \ j = 1,\dots,m;\\
&\mathscr{M} = S_R \otimes_{\varphi,\mathfrak{S}_R} \mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S_R);\\
&\mathscr{M}_j = S_{L_j} \otimes_{\varphi,\mathfrak{S}_{L_j}} \mathfrak{M}_j \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S_{L_j}),\ \ j = 1,\dots,m.
\end{aligned}$$

*We have*

$$\mathfrak{M}^* = \bigcap_{j=1}^{m} (\mathfrak{M}_j^* \cap \mathscr{M}), \quad \textit{and} \quad \mathrm{Fil}^h \mathfrak{M}^* = \bigcap_{j=1}^{m} (\mathrm{Fil}^h \mathfrak{M}_j^* \cap \mathscr{M}).$$

*Proof.* Since $\mathfrak{M}_j^* = \mathfrak{S}_{L_j} \otimes_{\varphi,\mathfrak{S}_R} \mathfrak{M}$ and $\mathscr{M} = S_R \otimes_{\varphi,\mathfrak{S}_R} \mathfrak{M}$, the first identity follows at once from Lemma 2.17. For the second, note that since $\mathrm{Fil}^h \mathfrak{M}^*$ is projective and $\mathrm{Fil}^h \mathfrak{M}_j^* \simeq \mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j}$ by Lemma 4.7, we have

$$\mathrm{Fil}^h \mathfrak{M}^* = \bigcap_{j=1}^{m} (\mathrm{Fil}^h \mathfrak{M}_j^* \cap \mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} S_R).$$

For the reverse inclusion, take $x \in \bigcap_{j=1}^m (\mathrm{Fil}^h \mathfrak{M}_j^* \cap \mathscr{M})$. In particular, $x \in \bigcap_{j=1}^m (\mathfrak{M}_j^* \cap \mathscr{M}) = \mathfrak{M}^*$, so $y = \Phi_{\mathfrak{M}}(x) \in \mathfrak{M}$, and the image $x_j$ of $x$ in $\mathfrak{M}_j^*$ lies in each $\mathrm{Fil}^h \mathfrak{M}_j^*$, hence $y_j := \Phi_{\mathfrak{M}_j}(x_j) \in E^h \mathfrak{M}_j$ for each $j$. Set $z := E^{-h} y \in \mathcal{O}_{\mathcal{E}_R} \otimes_{\mathfrak{S}_R} \mathfrak{M}$. By what we just observed, the images $z_j = E^{-h} y_j$ still belong to $\mathfrak{M}_j$ for every $j$. Applying Lemma 2.9, we deduce that $z \in \mathfrak{M}$, hence $y = E^h z \in E^h \mathfrak{M}$, and thus $x \in \mathrm{Fil}^h \mathfrak{M}^*$ as needed. □

**Lemma 4.15** (cf. [Kim15, Lem. 6.2])**.** *The functor*

$$\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R) \longrightarrow \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S_R)$$

*is fully faithful.*

*Proof.* Let $\mathfrak{M}_1, \mathfrak{M}_2 \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R)$, put $\mathscr{M}_i = S_R \otimes_{\varphi,\mathfrak{S}_R} \mathfrak{M}_i$, and let $f : \mathscr{M}_1 \to \mathscr{M}_2$ be a morphism of Breuil modules. By the local equivalence [Gao20, Thm. 5.1.3(1)], its base change at each Shilov point preserves $\mathfrak{M}_{i,j}^*$, their $\mathrm{Fil}^h$-submodules, and their divided Frobenii. Lemma 4.14 therefore gives a map $g : \mathfrak{M}_1^* \to \mathfrak{M}_2^*$ by restricting $f$ to $\mathfrak{M}_1^*$. Moreover, let $x \in \mathrm{Fil}^h \mathfrak{M}_1^*$. For every $j$, the base change $f_j$ of $f$ is a morphism of Breuil modules and therefore carries the image of $x$ into $\mathrm{Fil}^h \mathfrak{M}_{2,j}^*$. Since $f(x) \in \mathscr{M}_2$, the second intersection identity in Lemma 4.14 gives

$$g(x) = f(x) \in \bigcap_{j=1}^{m} \big(\mathrm{Fil}^h \mathfrak{M}_{2,j}^* \cap \mathscr{M}_2\big) = \mathrm{Fil}^h \mathfrak{M}_2^*.$$

Thus $g$ restricts to a map $\mathrm{Fil}^h \mathfrak{M}_1^* \to \mathrm{Fil}^h \mathfrak{M}_2^*$.

Define

$$h := \Psi_{\mathfrak{M}_2}^{-1} \circ (g|_{\mathrm{Fil}^h \mathfrak{M}_1^*}) \circ \Psi_{\mathfrak{M}_1} : \mathfrak{M}_1 \longrightarrow \mathfrak{M}_2.$$

The identity $\varphi_h \Psi_{\mathfrak{M}_i}(m) = \varphi(a_0)^h (1 \otimes m)$ gives $1 \otimes h = g$. Applying $\Phi_{\mathfrak{M}_2}$ to $g\Psi_{\mathfrak{M}_1} = \Psi_{\mathfrak{M}_2} h$ and using $\Psi_{\mathfrak{M}_1} \Phi_{\mathfrak{M}_1} = E^h$ shows, by $E$-torsion-freeness, that $\Phi_{\mathfrak{M}_2}(1 \otimes h) = h \Phi_{\mathfrak{M}_1}$. Thus $h$ is a Kisin-module morphism inducing $f$. The displayed formula also proves uniqueness. □

**Lemma 4.16.** *Let $\mathfrak{S} = \mathfrak{S}_R$, $S = S_R$, and $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S)$, where $0 \le h \le p-2$. There is a finite-projective $\mathfrak{S}$-module $\mathfrak{N}$ with an identification $S \otimes_{\mathfrak{S}} \mathfrak{N} \simeq \mathscr{M}$. For any such choice, put*

$$\mathfrak{M} = \ker\big(\mathfrak{N} \longrightarrow \mathscr{M} / \mathrm{Fil}^h \mathscr{M}\big).$$

*Then $\mathfrak{N} \to \mathscr{M} / \mathrm{Fil}^h \mathscr{M}$ is surjective, $\mathfrak{M}$ is finite projective over $\mathfrak{S}$, and $E^h \mathfrak{N} \subseteq \mathfrak{M} \subseteq \mathfrak{N}$. The natural map $S \otimes_{\mathfrak{S}} \mathfrak{M} \to \mathscr{M}$ is injective,*

$$E^h \mathscr{M} \subseteq S \otimes_{\mathfrak{S}} \mathfrak{M} \subseteq \mathrm{Fil}^h \mathscr{M}, \quad \textit{and} \quad \mathrm{Fil}^h \mathscr{M} = (S \otimes_{\mathfrak{S}} \mathfrak{M}) + \mathrm{Fil}^p S \cdot \mathscr{M}.$$

*Moreover, the map*

$$\Xi_0 : S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M} \xrightarrow{\sim} \mathscr{M}, \quad f \otimes m \longmapsto c_1^{-h} f \varphi_{h,\mathscr{M}}(m),$$

*is an isomorphism.*

*Proof.* Lift the finite-projective $\mathfrak{S}/(E)$-module $\mathscr{M}/\operatorname{Fil}^1 S \cdot \mathscr{M}$ to the $E$-adically complete ring $\mathfrak{S}$. Since $\operatorname{Fil}^1 S$ is contained in the Jacobson radical of $S$, lifting the resulting isomorphism modulo $\operatorname{Fil}^1 S$ gives $S \otimes_{\mathfrak{S}} \mathfrak{N} \simeq \mathscr{M}$. This is the argument of [Kim15, Lem. 6.4.1].

The map $\mathfrak{N} \to \mathscr{M}/\operatorname{Fil}^h \mathscr{M}$ is surjective because $\mathscr{M}/\operatorname{Fil}^h \mathscr{M}$ is annihilated by $\operatorname{Fil}^h S$ and $\mathfrak{S} \to S/\operatorname{Fil}^h S$ is surjective. The hypothesis that $\mathscr{M}/\operatorname{Fil}^h \mathscr{M}$ is finitely presented with $\operatorname{pd}_{\mathfrak{S}}(\mathscr{M}/\operatorname{Fil}^h \mathscr{M}) \leq 1$ implies that its kernel $\mathfrak{M}$ is finite projective. Also $E^h\mathfrak{N} \subseteq \mathfrak{M}$. The inclusion $\iota : \mathfrak{M} \to \mathfrak{N}$ and the map $\varsigma : \mathfrak{N} \to \mathfrak{M}$ given by multiplication by $E^h$ satisfy $\varsigma\iota = E^h$ and $\iota\varsigma = E^h$. Since $E$ is regular on $S$, these identities prove injectivity of $S \otimes_{\mathfrak{S}} \mathfrak{M} \to S \otimes_{\mathfrak{S}} \mathfrak{N}$ and the asserted containment of $E^h\mathscr{M}$.

Reducing the presentation of $\mathscr{M}/\operatorname{Fil}^h \mathscr{M}$ modulo $\operatorname{Fil}^h S$ gives $\operatorname{Fil}^h \mathscr{M} = (S \otimes_{\mathfrak{S}} \mathfrak{M}) + \operatorname{Fil}^h S \cdot \mathscr{M}$. Since $\operatorname{Fil}^h S = E^h S + \operatorname{Fil}^p S$, this is the stated filtration formula. For $i \geq p$, the scalar divided Frobenius satisfies $\varphi_{S,h}(\operatorname{Fil}^p S) \subset pS$ because $i - h - v_p(i!) \geq 1$. The compatibility axiom for $\varphi_{h,\mathscr{M}}$ therefore gives $\varphi_{h,\mathscr{M}}(\operatorname{Fil}^p S \cdot \mathscr{M}) \subseteq p\mathscr{M}$. The generation axiom ($\varphi_h(\operatorname{Fil}^h \mathscr{M})$ generates $\mathscr{M}$) and Nakayama's lemma imply the surjectivity of $\Xi_0$. Finally, $\mathfrak{M}[1/E] = \mathfrak{N}[1/E]$, so $\mathfrak{M}$ and $\mathfrak{N}$ have the same local rank. Frobenius preserves connected components, as it reduces to absolute Frobenius modulo $p$. Thus $\Xi_0$ is a surjection between finite-projective modules of the same local rank, and is an isomorphism. □

*Proof of Proposition 4.12.* Full faithfulness was proved above. For essential surjectivity, write $\mathfrak{S} = \mathfrak{S}_R$ and $S = S_R$. Choose $\mathfrak{N}$ and $\mathfrak{M}$ as in Lemma 4.16, and let $\beta_0 : \mathfrak{M} \to \operatorname{Fil}^h \mathscr{M}$ be the inclusion. We adapt the successive-approximation argument of [Kim15, Lem. 6.4.3], keeping the projective module $\mathfrak{M}$ fixed and varying its map into $\operatorname{Fil}^h \mathscr{M}$.

We first record the ring-theoretic facts used in the construction. Put $J_n = p^n \operatorname{Fil}^{p+n} S$, set $\ell_n = p - 1 - h + n - \left\lfloor \frac{n}{p-1} \right\rfloor$, and $\mu_n = n + \ell_n$. Then $\ell_n \geq 1$, the sequence $\mu_n$ is strictly increasing,

$$\varphi_{h,\mathscr{M}}(J_n\mathscr{M}) \subseteq p^{\mu_n}\mathscr{M}, \quad \text{and} \quad p^{\mu_n} S \subseteq \mathfrak{S} + J_{n+1}. \tag{4.2}$$

When $h = 1$, the first inclusion is proved in [Kim15, Claim 6.4.5] and the second shortly afterward on the same page ([Kim15, p. 37]). The argument is essentially the same for $h < p$.

Set $\mathfrak{M}^* = \mathfrak{S} \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}$, and write $\mathfrak{L}_n = \Xi_n(\mathfrak{M}^*) \subseteq \mathscr{M}$. We inductively construct $\mathfrak{S}$-linear maps $\beta_n : \mathfrak{M} \to \operatorname{Fil}^h \mathscr{M}$ whose normalized Frobenius linearizations

$$\Xi_n : S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M} \longrightarrow \mathscr{M}, \quad s \otimes m \longmapsto s c_1^{-h} \varphi_{h,\mathscr{M}}(\beta_n(m)),$$

are isomorphisms, and further require that

$$\beta_{n+1}(\mathfrak{M}) \subseteq \mathfrak{L}_n, \quad \text{and} \quad (\beta_{n+1} - \beta_n)(\mathfrak{M}) \subseteq J_n\mathscr{M}. \tag{4.3}$$

The initial map $\Xi_0$ is the isomorphism of the lemma. Since $S = \mathfrak{S} + \operatorname{Fil}^p S$, we have $\mathscr{M} = \mathfrak{L}_0 + \operatorname{Fil}^p S \cdot \mathscr{M}$, hence the natural reduction map

$$\mathfrak{L}_0 \to \mathscr{M}/\operatorname{Fil}^p S\mathscr{M}$$

is surjective. Since $\mathfrak{M}$ is projective, the composite

$$\overline{\beta_0} : \mathfrak{M} \to \operatorname{Fil}^h \mathscr{M} \to \mathscr{M}/\operatorname{Fil}^p S\mathscr{M}$$

lifts to $\beta_1 : \mathfrak{M} \to \mathfrak{L}_0$ and $(\beta_1 - \beta_0)(\mathfrak{M}) \subset \operatorname{Fil}^p S\mathscr{M} = J_0\mathscr{M}$. In particular, $\beta_1$ takes values in $\operatorname{Fil}^h \mathscr{M}$. Since $\varphi_{h,\mathscr{M}}(\operatorname{Fil}^p S\mathscr{M}) \subset p\mathscr{M}$ by (4.2) for $n = 0$, the argument below shows that $\Xi_1$ is an isomorphism.

Now suppose that $\beta_n$ and $\beta_{n+1}$ have been constructed. Their difference has image in $J_n\mathscr{M} \subseteq \operatorname{Fil}^h \mathscr{M}$, so $\beta_{n+1}$ takes values in $\operatorname{Fil}^h \mathscr{M}$. Fix $m \in \mathfrak{M}$, and write $\Delta_n := \Xi_{n+1} - \Xi_n$. By (4.2), we have

$$\Delta_n(1 \otimes m) = (\Xi_{n+1} - \Xi_n)(1 \otimes m) = c_1^{-h} \varphi_{h,\mathscr{M}}\big((\beta_{n+1} - \beta_n)(m)\big) \in p^{\mu_n}\mathscr{M},$$

and thus by $S$-linearity, $\Delta_n(S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}) \subset p^{\mu_n}\mathscr{M}$. Since $\Xi_n$ is an isomorphism, we may define

$$\vartheta_n := \Xi_{n+1}\Xi_n^{-1} = (\Xi_{n+1} - \Xi_n)\Xi_n^{-1} + 1 =: 1 + \delta_n,$$

where $\delta_n \in p^{\mu_n} \operatorname{End}_S(\mathscr{M})$. Thus $\vartheta_n$ and $\Xi_{n+1}$ are isomorphisms, and $\mathfrak{L}_{n+1} = \vartheta_n \mathfrak{L}_n$. We now construct $\beta_{n+2}$. Observe that $\vartheta_n\beta_{n+1}$ takes values in $\mathfrak{L}_{n+1}$. By the scalar inclusion $p^{\mu_n} S \subseteq \mathfrak{S} + J_{n+1}$ in (4.2), we have

$$p^{\mu_n}(S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}) \subseteq \mathfrak{M}^* + J_{n+1}(S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}).$$

Applying $\Xi_{n+1}$, we obtain

$$p^{\mu_n}\mathscr{M} \subseteq \mathfrak{L}_{n+1} + J_{n+1}\mathscr{M},$$

and $\delta_n\beta_{n+1}(\mathfrak{M}) \subset p^{\mu_n}\mathscr{M}$. Projectivity of $\mathfrak{M}$ gives an $\mathfrak{S}$-linear map $\alpha_n : \mathfrak{M} \to \mathfrak{L}_{n+1}$ congruent to $\delta_n\beta_{n+1}$ modulo $J_{n+1}\mathscr{M}$. Set $\beta_{n+2} = \vartheta_n\beta_{n+1} - \alpha_n$. This proves (4.3) at the next stage.

The maps $\beta_n$ converge $p$-adically to an $\mathfrak{S}$-linear map $\beta : \mathfrak{M} \to \operatorname{Fil}^h \mathscr{M}$, with $\beta \equiv \beta_0$ modulo $\operatorname{Fil}^p S \cdot \mathscr{M}$, and the $\Xi_n$ converge to an isomorphism $\Xi : S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M} \xrightarrow{\sim} \mathscr{M}$, which is the linearization of $c_1^{-h}\varphi_{h,\mathscr{M}} \circ \beta$. Indeed, the differences $\beta_{n+1} - \beta_n$ lie in $p^n \operatorname{Fil}^p S \cdot \mathscr{M}$, and $\varphi_{h,\mathscr{M}}$ is continuous on these convergent series. Each map $\Xi_n^{-1}\beta_{n+1}$ takes values in $\mathfrak{M}^*$. The $p$-adic closedness of $\mathfrak{S}$ in $S$, applied to the finite-projective module $\operatorname{Hom}_{\mathfrak{S}}(\mathfrak{M}, \mathfrak{M}^*)$, shows that their limit is an $\mathfrak{S}$-linear map

$$\Psi := \Xi^{-1}\beta : \mathfrak{M} \longrightarrow \mathfrak{M}^*.$$

We next show that $\Psi$ is injective with cokernel annihilated by $E^h$. Modulo $\operatorname{Fil}^p S$, the image of $\beta$ agrees with that of $\beta_0$ and contains the image of $E^h\mathscr{M}$. Using $S/\operatorname{Fil}^p S = \mathfrak{S}/E^p\mathfrak{S}$ and the fact that

$$\operatorname{Tor}_1^{\mathfrak{S}}(\mathfrak{M}^*, S/\operatorname{Fil}^p S) = \operatorname{Tor}_1^{\mathfrak{S}}(\mathfrak{M}^*, \mathfrak{S}/(E^p)) = 0$$

since $\mathfrak{M}^*$ is a projective $\mathfrak{S}$-module, we obtain a sequence of $\mathfrak{S}$-linear maps

$$\mathfrak{M} \xrightarrow{\beta} \operatorname{Fil}^h \mathscr{M} \xrightarrow{\Xi^{-1}} S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M} \to \frac{S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M}}{\operatorname{Fil}^p S \cdot (S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M})} \simeq \mathfrak{M}^*/E^p\mathfrak{M}^*,$$

thus

$$E^h(\mathfrak{M}^*/E^p\mathfrak{M}^*) \subseteq \operatorname{im}(\Psi \bmod E^p).$$

Projectivity of $\mathfrak{M}^*$ gives a map $\Upsilon : \mathfrak{M}^* \to \mathfrak{M}$ with $\Psi\Upsilon = E^h + E^pT$ for some $T \in \operatorname{End}_{\mathfrak{S}}(\mathfrak{M}^*)$. Since $E$ is in the Jacobson radical and $p - h > 0$, the endomorphism $V = 1 + E^{p-h}T$ is invertible. Consequently

$$\Phi := \Upsilon V^{-1} : \mathfrak{M}^* \longrightarrow \mathfrak{M} \quad \text{satisfies} \quad \Psi\Phi = E^h.$$

The modules $\mathfrak{M}$ and $\mathfrak{M}^*$ have the same local rank, so $\Psi[1/E]$ is an isomorphism. Since $\mathfrak{M}$ is $E$-torsion-free, $\Psi$ is injective, and hence also $\Phi\Psi = E^h$. It follows that $\Phi$ is injective and has cokernel annihilated by $E^h$.

Put $\Phi_{\mathfrak{M}} = \Phi$. This makes $\mathfrak{M}$ a Breuil–Kisin module of height at most $h$, and the two identities just established give $\operatorname{Fil}^h \mathfrak{M}^* = \Psi(\mathfrak{M})$. Under $\Xi$, the filtration on its associated Breuil module is therefore

$$S \cdot \beta(\mathfrak{M}) + \operatorname{Fil}^p S \cdot \mathscr{M} = S \cdot \beta_0(\mathfrak{M}) + \operatorname{Fil}^p S \cdot \mathscr{M} = \operatorname{Fil}^h \mathscr{M}.$$

By definition, the divided Frobenius on the associated Breuil module is the composite of $1 \otimes \Phi_{\mathfrak{M}}$ with $\varphi_{S,h} \otimes 1$. Since $\Phi_{\mathfrak{M}}\Psi(m) = E^h m$, this composite sends $\Psi(m)$ first to $E^h \otimes m$ and then to $\varphi_{S,h}(E^h) \otimes m = c_1^h(1 \otimes m)$. Thus, writing $\varphi_{h,\mathrm{new}}$ for this divided Frobenius transported by $\Xi$, and using $\Xi\Psi = \beta$, we obtain

$$\varphi_{h,\mathrm{new}}(\beta(m)) = c_1^h \Xi(1 \otimes m) = c_1^h c_1^{-h} \varphi_{h,\mathscr{M}}(\beta(m)) = \varphi_{h,\mathscr{M}}(\beta(m)).$$

Thus the two divided Frobenii agree on $S \cdot \beta(\mathfrak{M})$ which contains $E^h\mathscr{M}$ because $\Psi\Phi = E^h$, and the compatibility axiom also gives agreement on $\operatorname{Fil}^p S \cdot \mathscr{M}$. They agree on all of $\operatorname{Fil}^h \mathscr{M}$, and $\Xi$ is the required isomorphism of Breuil modules. This proves essential surjectivity. □

## 5. Relative Breuil modules and associated Galois representations

Our ultimate goal is to describe an equivalence of categories between semistable Galois representations (or equivalently, local systems) and an "integral crystal" category, namely relative Breuil modules. In this section we define the category of relative Breuil modules and construct the Breuil realization from prismatic $F$-crystals to relative Breuil modules.

5.1. **Relative Breuil modules.** We introduce our category of relative Breuil modules. Let $(X, M_X)$ be a small affine or formal spectrum of a CDVR. Let $R$ denote the ring of global sections in the small affine case, in which case $R$ is a small affine base ring, and let $\mathcal{O}_L$ denote the underlying CDVR in the other case. Set $\mathfrak{S} = \mathfrak{S}_R$ or $\mathfrak{S}_L$, and $S = S_R$ or $S_L$. We retain the notation

$$\omega_S^1 := S \otimes_{\mathfrak{S}} \omega^1_{\mathfrak{S}/\mathbb{Z}_p}$$

for the module of $p$-complete logarithmic differentials introduced in Subsection 2.4. We now refine the category ${}'\mathrm{Mod}^{\varphi,h}(S)$ from Section 4, specializing $B$ to $S$.

**Definition 5.1.** Retain the projective and torsion categories $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S)$ and $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(S)$ defined in Subsection 4.3. Let

$$\mathrm{Mod}^{\varphi,h}(S) \subset {}'\mathrm{Mod}^{\varphi,h}(S)$$

be the exact extension-closed subcategory generated by $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S)$ and $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(S)$. Its objects are called *Breuil modules of weight $h$*; the two distinguished full subcategories consist of the projective and torsion Breuil modules.

The category ${}'\mathrm{Mod}^{\varphi,h,\nabla}(S)$ consists of tuples $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h, \nabla)$ where the triple $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h)$ is an object of ${}'\mathrm{Mod}^{\varphi,h}(S)$ and

$$\nabla : \mathscr{M} \to \mathscr{M} \otimes_S \omega_S^1$$

is a quasi-nilpotent integrable connection such that

(1) $\nabla$ satisfies *Griffiths transversality* with respect to the filtration, i.e.

$$(\mathrm{Fil}^1 S)\nabla(\mathrm{Fil}^h \mathscr{M}) \subset \mathrm{Fil}^h \mathscr{M} \otimes_S \omega_S^1;$$

(2) the divided Frobenius $\varphi_h$ is *horizontal* with respect to $\nabla$, meaning that the diagram

$$\begin{array}{ccc} \mathrm{Fil}^h \mathscr{M} & \xrightarrow{E(u)\nabla} & \mathrm{Fil}^h \mathscr{M} \otimes_S \omega_S^1 \\ \downarrow{\scriptstyle \varphi_h} & & \downarrow{\scriptstyle \varphi_h \otimes \frac{d\varphi_S}{p}} \\ \mathscr{M} & \xrightarrow{c_1 \nabla} & \mathscr{M} \otimes_S \omega_S^1 \end{array}$$

commutes.

Morphisms $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_{h,\mathscr{M}}, \nabla_{\mathscr{M}}) \to (\mathscr{N}, \mathrm{Fil}^h \mathscr{N}, \varphi_{h,\mathscr{N}}, \nabla_{\mathscr{N}})$ are morphisms of the underlying objects of ${}'\mathrm{Mod}^{\varphi,h}(S)$ which commute with the connections. We let $\mathrm{Mod}^{\varphi,h,\nabla}(S)$ be the full subcategory whose underlying triple belongs to $\mathrm{Mod}^{\varphi,h}(S)$. For $\bullet \in \{\mathrm{proj}, \mathrm{tor}\}$, we similarly define $\mathrm{Mod}^{\varphi,h,\nabla,\bullet}(S)$ by requiring the underlying triple to belong to $\mathrm{Mod}^{\varphi,h,\bullet}(S)$.

The category $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S)$ is the category of projective Breuil modules with connection appearing in Theorem 1.7. In the CDVR case, Remark 4.11 identifies the quotient condition with $p$-torsion-freeness. The underlying data without connection thus satisfies the quasi-strong divisibility conditions of [Moo24, Def. 4.10]; the comparison with strongly divisible lattices with connection is established in Appendix B.

5.2. **The Breuil realization functor.** Let $(X, M_X)$ be a small affine with $X = \mathrm{Spf}\, R$ and framing $\square : R_0 \to R$, where

$$R_0 = \mathcal{O}_K \left\langle T_1, \ldots, T_m, T_{m+1}^{\pm 1}, \ldots, T_d^{\pm 1} \right\rangle / (T_1 \cdots T_m - \pi), \quad \text{for } 1 \le m \le d.$$

As in Subsection 2.1, the log structure is induced by $\mathbb{N}^d \to R$, $e_j \mapsto T_j$, and the framing lifts to the Breuil–Kisin ring $\mathfrak{S}$, with $u = T_1 \cdots T_m$ and $\varphi(T_j) = T_j^p$. We also allow $(X, M_X)$ to be the log

formal spectrum of a CDVR; in that case use the framed logarithmic coordinates $u, X_1, \dots, X_b$ in the constructions below. Fix $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}})$. We wish to define a functor

$$\mathrm{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}}) \to \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S).$$

Let $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon)$ be the Breuil–Kisin module with descent datum obtained from $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}})$. Let $\jmath : \mathfrak{S} \to S$ denote the inclusion, write $\varphi = \jmath\varphi_{\mathfrak{S}} : \mathfrak{S} \to S$, and put $c_1 = \varphi(E)/p \in S^\times$. Set

$$\mathscr{M} := S \otimes_{\varphi,\mathfrak{S}} \mathfrak{M} = S \otimes_{\jmath,\mathfrak{S}} \mathfrak{M}^* \quad \text{and} \quad \mathscr{M}^\varphi := S \otimes_{\jmath,\mathfrak{S}} \mathfrak{M}.$$

The map $\Phi = 1 \otimes \Phi_{\mathfrak{M}} : \mathscr{M} \to \mathscr{M}^\varphi$ is $S$-linear and becomes an isomorphism after inverting $E$. For any $r \geq 0$, set

$$\mathrm{Fil}^r \mathscr{M} = \Phi^{-1}(\mathrm{Fil}^r S\mathscr{M}^\varphi).$$

The ordinary semilinear Frobenius and its divided maps are

$$\varphi_{\mathscr{M}}(s \otimes m) = \varphi(s) \otimes \varphi_{\mathfrak{M}}(m) \quad \text{and} \quad \varphi_r = p^{-r}\varphi : \mathrm{Fil}^r \mathscr{M} \longrightarrow \mathscr{M}, \quad \text{for } 0 \leq r \leq h.$$

By Subsection 4.3, the triple $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h)$ is a projective Breuil module of weight $h$.

Let $\mathfrak{S}^{(1)}$ be the prismatic self-product, with coface maps $p_0, p_1 : \mathfrak{S} \to \mathfrak{S}^{(1)}$. Recall that $S^{(1)}$ denotes the log crystalline self-product of the Breuil $S$, and there is a canonical Frobenius map $\varphi^{(1)} : \mathfrak{S}^{(1)} \longrightarrow S^{(1)}$ satisfying $\varphi^{(1)} p_i = s_i \varphi$ for $i = 0, 1$. Base change of $\varepsilon$ along this map is the crystalline HPD stratification

$$\eta : S^{(1)} \otimes_{s_0,S} \mathscr{M} \xrightarrow{\sim} S^{(1)} \otimes_{s_1,S} \mathscr{M}.$$

Let $t_j$ run through the framed logarithmic coordinates. For a small semistable affine these are $T_1, \dots, T_d$, with $u = T_1 \cdots T_m$. For a CDVR use $u, X_1, \dots, X_b$. As in Subsection 3.5, its associated integrable quasi-nilpotent log connection is of the form

$$\nabla(x) = \sum_j N_j(x) \otimes d\log t_j.$$

The goal of this section is to prove that $(\mathscr{M}, \mathrm{Fil}^\bullet, \varphi_\bullet, \nabla)$ is an object of $\mathrm{Mod}^{\varphi,h,\nabla}(S_R)$. In particular, we need the following compatibility statement.

**Proposition 5.2.** *For $1 \leq r \leq h$ one has*

$$\nabla(\mathrm{Fil}^r \mathscr{M}) \subset \mathrm{Fil}^{r-1} \mathscr{M} \otimes_S \omega_S^1.$$

*Consequently,*

$$(\mathrm{Fil}^1 S)\nabla(\mathrm{Fil}^h \mathscr{M}) \subset \mathrm{Fil}^h \mathscr{M} \otimes_S \omega_S^1,$$

*and the diagram*

$$\begin{array}{ccc} \mathrm{Fil}^h \mathscr{M} & \xrightarrow{E\nabla} & \mathrm{Fil}^h \mathscr{M} \otimes_S \omega_S^1 \\ {\scriptstyle \varphi_h}\downarrow & & \downarrow{\scriptstyle \varphi_h \otimes d\varphi/p} \\ \mathscr{M} & \xrightarrow[c_1\nabla]{} & \mathscr{M} \otimes_S \omega_S^1 \end{array}$$

*commutes. Thus $\nabla$ satisfies conditions 1 and 2 of Definition 5.1.*

To prove transversality, we need to control $\Phi(N_j x)$ when $\Phi(x)$ lies in $\mathrm{Fil}^r S\mathscr{M}^\varphi$. To this end, we define

$$\Theta_j := E\Phi N_j \Phi^{-1} : \mathscr{M}^\varphi[1/(pE)] \to \mathscr{M}^\varphi[1/(pE)].$$

Note that since $\gamma_p(E) \in S$, inverting $E$ automatically inverts $p$. For each coordinate $t_j$, let $D_j : S \to S$ be the logarithmic derivation satisfying $D_j(t_k) = \delta_{jk} t_k$. Let us give some setup for the strategy. These satisfy the Lebniz rule $N_j(ax) = aN_j(x) + D_j(a)x$, as well as $D_j\varphi = p\varphi D_j$ and $D_j(\mathrm{Fil}^r S) \subset \mathrm{Fil}^{r-1} S$. The operator $\Theta_j$ as defined above satisfies

$$\Theta_j(fm) = f\Theta_j(m) + ED_j(f)m \tag{5.1}$$

The key to Proposition 5.2 is the following.

**Lemma 5.3.** *The operator $\Theta_j$ preserves $\mathscr{M}^\varphi[1/p]$.*

The is the main difficulty. We show that $\Theta_j$ is the scalar multiplying the "first-order corrections" of a certain base change of the descent datum $\varepsilon$. A technical obstacle is the lack of a map $\jmath^{(1)} : \mathfrak{S}^{(1)} \to S^{(1)}$ satisfying $\jmath^{(1)} p_i = s_i \jmath$. Instead, we define maps $\theta_j$ from $\mathfrak{S}^{(1)}$ to an $E$-twisted variation of the first-order thickening $S^{(1)}/J_\Delta^{[2]}$ of the codiagonal (Lemma 5.4 below).

We give some preparations. First, we recall the descriptions of self-products of the Breuil–Kisin prism. We do this for the small affine case, using $j, k$ for coordinate indices and $i \in \{0, 1\}$ for coface indices. The CDVR case has the same description with coordinates $u, X_1, \dots, X_b$. Set $T_{j,0} := T_j$ for $1 \le j \le d$. Exactification introduces units $U_j$ with relations $T_{j,1} = T_{j,0} U_j$. Set $\rho_j = 1 - U_j$, let $w_j$ satisfy $E w_j = \rho_j$, and set

$$A_0 := \mathfrak{S}[T_{1,1}, \dots, T_{m,1}][T_{m+1,1}^{\pm 1}, \dots, T_{d,1}^{\pm 1}].$$

Let $A^{(1)}$ be the $(\rho_j)_{1 \le j \le d}$-completion of

$$A_0[U_1^{\pm 1}, \dots, U_d^{\pm 1}]/(T_{j,1} - T_{j,0} U_j \mid 1 \le j \le d).$$

Thus $A^{(1)} \simeq \mathfrak{S}[[\rho_1, \dots, \rho_d]]$; for the formally étale extension of the framing, the second coefficient map is obtained by the unique lift of the diagonal map. Give $A^{(1)}$ the Frobenius extending that of the first coefficient ring and satisfying $\varphi(U_j) = U_j^p$. There is a natural surjection $A^{(1)} \to R$ with kernel $J^{(1)} := (E(u), \rho_j)_{1 \le j \le d}$, and we have

$$\mathfrak{S}^{(1)} = A^{(1)} \left\{ \frac{\rho_1}{E(u)}, \dots, \frac{\rho_d}{E(u)} \right\}^\wedge_{(p,E)}.$$

Concretely, this is the $(p, E)$-completed quotient of the free $\delta$-ring $A^{(1)}\{w_1, \dots, w_d\}_\delta$ by the $\delta$-ideal generated by $E w_j - \rho_j$ for $1 \le j \le d$. The projection maps $p_i : \mathfrak{S} \to A^{(1)}$ send $T_j \mapsto T_{j,i}$ for $i = 0, 1$, and these naturally extend to $\mathfrak{S} \to \mathfrak{S}^{(1)}$. We also set $u_i := p_i(u) \in \mathfrak{S}^{(1)}$. Above, $E(u) = E(u_0)$.

We now construct the log crystalline self-product $S^{(1)}$. The projection $A^{(1)} \to R/p$ has kernel $(p, E(u_0), \rho_1, \dots, \rho_d)$. Set

$$B^{(1)} := A^{(1)} \left\{ \frac{\varphi(E)}{p}, \frac{\varphi(\rho_1)}{p}, \dots, \frac{\varphi(\rho_d)}{p} \right\}^\wedge_p$$

We have that $A^{(1)}$ is $p$-torsion-free, and we claim that the sequence $(p, E, \rho_1, \dots, \rho_d)$ is a regular. Indeed, $E \equiv u^e \bmod p$ is a regular element of $\mathfrak{S}/p$ (Lemma 4.1) and then the $\rho_j$ remain independent formal variables after quotienting by $E$. Hence by [BS22, Cor. 2.39],

$$S^{(1)} \simeq D_{(E, \rho_1, \dots, \rho_d)}(A^{(1)})^\wedge$$

where the right-hand-side denotes the $p$-completed PD-envelope, and these are both isomorphic to $D_{(E)}(\mathfrak{S}^{(1)})^\wedge$ (cf. *loc. cit.* and [DLMS24b, Lem. 3.32]). Let $s_i : S \to S^{(1)}$ denote the co

Write $\sigma_j = 1 - s_1(T_j)/s_0(T_j)$ for the image of $\rho_j$ under this PD-envelope map. Let $J_\Delta$ denote the ideal of the codiagonal $S^{(1)} \to S$. We have

$$Q := S^{(1)}/J_\Delta^{[2]} \simeq S[\bar\sigma_1, \dots, \bar\sigma_d]/(\bar\sigma_j \bar\sigma_k \mid 1 \le j, k \le d).$$

Here $\bar\sigma_j$ is the class of $\sigma_j$, and all products of degree-one classes vanish. As noted above, we cannot map $\mathfrak{S}^{(1)} \to S^{(1)}$ compatibly with the untwisted coface maps, and the same obstruction occurs for $Q$: the relation $E w_j = \rho_j$ would require dividing $\bar\sigma_j$ by $E$. Instead, let

$$C_j := S[1/p][z_j]/(z_j^2) = S[1/p] \oplus S[1/p] z_j, \quad \text{for } 1 \le j \le d,$$

and consider the map

$$Q \to C_j : \bar\sigma_k \mapsto \delta_{jk} E z_j.$$

The two coefficient maps send $T_k$ to $T_k$ and $T_k(1 - \delta_{jk} E z_j)$, respectively. On the $j$th square-zero summand the map is multiplication by $E$:

$$S[1/p]\bar\sigma_j \longrightarrow S[1/p] z_j : \bar\sigma_j \longmapsto E z_j.$$

This is precisely the "$E$-twisted derivation" alluded to above. Since $\varphi(\sigma_k) = 1-(1-\sigma_k)^p$, the induced Frobenius on $Q$ satisfies $\varphi(\bar{\sigma}_k) = p\bar{\sigma}_k$. To make $Q \to C_j$ Frobenius-compatible, we need $\varphi(Ez_j) = pEz_j$. Thus we give $C_j$ the $\delta$-structure determined by

$$\varphi(z_j) = \frac{pE}{\varphi(E)} z_j = Ec_1^{-1} z_j.$$

The map

$$\xi_j : S^{(1)} \longrightarrow Q \longrightarrow C_j, \quad \bar{\sigma}_k \longmapsto \delta_{jk} Ez_j$$

commutes with Frobenius, since $\varphi(Ez_j) = pEz_j$. Composing $\xi_j$ with the coface maps gives $\xi_j \circ s_i : S \to C_j$ for $i = 0, 1$. Since $p$ is invertible in $C_j$, these composites extend uniquely to continuous ring maps

$$\widetilde{\xi}_{i,j} : S[1/p] \longrightarrow C_j.$$

The formulas for $s_i$ in $Q$ and the assignment $\bar{\sigma}_k \mapsto \delta_{jk} Ez_j$ give

$$\widetilde{\xi}_{0,j}(f) = f \quad \text{and} \quad \widetilde{\xi}_{1,j}(f) = f - ED_j(f) z_j \quad \text{for } f \in S[1/p].$$

Here $D_j$ is extended to $S[1/p]$ by $D_j(p^{-1}) = 0$. Write

$$\xi_{i,j} := \widetilde{\xi}_{i,j} \jmath : \mathfrak{S} \longrightarrow C_j, \quad \text{satisfying } \xi_j s_i = \widetilde{\xi}_{i,j}|_S.$$

The maps $\xi_{i,j}$ are the structural maps associated with the two cofaces; $C_j$ itself is indexed only by the coordinate.

**Lemma 5.4.** *For each $1 \le j \le d$, there is a continuous $\delta$-ring map*

$$\theta_j : \mathfrak{S}^{(1)} \longrightarrow C_j$$

*satisfying $\theta_j p_i = \xi_{i,j}$ for $i = 0, 1$ and $\xi_j \varphi^{(1)} = \theta_j \varphi$. Moreover, let $\vartheta_j := \theta_j^* \varepsilon$. Then*

$$\vartheta_j(1 \otimes y) = 1 \otimes y + z_j \otimes \Theta_j(y), \quad \text{for } y \in \mathscr{M}^{\varphi}[1/p]. \tag{5.2}$$

*Proof.* Fix $1 \le j \le d$. Define a ring map

$$\theta_j^0 : A^{(1)} := \mathfrak{S}[[\rho_1, \dots, \rho_d]] \longrightarrow C_j, \quad \theta_j^0(\rho_k) = \delta_{jk} Ez_j \quad \text{for } 1 \le k \le d.$$

We have that $\theta_j^0$ is compatible with Frobenius:

$$\varphi(1 - \delta_{jk} Ez_j) = 1 - \delta_{jk} pEz_j = (1 - \delta_{jk} Ez_j)^p.$$

Since $C_j$ is $p$-torsion-free, $\theta_j^0$ gives a map of $\delta$-rings, and by the left-adjoint property of the free $\delta$-ring construction, $\theta_j^0$ extends uniquely to a $\delta$-ring map

$$\theta_j^{\delta} : A^{(1)}\{w_1, \dots, w_d\}_{\delta} \longrightarrow C_j, \quad w_k \longmapsto \delta_{jk} z_j.$$

Moreover $\theta_j^{\delta}(Ew_k - \rho_k) = E\delta_{jk} z_j - \delta_{jk} Ez_j = 0$, so passing to the $\delta$-quotient gives

$$\theta_j^{\mathrm{nc}} : \mathfrak{S}^{(1),\mathrm{nc}} \longrightarrow C_j, \quad \text{where} \quad \mathfrak{S}^{(1),\mathrm{nc}} := A^{(1)}\{w_1, \dots, w_d\}_{\delta} / (Ew_k - \rho_k \mid 1 \le k \le d)_{\delta}.$$

By the presentation above, $\mathfrak{S}^{(1)}$ is the $(p, E)$-adic completion of $\mathfrak{S}^{(1),\mathrm{nc}}$. We verify that $\theta_j^{\mathrm{nc}}$ is $(p, E)$-adically continuous, and thus uniquely extends to $\mathfrak{S}^{(1)}$. For $f \in S[1/p]$, the relation $z_j^2 = 0$ gives $\delta(fz_j) = \varphi(f)\varphi(z_j)/p$. In particular, $\delta(z_j) = E\varphi(E)^{-1} z_j$, and iterating this gives

$$\delta^{n+1}(z_j) = \frac{E}{\varphi^{n+1}(E)} z_j$$

for all $n \ge 1$. Since $\varphi^n(E) = p\varphi^{n-1}(c_1)$ with $\varphi^{n-1}(c_1)$ a unit, these images lie in $p^{-1} S z_j$. The image of $A^{(1)}$ under $\theta_j^0$ lies in $S \oplus Sz_j$, and the underlying algebra of the free $\delta$-ring is generated over $A^{(1)}$ by $\delta^n(w_k)$ for $n \ge 0$. Therefore $\theta_j^{\mathrm{nc}}$ takes values in the fixed $p$-adically complete subring $\Lambda_j = S \oplus p^{-1} S z_j \subset C_j$. The divided-power identity $E^b = b! \gamma_b(E)$ implies $p^a E^b \Lambda_j \subseteq p^{a + v_p(b!)} \Lambda_j$, and $\min\{a + v_p(b!) : a + b = n,\ a, b \ge 0\} \to \infty$ as $n \to \infty$. Hence $\theta_j^{\mathrm{nc}}((p, E)^n \mathfrak{S}^{(1),\mathrm{nc}})$ decays to zero $p$-adically in $\Lambda_j$. The map $\theta_j^{\mathrm{nc}}$ consequently extends uniquely to a continuous ring map

$$\theta_j : \mathfrak{S}^{(1)} \longrightarrow \Lambda_j \hookrightarrow C_j.$$

Compatibility with $\delta$ passes to the completion by continuity. The restrictions of $\theta_j$ to the two copies of $\mathfrak{S}$ are $\xi_{0,j}$ and $\xi_{1,j}$: this holds on the framing, and then on the formally étale extension by uniqueness of lifting.

We verify $\xi_j\varphi^{(1)} = \theta_j\varphi$ on the generators of $\mathfrak{S}^{(1),\mathrm{nc}}$ and then pass to its completion. Both composites restrict to $\jmath\varphi$ on the first copy of $\mathfrak{S}$, and send $U_k$ to $1-\delta_{jk}pEz_j$. Applying the composites to $Ew_k = \rho_k$ gives

$$\varphi(E)\,\xi_j\varphi^{(1)}(w_k) = \delta_{jk}pEz_j = \varphi(E)\,\theta_j\varphi(w_k).$$

Since $\varphi(E)$ is invertible in $C_j$, the composites agree on $w_k$. They agree on its $\delta$-iterates by $\delta$-compatibility, and on the completion by continuity. Thus

$$\xi_j\varphi^{(1)} = \theta_j\varphi. \tag{5.3}$$

The pullback $\vartheta_j = \theta_j^*\varepsilon$ defines an isomorphism

$$\vartheta_j : C_j \otimes_{\widetilde{\xi}_{0,j},S[1/p]} \mathscr{M}^{\varphi}[1/p] \xrightarrow{\sim} C_j \otimes_{\widetilde{\xi}_{1,j},S[1/p]} \mathscr{M}^{\varphi}[1/p].$$

which reduces to the identity modulo $z_j$. The $z_j$-part of the target is canonically a copy of $\mathscr{M}^{\varphi}[1/p]$, since $z_j\widetilde{\xi}_{1,j}(f) = z_jf$. Consequently there is a unique continuous additive operator $\Theta'_j$ on $\mathscr{M}^{\varphi}[1/p]$ such that

$$\vartheta_j(1\otimes y) = 1\otimes y + z_j\otimes\Theta'_j(y).$$

We identify this coefficient $\Theta_j$. First, $C_j$-linearity and $\widetilde{\xi}_{0,j}(f) = \widetilde{\xi}_{1,j}(f) + ED_j(f)z_j$ give

$$\Theta'_j(fy) = f\Theta'_j(y) + ED_j(f)y.$$

Thus $\Theta'_j$ extends to the localization $\mathscr{M}^{\varphi}[1/(pE)]$ by this Leibniz rule.

Put $\mathfrak{M}_i^{(1)} = \mathfrak{S}^{(1)}\otimes_{p_i,\mathfrak{S}}\mathfrak{M}$. Frobenius compatibility of descent (see Definition 3.4) gives the square

$$\begin{array}{ccc} \varphi^*\mathfrak{M}_0^{(1)} & \xrightarrow{\varphi^*\varepsilon} & \varphi^*\mathfrak{M}_1^{(1)} \\ {\scriptstyle\Phi_0}\downarrow & & \downarrow{\scriptstyle\Phi_1} \\ \mathfrak{M}_0^{(1)} & \xrightarrow[\varepsilon]{} & \mathfrak{M}_1^{(1)}. \end{array}$$

We identify how this diagram changes after pullback along $\theta_j$. For $i = 0, 1$, write

$$\alpha_i := \theta_j p_i = \widetilde{\xi}_{i,j}\jmath, \qquad \beta_i := \theta_j\varphi_{\mathfrak{S}^{(1)}}p_i = \widetilde{\xi}_{i,j}\varphi : \mathfrak{S}\longrightarrow C_j.$$

The equality for $\beta_i$ uses that $p_i$ commutes with Frobenius. Thus there are canonical identifications

$$C_j\otimes_{\theta_j,\mathfrak{S}^{(1)}}\mathfrak{M}_i^{(1)} \simeq C_j\otimes_{\alpha_i,\mathfrak{S}}\mathfrak{M} \simeq C_j\otimes_{\widetilde{\xi}_{i,j},S[1/p]}\mathscr{M}^{\varphi}[1/p],$$
$$C_j\otimes_{\theta_j,\mathfrak{S}^{(1)}}\varphi^*_{\mathfrak{S}^{(1)}}\mathfrak{M}_i^{(1)} \simeq C_j\otimes_{\beta_i,\mathfrak{S}}\mathfrak{M} \simeq C_j\otimes_{\widetilde{\xi}_{i,j},S[1/p]}\mathscr{M}[1/p].$$

Under these identifications the vertical arrows are the base changes of $\Phi : \mathscr{M}\to\mathscr{M}^{\varphi}$, and the lower horizontal arrow is $\vartheta_j = \theta_j^*\varepsilon$. By (5.3) and the definition $\eta = (\varphi^{(1)})^*\varepsilon$, the upper horizontal arrow is

$$\theta_j^*(\varphi^*_{\mathfrak{S}^{(1)}}\varepsilon) = (\theta_j\varphi_{\mathfrak{S}^{(1)}})^*\varepsilon = (\xi_j\varphi^{(1)})^*\varepsilon = \xi_j^*\eta.$$

In full, the pulled-back square is

$$\begin{array}{ccc} C_j\otimes_{\widetilde{\xi}_{0,j},S[1/p]}\mathscr{M}[1/p] & \xrightarrow{\xi_j^*\eta} & C_j\otimes_{\widetilde{\xi}_{1,j},S[1/p]}\mathscr{M}[1/p] \\ {\scriptstyle 1\otimes\Phi}\downarrow & & \downarrow{\scriptstyle 1\otimes\Phi} \\ C_j\otimes_{\widetilde{\xi}_{0,j},S[1/p]}\mathscr{M}^{\varphi}[1/p] & \xrightarrow[\vartheta_j]{} & C_j\otimes_{\widetilde{\xi}_{1,j},S[1/p]}\mathscr{M}^{\varphi}[1/p]. \end{array}$$

Pulling back along $\theta_j$ identifies the upper modules with $C_j \otimes_{\widetilde{\xi}_{i,j},S[1/p]} \mathscr{M}[1/p]$, the lower modules with $C_j \otimes_{\widetilde{\xi}_{i,j},S[1/p]} \mathscr{M}^{\varphi}[1/p]$, and the vertical maps with the pullbacks of $\Phi$. The lower horizontal map is $\vartheta_j$. By (5.3), the upper horizontal map is

$$\theta_j^*(\varphi^*\varepsilon) = (\theta_j\varphi)^*\varepsilon = (\xi_j\varphi^{(1)})^*\varepsilon = \xi_j^*\eta.$$

The first-order Taylor term of $\eta$ is $1 \otimes x + \sum_k \bar{\sigma}_k \otimes N_k(x)$, since $s_0(t_k)/s_1(t_k) - 1$ agrees with $\sigma_k$ to first order. Hence the upper map sends

$$1 \otimes x \longmapsto 1 \otimes x + Ez_j \otimes N_j(x).$$

Comparing the two paths around the pulled-back square gives

$$\Theta_j'(\Phi(x)) = E\Phi(N_j(x)) \quad \text{for } x \in \mathscr{M}[1/p].$$

Both sides satisfy the same Leibniz rule relative to $\Phi$, so this equality extends after inverting $E$. There $\Phi$ is an isomorphism, and therefore $\Theta_j' = E\Phi N_j\Phi^{-1} = \Theta_j$. Since $\Theta_j'$ is defined on $\mathscr{M}^{\varphi}[1/p]$, this proves both the regularity of $\Theta_j$ and the formula (5.2). □

*Proof of Lemma 5.3.* By Lemma 5.4, $\Theta_j(y)$ is the first-order coefficient of $\vartheta_j(1 \otimes y)$ for $y \in \mathscr{M}^{\varphi}[1/p]$, and hence belongs to $\mathscr{M}^{\varphi}[1/p]$. □

*Proof of Proposition 5.2.* As noted above, $ED_j$ preserves $\operatorname{Fil}^r S$. Thus (5.1) gives

$$\Theta_j(\operatorname{Fil}^r S\mathscr{M}^{\varphi}[1/p]) \subseteq \operatorname{Fil}^r S\mathscr{M}^{\varphi}[1/p]$$

for any $r \geq 0$. We next justify cancellation of $E$ integrally. For $1 \leq r < p$, multiplication by $E$ induces an injection $\mathfrak{S}/(E^{r-1}) \hookrightarrow \mathfrak{S}/(E^r)$. Tensoring with finite projective $S$-module $\mathscr{M}^{\varphi}$ along $S \to S/\operatorname{Fil}^r S \simeq \mathfrak{S}/(E^r)$ therefore gives an injection $\mathscr{M}^{\varphi}/\operatorname{Fil}^{r-1} S\mathscr{M}^{\varphi} \xrightarrow{E} \mathscr{M}^{\varphi}/\operatorname{Fil}^r S\mathscr{M}^{\varphi}$. Moreover, $\mathscr{M}^{\varphi}/\operatorname{Fil}^{r-1} S\mathscr{M}^{\varphi}$ is $p$-torsion-free, as it is obtained from tensoring $\mathscr{M}^{\varphi}$ along the same map $S \to \mathfrak{S}/(E^r)$ above, and the target is $p$-torsion-free (Lemma 4.1). In particular, if $y \in \mathscr{M}^{\varphi}$ and $Ey \in \operatorname{Fil}^r S\mathscr{M}^{\varphi}[1/p]$, the map above implies that the class of $y$ in $(\mathscr{M}^{\varphi}/\operatorname{Fil}^{r-1} S\mathscr{M}^{\varphi})[1/p]$ is zero, so $p$-torsion-freeness implies $y \in \operatorname{Fil}^{r-1}\mathscr{M}^{\varphi}$.

Now take $x \in \operatorname{Fil}^r \mathscr{M}$, with $1 \leq r \leq h$, and fix $j$. By definition $\Phi(x) \in \operatorname{Fil}^r S\mathscr{M}^{\varphi}$. The identities above give

$$E\Phi(N_j(x)) = \Theta_j(\Phi(x)) \in \operatorname{Fil}^r S\mathscr{M}^{\varphi}[1/p].$$

The connection is integral, so $N_j(x) \in \mathscr{M}$ and $\Phi(N_j(x)) \in \mathscr{M}^{\varphi}$. The preceding argument implies $\Phi(N_j(x)) \in \operatorname{Fil}^{r-1} S\mathscr{M}^{\varphi}$, i.e. $N_j(x) \in \operatorname{Fil}^{r-1}\mathscr{M}$. We deduce Griffiths transversality:

$$\nabla(\operatorname{Fil}^r \mathscr{M}) \subseteq \operatorname{Fil}^{r-1}\mathscr{M} \otimes_S \omega_S^1.$$

Multiplicativity gives $(\operatorname{Fil}^1 S)\operatorname{Fil}^{h-1}\mathscr{M} \subset \operatorname{Fil}^h \mathscr{M}$, proving the first assertion for $h \geq 1$; for $h = 0$ this is immediate.

Base-changing the Frobenius-compatible prismatic descent square along $\varphi^{(1)}$ shows that the linearized ordinary Frobenius $S \otimes_{\varphi,S} \mathscr{M} \to \mathscr{M}$ is compatible with the HPD stratifications. Thus ordinary Frobenius is compatible with $\eta$. Modulo $J_{\Delta}^{[2]}$, we have

$$\eta(1 \otimes x) = 1 \otimes x + \sum_j \bar{\sigma}_j \otimes N_j(x) \quad \text{and} \quad \varphi(\bar{\sigma}_j) = p\bar{\sigma}_j.$$

Comparing the coefficients of $\bar{\sigma}_j$ in this compatibility gives $N_j\varphi = p\varphi N_j$. For $x \in \operatorname{Fil}^h \mathscr{M}$, transversality gives $EN_j(x) \in \operatorname{Fil}^h \mathscr{M}$. In $\mathscr{M}[1/p]$,

$$\varphi_h(EN_j(x)) = p^{-h}\varphi(EN_j(x)) = p^{1-h}c_1\varphi(N_j(x)) = c_1N_j(p^{-h}\varphi(x)) = c_1N_j(\varphi_h(x)).$$

Both sides are integral and $\mathscr{M}$ is $p$-torsion-free, so this is an identity in $\mathscr{M}$. Finally, $(d\varphi/p)(d\log t_j) = d\log t_j$. The component identities therefore give the diagram in the statement. □

**Definition 5.5.** The *Breuil realization functor* is given by

$$\mathbb{M}_{\mathrm{Breuil}} : \mathrm{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}}) \to \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S)$$

mapping a prismatic $F$-crystal $(\mathcal{E}, \varphi_{\mathcal{E}})$ to the quadruple

$$(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h, \nabla)$$

which is an object of the target category by Proposition 5.2 and Subsection 4.3. The construction is functorial: morphisms preserve the filtration and divided Frobenius, and compatibility with descent makes them horizontal.

5.3. **Base change to Shilov points.** For each $1 \le j \le m$, write $\iota_j : (\Delta_j, M_{\Delta_j}) \to (X, M_X)$ for the inclusion of each Shilov point (cf. notation of § 3.4). Fix $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h, \nabla) \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$, and let

$$(S_R, (p), M_{\mathrm{Spf}\, S_R}) \longrightarrow (S_{L_j}, (p), M_{\mathrm{Spf}\, S_{L_j}})$$

be the induced morphism of Breuil log prisms. Write

$$\begin{aligned}
\mathscr{M}_j &:= \mathscr{M} \otimes_{S_R} S_{L_j}, \\
\mathrm{Fil}^h \mathscr{M}_j &:= \mathrm{Fil}^h \mathscr{M} \widehat{\otimes}_{S_R} S_{L_j}, \\
\varphi_{h,j}(m \otimes s) &:= \varphi_h(m) \otimes \varphi(s), \\
\nabla_j(m \otimes s) &:= (\nabla m) \otimes s + m \otimes ds.
\end{aligned}$$

Here the third formula is taken for $m \in \mathrm{Fil}^h \mathscr{M}$ and extends continuously to the completed tensor product. Lemma 4.10 identifies $\mathrm{Fil}^h \mathscr{M}_j$ with a closed, $p$-saturated submodule of $\mathscr{M}_j$. The natural map on logarithmic differentials is an isomorphism

$$S_{L_j} \otimes_{S_R} \omega^1_{S_R} \xrightarrow{\sim} \omega^1_{S_{L_j}}.$$

Indeed, the source has basis $d\log T_1, \ldots, d\log T_d$, while the target has basis $d\log u$ together with the $d\log T_i$ for $i \neq j$; under this map

$$d\log T_j = d\log u - \sum_{\substack{1 \le i \le m \\ i \neq j}} d\log T_i.$$

Thus the two displayed bases are related by an invertible change of basis. In the last formula defining $\nabla_j$, we use this isomorphism to view the first summand in $\mathscr{M}_j \otimes_{S_{L_j}} \omega^1_{S_{L_j}}$.

**Lemma 5.6.** *Let*

$$\eta : \mathscr{M} \otimes_{S_R, p_0} S_R^{(1)} \xrightarrow{\sim} \mathscr{M} \otimes_{S_R, p_1} S_R^{(1)}$$

*be an HPD-stratification with associated connection $\nabla$. Then the base-changed HPD-stratification*

$$\eta_j : \mathscr{M}_j \otimes_{S_{L_j}, p_0} S_{L_j}^{(1)} \xrightarrow{\sim} \mathscr{M}_j \otimes_{S_{L_j}, p_1} S_{L_j}^{(1)}$$

*has associated connection $\nabla_j$.*

This is essentially the same argument as [Tsu20, Prop. 23], but we provide a proof customized to our notation for the convenience of the reader.

*Proof.* Set $S := S_R$, $S' := S_{L_j}$, and $\mathscr{M}' := \mathscr{M}_j$, and write $p'_0, p'_1 : S' \to S'^{(1)}$ for the two projections. Let $J' \subset S'^{(1)}$ be the diagonal PD-ideal. Under the canonical identification

$$J'/J'^{[2]} \simeq \omega^1_{S'},$$

the connection associated to $\eta_j$ is obtained by subtracting the identity from $\eta_j$ and reducing modulo $J'^{[2]}$. Fix $m \in \mathscr{M}$ and $s' \in S'$. Since

$$(m \otimes s') \otimes 1 = (m \otimes 1) \otimes p'_0(s'),$$

the $p'_0$-linearity of $\eta_j$ gives

$$\eta_j((m \otimes s') \otimes 1) = \eta_j((m \otimes 1) \otimes 1) p'_0(s').$$

By the definition of the connection attached to $\eta$, after base change to $S'$ we have

$$\eta_j((m\otimes 1)\otimes 1) \equiv (m\otimes 1)\otimes 1 + (\nabla m)\otimes 1 \pmod{J'^{[2]}}.$$

The second summand already has coefficients in $J'/J'^{[2]}$, so in that summand we may replace $p_0'(s')$ by $p_1'(s')$. If

$$ds' := p_0'(s') - p_1'(s') \pmod{J'^{[2]}}$$

is viewed as an element of $\omega^1_{S'}$, it follows that

$$\eta_j((m\otimes s')\otimes 1) \equiv (m\otimes s')\otimes 1 + (\nabla m)\otimes s' + m\otimes ds' \pmod{J'^{[2]}}.$$

Consequently the connection associated to $\eta_j$ is

$$m\otimes s' \longmapsto (\nabla m)\otimes s' + m\otimes ds',$$

which is precisely $\nabla_j$. □

**Lemma 5.7.** *The quadruple $(\mathscr{M}_j, \mathrm{Fil}^h \mathscr{M}_j, \varphi_{h,j}, \nabla_j)$ is an object of $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_{L_j})$.*

We write the resulting functor $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h, \nabla) \mapsto (\mathscr{M}_j, \mathrm{Fil}^h \mathscr{M}_j, \varphi_{h,j}, \nabla_j)$ as

$$\iota_j^* : \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R) \to \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_{L_j}).$$

*Proof.* The module $\mathscr{M}_j$ is finite projective. Put $Q = \mathscr{M}/\mathrm{Fil}^h \mathscr{M}$. Lemma 4.10 gives

$$\mathscr{M}_j/\mathrm{Fil}^h \mathscr{M}_j \simeq S_{L_j}\widehat{\otimes}_{S_R} Q \simeq \mathfrak{S}_{L_j}\otimes_{\mathfrak{S}_R} Q.$$

Tensoring a finite-projective resolution of $Q$ along the flat map $\mathfrak{S}_R \to \mathfrak{S}_{L_j}$ proves the required projective-dimension bound. Lemma 2.15, together with closedness of $\mathrm{Fil}^h \mathscr{M}_j$, gives $(\mathrm{Fil}^h S_{L_j})\mathscr{M}_j \subset \mathrm{Fil}^h \mathscr{M}_j$.

The image generation of $\varphi_{h,j}(\mathrm{Fil}^h \mathscr{M}_j)$ is easily checked on a finite set of geenrators, and compatibility axiom for $\varphi_{h,j}$ with $\mathrm{Fil}^h S\mathscr{M}$ can be checked on simple tensors and extends by continuity. Integrability follows from the base-change formula for the connection, and quasi-nilpotence follows by pulling back the HPD-stratification, whose associated connection is $\nabla_j$ by the previous lemma. Griffiths transversality and Frobenius horizontality can also be checked on elementary tensors by the corresponding identities for $\mathscr{M}$ and the scalar Breuil rings; for transversality, this also follows from the density of the inclusion $S_{L_j}\otimes_{S_R} \mathrm{Fil}^1 S_R \subset \mathrm{Fil}^1 S_{L_j}$. □

Retain the Shilov-point notation $(\Delta_j, M_{\Delta_j})$ and $\iota_j : (\Delta_j, M_{\Delta_j}) \to (X, M_X)$ from Definition 2.7. Pullback along $\iota_j$ defines functors on prismatic crystals and on the corresponding Breuil realizations; we denote both functors by $\iota_j^*$.

**Lemma 5.8.** *The Breuil realization functor commutes with base change along Shilov points, i.e. the diagram*

$$\begin{array}{ccc} \mathrm{Vect}^{\varphi,h}((X,M_X)_{\mathbb{\Delta}}) & \xrightarrow{\mathbb{M}_{\mathrm{Breuil}}} & \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R) \\ \downarrow{\scriptstyle \iota_j^*} & & \downarrow{\scriptstyle \iota_j^*} \\ \mathrm{Vect}^{\varphi,h}((\Delta_j,M_{\Delta_j})_{\mathbb{\Delta}}) & \xrightarrow{\mathbb{M}_{\mathrm{Breuil}}} & \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_{L_j}) \end{array}$$

*Proof.* This follows from Lemmas 4.7, 5.7, and 5.6; compatibility with Frobenius is immediate from the base-change formula. □

**Definition 5.9.** Let $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h, \nabla) \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S)$. Set $\mathscr{D}(\mathscr{M}) := \mathscr{M}$, $\varphi_{\mathscr{D}(\mathscr{M})} := c_1^{-h}\varphi_h(E(u)^h -)$, and $\eta_{\mathscr{D}(\mathscr{M})} := \eta$ where $\eta : p_0^*\mathscr{M} \to p_1^*\mathscr{M}$ is the HPD-stratification associated to the integrable connection $\nabla$. The triple

$$(\mathscr{D}(\mathscr{M}), \varphi_{\mathscr{D}(\mathscr{M})}, \eta_{\mathscr{D}(\mathscr{M})})$$

defines an object of $\mathrm{Vect}^{\varphi}((X_1, M_{X_1})_{\mathrm{CRIS}})$. Write $\mathscr{M}$ for the object of $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S)$ and $\mathscr{D}(\mathscr{M})$ for the resulting triple as described. We denote the resulting functor $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S) \to \mathrm{Vect}^{\varphi}((X_1, M_{X_1})_{\mathrm{CRIS}})$ by $\mathscr{D}$.

**Proposition 5.10.** *On* $\mathrm{Vect}^{\varphi,h}((X,M_X)_{\mathbb{\Delta}})$*, there is a natural isomorphism of functors*

$$\mathscr{D}\circ\mathbb{M}_{\mathrm{Breuil}}\xrightarrow{\sim}D_{\mathrm{cris}}$$

*with values in* $\mathrm{Vect}^{\varphi}((X_1,M_{X_1})_{\mathrm{CRIS}})$*. Consequently, there is a natural isomorphism*

$$\big(\mathscr{D}\circ\mathbb{M}_{\mathrm{Breuil}}\big)[1/p]\xrightarrow{\sim}D_{\mathrm{isoc}}$$

*with values in* $\mathrm{Vect}^{\varphi}_{\mathbb{Q}}((X_1,M_{X_1})_{\mathrm{CRIS}})$*.*

*Proof.* Fix an effective prismatic $F$-crystal $\mathcal{E}_{\mathbb{\Delta}}$ of height at most $h$, and write

$$\mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}})=(\mathscr{M},\mathrm{Fil}^h\mathscr{M},\varphi_h,\nabla).$$

The integral crystalline realization $D_{\mathrm{cris}}(\mathcal{E}_{\mathbb{\Delta}})$ is the $F$-crystal determined by the triple $(\mathscr{M},\varphi_{\mathscr{M}},\eta)$, where $\eta: S^{(1)}\otimes_{p_0,S}\mathscr{M}\xrightarrow{\sim}S^{(1)}\otimes_{p_1,S}\mathscr{M}$ is obtained by evaluating $\mathcal{E}_{\mathbb{\Delta}}$ on the two projections. By construction of the Breuil realization, the connection $\nabla$ corresponds to this same HPD-stratification $\eta$. It remains to compare the integral Frobenii. By the definition of the divided Frobenius on $\mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}})$, for $m\in\mathscr{M}$ one has

$$\varphi_h(E(u)^h m)=p^{-h}\varphi_{\mathscr{M}}(E(u)^h m)=p^{-h}\varphi(E(u))^h\varphi_{\mathscr{M}}(m)=c_1^h\varphi_{\mathscr{M}}(m).$$

Therefore the Frobenius $c_1^{-h}\varphi_h(E(u)^h-)$ defining $\mathscr{D}(\mathscr{M})$ is precisely $\varphi_{\mathscr{M}}$. This proves the first isomorphism as an isomorphism of $F$-crystals. The second follows by inverting $p$ and using the natural identification $D_{\mathrm{cris}}(\mathcal{E}_{\mathbb{\Delta}})[1/p]\simeq D_{\mathrm{isoc}}(\mathcal{E}_{\mathbb{\Delta}})$. □

## 6. Functors to Galois representations

Let $(X,M_X)$ be a small affine or the log formal spectrum of a CDVR. We write

$$(\mathfrak{S},(E),M_{\mathrm{Spf}\,\mathfrak{S}}),\quad (S,(p),M_{\mathrm{Spf}\,S}),\quad (\mathbb{A}_{\mathrm{inf}},(E),M_{\mathrm{Spf}\,\mathbb{A}_{\mathrm{inf}}}),\quad\text{and}\quad (\mathbb{A}_{\mathrm{cris}},(p),M_{\mathrm{Spf}\,\mathbb{A}_{\mathrm{cris}}})$$

for the associated Breuil–Kisin, Breuil, $\mathbb{A}_{\mathrm{inf}}$, and $\mathbb{A}_{\mathrm{cris}}$ log prisms, respectively. Here $E$ denotes $E(u)$ in the small affine case and $E_L(u)$ in the CDVR case. We drop the subscripts and arguments to treat these two cases simultaneously, but will include them when necessary. Correspondingly, we write $T_{\mathfrak{S}}$ when a discussion applies to both the small affine and CDVR cases, and reserve $T_{\mathfrak{S}_R}$ for discussions specific to the small affine case. Results in the CDVR case are essentially known by [Moo24] and [DLMS24b], so the novelty applies to the small affine case. We include the CDVR case in our consideration to maintain notational consistency, as well as to make it clear that we can apply our results along the Shilov points which will be necessary to use purity.

6.1. **The $T_{\mathfrak{S}}$ functor.** Let $(\mathfrak{M},\varphi_{\mathfrak{M}})\in\mathrm{Mod}^{\varphi,h}(\mathfrak{S}_R)$. Endow the module $\mathfrak{M}\otimes_{\mathfrak{S}}W(\overline{R}^\flat)$ with the $\varphi$-action $\varphi(m\otimes a):=\varphi_{\mathfrak{M}}(m)\otimes\varphi(a)$. We define

$$T_{\mathfrak{S}}(\mathfrak{M}):=(\mathfrak{M}\otimes_{\mathfrak{S}}W(\overline{R}^\flat[(\pi^\flat)^{-1}]))^{\varphi=1}.$$

We also extend the $\varphi_h:\mathrm{Fil}^h\mathfrak{M}^*\to\mathfrak{M}^*$ map semilinearly to $\varphi_h:\mathrm{Fil}^h\mathfrak{M}^*\otimes_{\mathfrak{S}_R}W(\overline{R}^\flat)\to\mathfrak{M}^*\otimes_{\mathfrak{S}_R}W(\overline{R}^\flat)$ via $\varphi_h(m\otimes a)=\varphi_h(m)\otimes\varphi(a)$, and set

$$T^h_{\mathfrak{S}}(\mathfrak{M}):=(\mathrm{Fil}^h\mathfrak{M}^*\otimes_{\mathfrak{S}_R}W(\overline{R}^\flat))^{\varphi_h=1}.$$

We have $G_{R_\infty}$-actions on $T_{\mathfrak{S}}(\mathfrak{M})$ and $T^h_{\mathfrak{S}}(\mathfrak{M})$ given by $g\cdot(m\otimes a):=m\otimes g(a)$ for $g\in G_{R_\infty}$. This action preserves the image of $\mathfrak{S}_R$ in $W(\overline{R}^\flat)$.

**Lemma 6.1.** *Let* $\mathfrak{M}\in\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R)$*, and set*

$$\mathcal{M}:=\mathcal{O}_{\mathcal{E}_R}\otimes_{\mathfrak{S}_R}\mathfrak{M}.$$

*There are natural $G_{R_\infty}$-equivariant isomorphisms*

$$T^h_{\mathfrak{S}_R}(\mathfrak{M})(-h)\xrightarrow{\sim}T_{\mathfrak{S}_R}(\mathfrak{M})\xrightarrow{\sim}T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M}).$$

*The analogous statement holds over a CDVR $\mathcal{O}_L$, with $\mathfrak{S}_R$, $\mathcal{O}_{\mathcal{E}_R}$, and $G_{R_\infty}$ replaced by $\mathfrak{S}_L$, $\mathcal{O}_{\mathcal{E}_L}$, and $G_{L_\infty}$, respectively.*

*Proof.* The second isomorphism follows immediately from the definitions. We prove the first. The CDVR case is just the argument below on a single local factor, so we only address the small affine case explicitly. Retain the notation

$$A = W(\overline{R}^\flat), \quad B = W\big(\overline{R}^\flat[(\pi^\flat)^{-1}]\big), \quad C = \prod_{\mathfrak{P}\in\mathcal{P}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}}), \quad \text{and} \quad D = \prod_{\mathfrak{P}\in\mathcal{P}} W(\overline{L(\mathfrak{p})}^\flat)$$

introduced before Lemma 2.11. Since $\mathrm{Fil}^h \mathfrak{M}^*$ is finite projective over $\mathfrak{S}_R$ and $E^h\mathfrak{M}^* \subset \mathrm{Fil}^h \mathfrak{M}^*$, we have that

$$\mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B = \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B. \tag{6.1}$$

The linearized Frobenius induces a Frobenius-equivariant isomorphism $1 \otimes \Phi_{\mathfrak{M}} : \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B \xrightarrow{\sim} \mathfrak{M} \otimes_{\mathfrak{S}_R} B$. We may therefore identify $T_{\mathfrak{S}_R}(\mathfrak{M})$ with $(\mathfrak{M}^* \otimes_{\mathfrak{S}_R} B)^{\varphi=1}$. As in the proof of [LL23, Lem. 6.11], choose $\mathfrak{t} \in W(\overline{K}^\flat)$ satisfying $\varphi(\mathfrak{t}) = a_0^{-1}E(u)\mathfrak{t}$, and put $\beta := \varphi(\mathfrak{t})$. In particular, $\varphi(\beta) = \varphi(a_0^{-1}E(u))\beta$. The element $\beta$ is a unit in $B$. Indeed, reduction modulo $p$ gives $\bar{\mathfrak{t}}^{p-1} = \overline{a_0}^{-1}u^e$. Since $u = \pi^\flat$ is invertible in $\overline{R}^\flat[(\pi^\flat)^{-1}]$, both $\bar{\mathfrak{t}}$ and $\overline{\beta} = \bar{\mathfrak{t}}^p$ are units there.

Now fix $x \in \mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B$. Then by definition of $\varphi_h$, we have

$$\varphi_h(x) = x \quad \Longleftrightarrow \quad \varphi(x) = \left(\frac{\varphi(\beta)}{\beta}\right)^h x \quad \Longleftrightarrow \quad \varphi\left(\frac{x}{\beta^h}\right) = \frac{x}{\beta^h}.$$

Hence, dividing by $\beta^h$ defines an injective map

$$T^h_{\mathfrak{S}_R}(\mathfrak{M}) \longrightarrow (\mathfrak{M}^* \otimes_{\mathfrak{S}_R} B)^{\varphi=1}(h). \tag{6.2}$$

We claim this is surjective. Fix $y \in (\mathfrak{M}^* \otimes_{\mathfrak{S}_R} B)^{\varphi=1}$ and set $x = \beta^h y$. By (6.1), $x \in \mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B$, and the preceding calculation shows that $\varphi_h(x) = x$. For $\mathfrak{P} \in \mathcal{P}$, put

$$\mathfrak{M}_{\mathfrak{p}} := \mathfrak{S}_{L(\mathfrak{p})} \otimes_{\mathfrak{S}_R} \mathfrak{M}, \quad \text{and} \quad \mathrm{Fil}^h \mathfrak{M}^*_{\mathfrak{p}} := \mathfrak{S}_{L(\mathfrak{p})} \otimes_{\mathfrak{S}_R} \mathrm{Fil}^h \mathfrak{M}^*.$$

Then [LL23, Lem. 6.11(1)], whose proof applies verbatim in the imperfect residue field case, is the isomorphism

$$\Big(\mathrm{Fil}^h \mathfrak{M}^*_{\mathfrak{p}} \otimes_{\mathfrak{S}_{L(\mathfrak{p})}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}})\Big)^{\varphi_h=1} \xrightarrow{\sim} \Big(\mathfrak{M}^*_{\mathfrak{p}} \otimes_{\mathfrak{S}_{L(\mathfrak{p})}} W(\overline{L(\mathfrak{p})}^\flat)\Big)^{\varphi=1}(h), \quad z \longmapsto \frac{z}{\beta^h}.$$

In particular, the image of $\beta^h y = x$ in the $\mathfrak{P}$-factor of $\mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} D$ belongs to $\mathrm{Fil}^h \mathfrak{M}^*_{\mathfrak{p}} \otimes_{\mathfrak{S}_{L(\mathfrak{p})}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}})$. Hence

$$x \in \Big(\mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B\Big) \cap \Big(\mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} C\Big) \subset \mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} D.$$

Lemma 2.11 then implies $x \in \mathrm{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} A$. Since $\varphi_h(x) = x$, we have $x \in T^h_{\mathfrak{S}_R}(\mathfrak{M})$, and its image under (6.2) is $y$, proving that (6.2) is an isomorphism.

It remains to check the Tate twist. As in [LL23, Lem. 6.11], we set $\mathfrak{c} = \prod_{n\geq 1} \varphi^n\left(\frac{a_0^{-1}E(u)}{p}\right)$, which is a unit in $\mathbb{A}_{\mathrm{cris}}$, and we have that $t = \log[\epsilon] = \mathfrak{c}\beta$. Every $g \in G_{R_\infty}$ fixes $u = [\pi^\flat]$ and the coefficients of $E(u)$, and its action commutes with Frobenius. Thus $g(\mathfrak{c}) = \mathfrak{c}$ and $\frac{g(\beta)}{\beta} = \frac{g(t)}{t} = \chi_{\mathrm{cyc}}(g)$. It follows that $x \mapsto x/\beta^h$ is $G_{R_\infty}$-equivariant after twisting the target by $(h)$, and thus

$$T^h_{\mathfrak{S}_R}(\mathfrak{M}) \xrightarrow{\sim} T_{\mathfrak{S}_R}(\mathfrak{M})(h),$$

which proves the first asserted isomorphism after twisting by $(-h)$. □

**Lemma 6.2.** *Let* $\mathfrak{M}_i \in \mathrm{Mod}^{\varphi,h}(\mathfrak{S})$ *for* $i = 1, 2, 3$ *and*

$$0 \to \mathfrak{M}_1 \to \mathfrak{M}_2 \to \mathfrak{M}_3 \to 0$$

*be an admissible short exact sequence in* $\mathrm{Mod}^{\varphi,h}(\mathfrak{S})$. *Suppose that* $\mathfrak{M}_3$ *lies in* $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S})$ *or* $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(\mathfrak{S})$. *Then the sequence*

$$0 \to \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}_1^* \to \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}_2^* \to \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}_3^* \to 0 \tag{6.3}$$

*is exact.*

*Proof.* The sequence $0 \to \mathrm{Fil}^h \mathfrak{M}_1^* \to \mathrm{Fil}^h \mathfrak{M}_2^* \to \mathrm{Fil}^h \mathfrak{M}_3^* \to 0$ is exact by [LL23, Lem. 6.9], which is an immediate consequence of the isomorphism $\mathrm{Fil}^h \mathfrak{M}_i^* \simeq \mathfrak{M}_i$. Thus the obstruction to exactness of (6.3) follows from the vanishing of $\mathrm{Tor}_1^{\mathfrak{S}}(\mathrm{Fil}^h \mathfrak{M}_3^*, \mathbb{A}_{\mathrm{inf}})$ which is clear when $\mathfrak{M}_3$ is projective, and vanishes in the second case because $\mathbb{A}_{\mathrm{inf}}$ is $p$-torsion-free. □

6.2. **The $T_{\mathrm{cris}}$-functor.** Let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h}(S)$. We extend the divided Frobenius $\varphi_h : \mathrm{Fil}^h \mathscr{M} \to \mathscr{M}$ to

$$\varphi_h : \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M} \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathscr{M} : a \otimes m \mapsto \varphi(a) \otimes \varphi_h(m).$$

We use the same notation as there should be no confusion in context. Set

$$T_{\mathrm{cris}}(\mathscr{M}) := (\mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M})^{\varphi_h = 1} = \{x \in \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M} \ : \ \varphi_h(x) = x\}.$$

**Lemma 6.3.** *The complex* $\mathbb{A}_{\mathrm{inf}} \otimes^{\mathbf{L}}_{\mathfrak{S}} (S/p)$ *is concentrated in degree zero, i.e.* $\mathrm{Tor}_i^{\mathfrak{S}}(\mathbb{A}_{\mathrm{inf}}, S/p) = 0$ *for any* $i \geq 1$.

*Proof.* We briefly recall the argument that $S/p$ is free over $\mathfrak{S}/(p, E^p)$. Since $E^p = p!\gamma_p(E) = 0$ in $S/p$, the map $\mathfrak{S}/p \to S/p$ factors through $\mathfrak{S}/(p, E^p)$. As a module over $\mathfrak{S}/(p, E^p)$, the ring $S/p$ has basis $\prod_{i \geq 1} \gamma_{p^i}(E)^{a_i}$ for $0 \leq a_i < p$ and all but finitely many $a_i$ vanish. Indeed, if $n = \sum_{i \geq 0} a_i p^i$ is the base-$p$ expansion of $n$, then the divided-power multiplication identities give $\gamma_n(E) = u_n E^{a_0} \prod_{i \geq 1} \gamma_{p^i}(E)^{a_i}$ where $u_n \in \mathbb{F}_p^\times$. This gives the stated basis.

Now $(p, E^p)$ is a regular sequence on both $\mathfrak{S}$ and $\mathbb{A}_{\mathrm{inf}}$ (Lemma 4.1). Thus the Koszul complex $K_{\mathfrak{S}}(p, E^p)$ is a finite free resolution of $\mathfrak{S}/(p, E^p)$, while

$$\mathbb{A}_{\mathrm{inf}} \otimes_{\mathfrak{S}} K_{\mathfrak{S}}(p, E^p) = K_{\mathbb{A}_{\mathrm{inf}}}(p, E^p)$$

is exact in positive degrees. Here we use that a regular sequence is Koszul-regular; see [Aut, Tag 062F]. It follows that

$$\mathbb{A}_{\mathrm{inf}} \otimes^{\mathbf{L}}_{\mathfrak{S}} \mathfrak{S}/(p, E^p) \simeq \mathbb{A}_{\mathrm{inf}}/(p, E^p),$$

and thus

$$\mathbb{A}_{\mathrm{inf}} \otimes^{\mathbf{L}}_{\mathfrak{S}} (S/p) \simeq \mathbb{A}_{\mathrm{inf}}/(p, E^p) \otimes^{\mathbf{L}}_{\mathfrak{S}/(p,E^p)} (S/p).$$

where the final derived tensor product is underived because $S/p$ is free over $\mathfrak{S}/(p, E^p)$. □

**Lemma 6.4.** *Let* $\mathscr{M}_i \in \mathrm{Mod}^{\varphi,h}(S)$, $i = 1, 2, 3$, *and suppose* $\mathscr{M}_3$ *fits into a short exact sequence*

$$0 \to \mathscr{M}_1 \to \mathscr{M}_2 \to \mathscr{M}_3 \to 0$$

*in* $\mathrm{Mod}^{\varphi,h}(S)$ *where* $\mathscr{M}_1, \mathscr{M}_2 \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S)$ *and* $\mathscr{M}_3 \in \mathrm{Mod}^{\varphi,h,\mathrm{tor}}(S)$ *is killed by* $p^n$. *Then* $\mathrm{Tor}_1^S(\mathrm{Fil}^h \mathscr{M}_3, \mathbb{A}_{\mathrm{cris}}) = 0$. *In particular, the sequence*

$$0 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}_1 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}_2 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}_3 \to 0$$

*is exact.*

*Proof.* Set $F_i = \mathrm{Fil}^h \mathscr{M}_i$, and $Q_i = \mathscr{M}_i/F_i$. Admissibility gives short exact sequences on both the underlying modules and their $F_i$-submodules. The snake lemma therefore gives

$$0 \longrightarrow Q_1 \longrightarrow Q_2 \longrightarrow Q_3 \longrightarrow 0.$$

For $i = 1, 2$, the definition gives a finite-projective $\mathfrak{S}$-resolution of $Q_i$ of length at most one. Since $E$ is regular on $\mathbb{A}_{\mathrm{inf}}$, Lemma 4.9 gives $\mathrm{Tor}_j^{\mathfrak{S}}(\mathbb{A}_{\mathrm{inf}}, Q_i) = 0$ for $i = 1, 2$ and every $j \geq 1$. The quotient sequence consequently gives $\mathrm{Tor}_j^{\mathfrak{S}}(\mathbb{A}_{\mathrm{inf}}, Q_3) = 0$ for $j \geq 2$. On the other hand, $\mathrm{Tor}_1^{\mathfrak{S}}(\mathbb{A}_{\mathrm{inf}}, \mathscr{M}_3) = 0$

by Lemma 6.3 and dévissage through the defining torsion filtration. Applying Tor to $0 \to F_3 \to \mathscr{M}_3 \to Q_3 \to 0$ now proves

$$\mathrm{Tor}_1^{\mathfrak{S}}(\mathbb{A}_{\mathrm{inf}}, F_3) = 0. \tag{6.4}$$

For clarity, the comparison with Tor over $S$ uses the finite-level form of the completed period-ring base change in Lemma 2.18. Write $S_n = S/p^n$. Lemma 6.3, induction, and Lemma 2.18,

$$\mathbb{A}_{\mathrm{inf}} \otimes^{\mathbf{L}}_{\mathfrak{S}} S_n \simeq \mathbb{A}_{\mathrm{inf}} \otimes_{\mathfrak{S}} S_n \simeq \mathbb{A}_{\mathrm{cris}}/p^n.$$

Since $\mathbb{A}_{\mathrm{cris}}$ is $p$-torsion-free, also $\mathbb{A}_{\mathrm{cris}} \otimes^{\mathbf{L}}_S S_n \simeq \mathbb{A}_{\mathrm{cris}}/p^n$. The module $F_3$ is killed by $p^n$, so associativity gives

$$\mathbb{A}_{\mathrm{cris}} \otimes^{\mathbf{L}}_S F_3 \simeq (\mathbb{A}_{\mathrm{cris}}/p^n) \otimes^{\mathbf{L}}_{S_n} F_3 \simeq \mathbb{A}_{\mathrm{inf}} \otimes^{\mathbf{L}}_{\mathfrak{S}} F_3.$$

Together with (6.4), this proves the asserted Tor vanishing. Tensoring $F_\bullet$ with $\mathbb{A}_{\mathrm{cris}}$ gives the short exact sequence

$$0 \to \mathbb{A}_{\mathrm{cris}} \otimes_S F_1 \to \mathbb{A}_{\mathrm{cris}} \otimes_S F_2 \to \mathbb{A}_{\mathrm{cris}} \otimes_S F_3 \to 0.$$

The third term is killed by $p^n$, hence completion preserves the exactness ([Aut, Tag 0BNG]), i.e.

$$0 \longrightarrow \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S F_1 \longrightarrow \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S F_2 \longrightarrow \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S F_3 \longrightarrow 0,$$

is exact as needed. □

The following lemma is a technical input to the subsequent proposition.

**Lemma 6.5.** *Set $B := W(\overline{R}^\flat[(\pi^\flat)^{-1}])$. Let $\mathcal{N}$ be a finite $B$-module and let $\Phi : \mathcal{N} \to \mathcal{N}$ be a $\varphi_B$-semilinear map whose linearization is an isomorphism. Suppose that either*

(a) *$\mathcal{N}$ is finite projective over $B$; or*
(b) *$\mathcal{N}$ is killed by a power of $p$ and admits a finite filtration by $\Phi$-stable submodules whose successive quotients are killed by $p$ and finite projective over $B/(p)$.*

*Then*

$$\Phi - 1 : \mathcal{N} \longrightarrow \mathcal{N}$$

*is surjective.*

*Proof.* Write $\overline{B} := B/p = \overline{R}^\flat[(\pi^\flat)^{-1}]$. We first treat the case where $p\mathcal{N} = 0$ and $\mathcal{N}$ is finite projective over $\overline{B}$. For any $y \in \mathcal{N}$, the solutions of

$$\Phi(x) - x = y$$

form a finite étale cover of $\mathrm{Spec}(\overline{B})$. Indeed, locally on $\mathrm{Spec}(\overline{B})$, after choosing a basis of $\mathcal{N}$, the equation has the form

$$AX^{[p]} - X = Y, \quad \text{for } A \in \mathrm{GL}_d(\overline{B}).$$

The corresponding algebra is finite because $A$ is invertible, and it is étale because its Jacobian matrix is $-I_d$. By construction, $\overline{B}$ is the tilt of the geometric perfectoid cover obtained from $\overline{R}$, so every finite étale cover splits. Hence the displayed equation has a solution in $\mathcal{N}$. Surjectivity is stable under extensions of $\varphi$-modules proving case (b). Case (a) follows from a standard lifting argument. □

The following is the direct analogue of [LL23, Lem. 6.11, Prop. 6.12]. For part (a), we reduce to the statement of *loc. cit.* using the ring-theoretic and module-theoretic purity statements in Lemmas 2.10 and 2.11; for part (b), we give the coefficient comparison and contraction argument explicitly.

**Proposition 6.6.** *Let $\mathfrak{M}$ belong to $\mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S})$ or $\mathrm{Mod}^{\varphi,h,\mathrm{tor}}(\mathfrak{S})$, let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h}(S)$ be its associated Breuil module, and let $\mathcal{M} = \mathfrak{M} \otimes_{\mathfrak{S}} \mathcal{O}_{\mathcal{E}_R}$ be the associated étale $\varphi$-module. The following hold:*

(a) *The sequence*

$$0 \to T^h_{\mathfrak{S}}(\mathfrak{M}) \to \mathbb{A}_{\inf}\widehat{\otimes}_{\mathfrak{S}} \operatorname{Fil}^h \mathfrak{M}^* \xrightarrow{\varphi_h - 1} \mathbb{A}_{\inf}\widehat{\otimes}_{\mathfrak{S}}\mathfrak{M}^* \to 0$$

*is exact.*

(b) *The sequence*

$$0 \to T_{\mathrm{cris}}(\mathscr{M}) \to \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M} \xrightarrow{\varphi_h - 1} \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\mathscr{M} \to 0$$

*is exact.*

*Proof.* (a) We use the notation of the proof of Lemma 6.1, and write $\delta_\Lambda := \varphi_{h,\Lambda} - 1$ for $\Lambda \in \{A, B, C, D, W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}}), W(\overline{L(\mathfrak{p})}^\flat)\}$, and $\Phi_{h,\Lambda}$ for the linearization of $\varphi_{h,\Lambda}$. We will sometimes drop the $\Lambda$-subscript if it is clear from context.

Now [LL23, Lem. 6.11(2)] gives the surjectivity of each

$$\delta : \operatorname{Fil}^h \mathfrak{M}^*_{\mathfrak{p}} \otimes_{\mathfrak{S}_{L(\mathfrak{p})}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}}) \to \mathfrak{M}^*_{\mathfrak{p}} \otimes_{\mathfrak{S}_{L(\mathfrak{p})}} W(\mathcal{O}^\flat_{\overline{L(\mathfrak{p})}}).$$

Moreover, the integrality argument in the proof of the same lemma, applied to each Shilov factor, gives

$$\ker\Big(\delta_D : \operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} D \longrightarrow \mathfrak{M}^* \otimes_{\mathfrak{S}_R} D\Big) \subset \operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} C.$$

For a torsion object these two conclusions follow by dévissage from the case killed by $p$: the defining Frobenius-stable filtration remains exact after each of the displayed base changes, its successive quotients are finite projective in characteristic $p$, and both surjectivity of $\delta$ and the asserted kernel containment are preserved under extensions. Thus the preceding assertions hold for both projective and torsion $\mathfrak{M}$. Since $E$ is invertible in $B$, we have

$$\operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B = \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B \simeq \mathcal{M} \otimes_{\mathcal{O}_{\mathcal{E}_R}} B.$$

Under this identification,

$$\varphi_h(m \otimes b) = \frac{\varphi_{\mathcal{M}}(m)}{\varphi((a_0^{-1}E(u))^h)} \otimes \varphi_B(b).$$

This is the Frobenius of the twisted étale $\varphi$-module defined above. Since $a_0$ and $E(u)$ are units in $B$, its linearization is an isomorphism. If $\mathfrak{M}$ is projective, then $\mathfrak{M}^* \otimes_{\mathfrak{S}_R} B$ is finite projective over $B$. If $\mathfrak{M}$ is torsion, then it admits a Frobenius-stable filtration with successive quotients finite projective over $\mathfrak{S}_R/(p)$, and after base change the successive quotients are finite projective over $B/(p)$. Lemma 6.5 therefore shows in either case that $\delta_B$ is surjective. Now take $y \in \mathfrak{M}^* \otimes_{\mathfrak{S}_R} A$. Choose $x_B \in \operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B$ and $x_C \in \operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} C$ with $\delta_B(x_B) = \delta_C(x_C) = y$. Then $z := x_C - x_B$ lies in $\ker(\delta_D)$, hence in $\operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} C$. Therefore $x_B = x_C - z$ lies in both $\operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B$ and $\operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} C$. The module $\operatorname{Fil}^h \mathfrak{M}^*$ is finite projective when $\mathfrak{M}$ is projective, and it has a filtration by finite projective $\mathfrak{S}_R/(p)$-modules when $\mathfrak{M}$ is torsion, since $\operatorname{Fil}^h \mathfrak{M}^* \simeq \mathfrak{M}$. Thus Lemma 2.11 gives

$$\Big(\operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} B\Big) \cap \Big(\operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} C\Big) = \operatorname{Fil}^h \mathfrak{M}^* \otimes_{\mathfrak{S}_R} A.$$

Thus $x_B$ belongs to the source of $\delta_A$ and satisfies $\delta_A(x_B) = y$, proving surjectivity.

(b) It remains to prove the surjectivity of $\varphi_{\mathscr{M},h} - 1$. We essentially give the argument of the last paragraph of [LL23, Prop. 6.12], but provide more details to make the $p$-adic lifting step clear in the projective case. Put

$$P := \mathbb{A}_{\inf}\widehat{\otimes}_{\mathfrak{S}} \operatorname{Fil}^h \mathfrak{M}^* \quad \text{and} \quad P_0 := \mathbb{A}_{\inf}\widehat{\otimes}_{\mathfrak{S}}\mathfrak{M}^*,$$

as well as

$$Q := \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M} \quad \text{and} \quad Q_0 := \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\mathscr{M}.$$

Let $\theta : P \to Q$ (resp. $\theta_0 : P_0 \to Q_0$) denote the maps induced by the inclusion $\mathrm{Fil}^h \mathfrak{M}^* \to \mathrm{Fil}^h \mathscr{M}$ (resp. $\mathfrak{M}^* \to \mathscr{M}$). Recall the elements $\mathfrak{c} \in \mathbb{A}_{\mathrm{cris}}^{\times}$ from the proof of Lemma 6.1, and $c = c_1/\varphi(a_0)$ from Remark 4.6. We have that $\varphi\left(\frac{a_0^{-1}E}{p}\right) = \frac{\varphi(E)}{p\varphi(a_0)} = c$, hence

$$\varphi(\mathfrak{c}) = \varphi\left(\prod_{n\geq 1} \varphi^n\left(\frac{a_0^{-1}E}{p}\right)\right) = \prod_{n\geq 2} \varphi^n\left(\frac{a_0^{-1}E}{p}\right) = c^{-1}\mathfrak{c}.$$

Therefore (using Remark 4.6),

$$\varphi_{h,\mathscr{M}}(\mathfrak{c}^h\theta(x)) = \varphi(\mathfrak{c})^h c^h \theta(\varphi_{h,\mathfrak{M}}(x)) = \mathfrak{c}^h\theta(\varphi_{h,\mathfrak{M}}(x))$$

and hence $\rho := \mathfrak{c}^h\theta : P \to Q$ (resp $\rho_0 := \mathfrak{c}^h\theta_0 : P_0 \to Q_0$) are compatible with divided Frobenii, i.e. $\varphi_{\mathscr{M},h} \circ \rho = \rho_0 \circ \varphi_{\mathfrak{M},h}$. In particular, setting $\delta_{\mathfrak{M}} := \varphi_{\mathfrak{M},h} - 1$ and $\delta_{\mathscr{M}} := \varphi_{\mathscr{M},h} - 1$, we have

$$\delta_{\mathscr{M}} \circ \rho = \rho_0 \circ \delta_{\mathfrak{M}}.$$

Since $\mathbb{A}_{\mathrm{cris}} = \mathbb{A}_{\mathrm{inf}} + \mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}}$, Lemma 4.7(b) and Lemma 2.18 give

$$Q_0 = \rho_0(P_0) + \mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}} \cdot Q_0. \tag{6.5}$$

Note that multiplication by the unit $\mathfrak{c}^h$ does not change the image modulo $\mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}}$. Fix $z \in \mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}} \cdot Q_0$. By Lemma 2.18, $\mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}} = (\mathrm{Fil}^p S)\mathbb{A}_{\mathrm{cris}}$, and furthermore $\mathrm{Fil}^p S \cdot \mathscr{M} \subseteq \mathrm{Fil}^h \mathscr{M}$. Thus, the multiplication map factors as

$$\mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S (\mathrm{Fil}^p S \cdot \mathscr{M}) \longrightarrow Q \longrightarrow Q_0,$$

and its image in $Q_0$ is exactly $\mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}} \cdot Q_0$. Thus $z$ admits a lift $\widetilde{z} \in Q$. Explicitly, for a simple tensor of the form $z = a\gamma_j(E([\pi^\flat]))\otimes m$ with $a \in \mathbb{A}_{\mathrm{cris}}$, $j \geq p$ and $m \in \mathscr{M}$, we can take $\widetilde{z} = a \otimes \gamma_j(E(u))m \in Q$. Now note that $\varphi_h(\gamma_j(E)) = \frac{p^{j-h}c_1^j}{j!}$ which has $p$-adic valuation $j - h - v_p(j!) \geq p - 1 - h$. Set $\lambda := p - 1 - h \geq 1$. Write $\delta_{\mathfrak{M}} := \varphi_{\mathfrak{M},h} - 1$ and $\delta_{\mathscr{M}} := \varphi_{\mathscr{M},h} - 1$, and fix $y \in Q_0$, and put $e_0 := y$. Suppose that we have constructed $x_0, \ldots, x_{r-1} \in Q$ so that $x_i \in p^{i\lambda}Q$ and

$$e_r := y - \delta_{\mathscr{M}}\left(\sum_{i=0}^{r-1} x_i\right) \in p^{r\lambda}Q_0,$$

In view of (6.5), we may write $y_r = \rho(p_r) + z_r$ with $p_r \in P_0$ and $z_r \in \mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}} \cdot Q_0$. By part (a), we may choose $\widetilde{p}_r \in P$ with $\delta_{\mathfrak{M}}(\widetilde{p}_r) = p_r$, and by the argument above, we may choose a lift $\widetilde{z}_r \in Q$ with $\widetilde{z}_r \mapsto z_r$ and by the computation above, $\varphi_h(\widetilde{z}_r) \in p^\lambda Q_0$. Set

$$x_r := p^{r\lambda}(\rho(\widetilde{p}_r) - \widetilde{z}_r) \in p^{r\lambda}Q.$$

Then

$$\begin{aligned}
\delta_{\mathscr{M}}(x_r) &= p^{r\lambda}(\rho_0(\delta_{\mathfrak{M}}(\widetilde{p}_r)) - (\varphi_h(\widetilde{z}_r) - z_r)) \\
&= p^{r\lambda}(\rho_0(p_r) + z_r - \varphi_h(\widetilde{z}_r)) \\
&= p^{r\lambda}(y_r - \varphi_h(\widetilde{z}_n)) \\
&= e_r - p^{(r+1)\lambda}w_r
\end{aligned}$$

for some $w_r \in Q_0$. In particular, $e_{r+1} := e_r - \delta_{\mathscr{M}}(x_r) \in p^{(r+1)\lambda}Q_0$ which establishes the inductive step. If $\mathfrak{M}$ is torsion, then the construction terminates. Otherwise the series $x := \sum_{r\geq 0} x_r$ converges in the $p$-adically complete module $Q$, and $e_r \to 0$ in $Q_0$. We deduce that $\delta_{\mathscr{M}}(x) = y$; in the projective case, this follows by $p$-adic continuity, and of course in the torsion case, the sum is finite and $e_r = 0$ for $r \gg 0$ and we obtain the same result. □

**Corollary 6.7.** *The following exactness statements hold.*

(a) *Let*

$$0 \longrightarrow \mathfrak{M}_1 \longrightarrow \mathfrak{M}_2 \longrightarrow \mathfrak{M}_3 \longrightarrow 0$$

*be an admissible short exact sequence of Breuil–Kisin modules such that each $\mathfrak{M}_i$ is projective or torsion. Then*

$$0 \longrightarrow T^h_{\mathfrak{S}}(\mathfrak{M}_1) \longrightarrow T^h_{\mathfrak{S}}(\mathfrak{M}_2) \longrightarrow T^h_{\mathfrak{S}}(\mathfrak{M}_3) \longrightarrow 0$$

*is exact.*

(b) *Let*

$$0 \longrightarrow \mathfrak{M}_1 \longrightarrow \mathfrak{M}_2 \longrightarrow \mathfrak{M}_3 \longrightarrow 0$$

*be an admissible short exact sequence of Breuil–Kisin modules such that $\mathfrak{M}_1$ and $\mathfrak{M}_2$ are projective and $\mathfrak{M}_3$ is torsion. Let*

$$0 \longrightarrow \mathscr{M}_1 \longrightarrow \mathscr{M}_2 \longrightarrow \mathscr{M}_3 \longrightarrow 0$$

*be the associated admissible short exact sequence of Breuil modules. Then*

$$0 \longrightarrow T_{\mathrm{cris}}(\mathscr{M}_1) \longrightarrow T_{\mathrm{cris}}(\mathscr{M}_2) \longrightarrow T_{\mathrm{cris}}(\mathscr{M}_3) \longrightarrow 0$$

*is exact.*

*Proof.* For part (a), Lemma 6.2 gives an exact sequence

$$0 \to \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}_1^* \to \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}_2^* \to \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}_3^* \to 0.$$

The corresponding sequence formed from the unfiltered modules $\mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathfrak{M}_i^*$ is also exact. Indeed, Frobenius on $\mathfrak{S}$ is flat by Proposition 2.5(a), and the possible obstruction before completion is

$$\mathrm{Tor}_1^{\mathfrak{S}}(\mathfrak{M}_3^*, \mathbb{A}_{\mathrm{inf}}).$$

This vanishes when $\mathfrak{M}_3$ is projective, and in the torsion case it vanishes by dévissage from the $p$-torsion case because $\mathbb{A}_{\mathrm{inf}}$ is $p$-torsion-free. The maps $\varphi_h - 1$ therefore give a morphism between these two short exact sequences. By Proposition 6.6(a), all three vertical maps are surjective. The snake lemma now gives the asserted short exact sequence of their kernels, which are the $T^h_{\mathfrak{S}}(\mathfrak{M}_i)$.

For part (b), Lemma 6.4 gives an exact sequence

$$0 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}_1 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}_2 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}_3 \to 0.$$

The corresponding sequence $0 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathscr{M}_1 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathscr{M}_2 \to \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathscr{M}_3 \to 0$ is exact as well. Since $\mathscr{M}_3$ is killed by a power of $p$, this may be checked modulo $p^n$, where the completed base-change identity of Lemma 2.18 is an ordinary tensor-product identity. The torsion Breuil module $\mathscr{M}_3$ admits a finite filtration whose successive quotients are finite projective over $S/p$, and the required Tor vanishing follows by dévissage from Lemma 6.3. We again obtain a morphism between two short exact sequences whose vertical maps are the maps $\varphi_h - 1$. These maps are surjective by Proposition 6.6(b), so the snake lemma identifies the resulting short exact sequence of kernels with the asserted sequence of $T_{\mathrm{cris}}$-modules. □

6.3. **The comparison in the torsion case.** We show that the functors to Galois representations in the previous sections are isomorphic when applied to corresponding Breuil–Kisin and Breuil modules via Proposition 4.12. This will imply, via comparison to étale $\varphi$-modules, that the $\mathbb{Z}_p$-module $T_{\mathrm{cris}}(\mathscr{M})$ is finite free. Recall the element $\mathfrak{c} = \prod_{n \geq 1} \varphi^n \left( \frac{a_0^{-1} E(u)}{p} \right) \in \mathbb{A}_{\mathrm{cris}}^{\times}$ from the proof of Lemma 6.1, as well as $c := \varphi(a_0)^{-1} c_1$ from Remark 4.6: we have $\varphi(\mathfrak{c}) = c^{-1}\mathfrak{c}$. Let $\mathfrak{M}$ be projective or torsion and let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h}(S)$ be its associated Breuil module. On $\mathrm{Fil}^h \mathfrak{M}^*$ one has $\varphi_{h,\mathscr{M}}(x) = c^h \varphi_{h,\mathfrak{M}^*}(x)$ (by Remark 4.6). Thus the map

$$\rho : \mathbb{A}_{\mathrm{inf}} \widehat{\otimes}_{\mathfrak{S}} \mathrm{Fil}^h \mathfrak{M}^* \longrightarrow \mathbb{A}_{\mathrm{cris}} \widehat{\otimes}_S \mathrm{Fil}^h \mathscr{M}, \quad x \longmapsto \mathfrak{c}^h x \tag{6.6}$$

is $\varphi_h$-compatible. Since $\mathfrak{c}$ is $G_{R_\infty}$-invariant, $\rho$ is $G_{R_\infty}$-equivariant. The main goal of this section is to show the following.

**Theorem 6.8.** *Let $\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S})$ and $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S)$ be the associated Breuil module via Proposition 4.12. Then the map induced by* (6.6)

$$\rho : T^h_{\mathfrak{S}}(\mathfrak{M}) \to T_{\mathrm{cris}}(\mathscr{M})$$

*is a $G_{R_\infty}$-equivariant isomorphism.*

We deduce this theorem from the $p$-torsion case.

**Proposition 6.9.** *Let $\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{tor}}(\mathfrak{S})$ be a $p$-torsion object, and let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{tor}}(S)$ be the associated Breuil module. Then the map induced by* (6.6)

$$\rho : T^h_{\mathfrak{S}}(\mathfrak{M}) \to T_{\mathrm{cris}}(\mathscr{M})$$

*is a $G_{R_\infty}$-equivariant isomorphism.*

*Proof.* Recall that $\mathfrak{S}_1 = \mathfrak{S}/p$, $S_1 = S/p$, and we use the notation $\mathbb{A}_{\mathrm{inf},1} := \mathbb{A}_{\mathrm{inf}}/p$, and $\mathbb{A}_{\mathrm{cris},1} := \mathbb{A}_{\mathrm{cris}}/p$ below. Set $J := \mathrm{im}(\mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}} \to \mathbb{A}_{\mathrm{cris},1})$. Since $\mathbb{A}_{\mathrm{cris}} = \mathbb{A}_{\mathrm{inf}} + \mathrm{Fil}^p \mathbb{A}_{\mathrm{cris}}$, the natural map $\mathbb{A}_{\mathrm{inf},1} \to \mathbb{A}_{\mathrm{cris},1}/J$ induces an isomorphism

$$\mathbb{A}_{\mathrm{inf}}/(p, u^{ep}) \simeq \mathbb{A}_{\mathrm{cris}}/(p, \mathrm{Fil}^p) \simeq \mathbb{A}_{\mathrm{cris},1}/J. \tag{6.7}$$

For simplicity, write

$$P_0 := \mathbb{A}_{\mathrm{inf},1} \otimes_{\mathfrak{S}_1} \mathfrak{M}^*, \quad \text{and} \quad P := \mathrm{im}(\mathbb{A}_{\mathrm{inf},1} \otimes_{\mathfrak{S}_1} \mathrm{Fil}^h \mathfrak{M}^* \to P_0)$$

as well as

$$Q_0 := \mathbb{A}_{\mathrm{cris},1} \otimes_{S_1} \mathscr{M}, \quad \text{and} \quad Q := \mathrm{im}(\mathbb{A}_{\mathrm{cris},1} \otimes_{S_1} \mathrm{Fil}^h \mathscr{M} \to Q_0).$$

Note that these are the mod $p$ reductions of the analogous modules appearing in the proof of Proposition 6.6(b). In particular, $T^h_{\mathfrak{S}}(\mathfrak{M}) = P^{\varphi_h=1}$ and $T_{\mathrm{cris}}(\mathscr{M}) = Q^{\varphi_h=1}$. We also consider their reductions by the image of $\mathrm{Fil}^p$; set

$$\overline{P} := \mathrm{im}(P \to P_0/u^{ep}P_0) = P/(P \cap u^{ep}P_0)$$

and

$$\overline{Q} := \mathrm{im}(Q \to Q_0/JQ_0).$$

The inclusion $\mathrm{Fil}^h \mathfrak{M}^* \to \mathrm{Fil}^h \mathscr{M}$ induces a map $\theta : P \to Q$, and in turn $\bar{\theta} : \overline{P} \to \overline{Q}$. By Lemma 4.7(b),

$$\mathrm{Fil}^h \mathscr{M} = \mathrm{Fil}^h \mathfrak{M}^* + (\mathrm{Fil}^p S)\mathscr{M} \tag{6.8}$$

inside $\mathscr{M}$, extending along $\mathfrak{S}_1 \to \mathbb{A}_{\mathrm{inf},1}$ and reducing by $J$ gives $\overline{\theta} : \overline{P} \to \overline{Q}$; in view of (6.7), we have that $\overline{\theta}$ is an isomorphism. We must modify $\theta$ slightly so that it commutes with divided Frobenii. Fix $x \in \mathrm{Fil}^h \mathfrak{M}^*$. Recall that $\Psi_{\mathfrak{M}} : \mathfrak{M} \xrightarrow{\sim} \mathrm{Fil}^h \mathfrak{M}^*$ satisfies $\Phi_{\mathfrak{M}}\Psi_{\mathfrak{M}}(m) = E^h m$, so we may write $x = \Psi_{\mathfrak{M}}(m)$ for some $m \in \mathfrak{M}$. We obtain a commutative diagram

$$\begin{array}{ccc} P^{\varphi_h=1} & \xrightarrow{\rho} & Q^{\varphi_h=1} \\ \downarrow & & \downarrow \\ \overline{P}^{\varphi_h=1} & \xrightarrow[\sim]{\overline{\rho}} & \overline{Q}^{\varphi_h=1} \end{array}$$

where the vertical arrows are the natural reduction maps. We show that both vertical arrows are bijective, implying that $\rho$ is bijective.

For the left vertical arrow, note that $u^{eh}P_0 \subset P$. If $r \geq p$ and $y \in P_0$, semilinearity gives

$$\varphi_h(u^{er}y) = u^{ep(r-h)}\varphi_h(u^{eh}y). \tag{6.9}$$

Thus $\varphi_h$ carries $u^{er}P_0$ into $u^{er'}P_0$, where $r' = p(r-h)$. Starting with $r_0 = p$ and defining $r_{n+1} = p(r_n - h)$, the assumption $h \leq p-2$ implies that $r_n \to \infty$.

For injectivity, suppose $x \in P^{\varphi_h=1}$ maps to zero in $\overline{P}$. Then $x \in u^{ep}P_0$, and repeated application of (6.9), together with $x = \varphi_h(x)$, gives $x \in u^{er_n}P_0$ for every $n$. Since $P_0$ is $u$-adically separated, $x = 0$. For surjectivity, take $\overline{x} \in \overline{P}^{\varphi_h=1}$ and choose a lift $x_0 \in P$. Inductively set $x_{n+1} = \varphi_h(x_n)$.

We have $x_1 - x_0 := \varphi_h(x_0) - x_0 \in u^{ep}P_0$, and thus (6.9) gives $x_{n+1} - x_n \in u^{er_n}P_0$ for every $n \geq 0$. Hence $(x_n)$ converges $u$-adically to an element $x \in P$, lifting $\overline{x}$. Thus the left vertical map is bijective.

Next we treat the right vertical map. Since $\overline{\mathfrak{c}}$ is a unit, extending (6.8) along $\mathfrak{S}_1 \to \mathbb{A}_{\mathrm{inf},1}$ gives

$$Q = \rho(P) + JQ_0$$

and we have $\varphi_h(JQ_0) = 0$. Indeed, for $i \geq p$, $i - h - v_p(i!) \geq 1$, so the divided Frobenius of the divided powers $JQ_0$ vanish modulo $p$. Now suppose $y \in Q^{\varphi_h=1}$ maps to zero in $\overline{Q}$. Then $y \in JQ_0$, and $y = \varphi_h(y) = 0$, giving injectivity. The commutativity of the diagram immediately gives that the right vertical is surjective, and putting this all together, we obtain that $\rho$ is an isomorphism. □

6.4. **Passage from torsion to projective modules.**

*Proof of Theorem 6.8.* Fix $\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S})$, and let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(S)$ be the associated Breuil module.

The modules $\mathbb{A}_{\mathrm{inf}}\widehat{\otimes}_{\mathfrak{S}}\mathfrak{M}^*$, $\mathbb{A}_{\mathrm{inf}}\widehat{\otimes}_{\mathfrak{S}}\operatorname{Fil}^h\mathfrak{M}^*$, and $\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\mathscr{M}$ are $p$-complete, since they are finite projective over $p$-complete rings. Put $Q = \mathscr{M}/\operatorname{Fil}^h\mathscr{M}$. By Lemmas 4.8 and 4.9, $\operatorname{Tor}_1^{\mathfrak{S}}(\mathbb{A}_{\mathrm{inf}}, Q) = 0$, and $\mathbb{A}_{\mathrm{inf}}\widehat{\otimes}_{\mathfrak{S}}Q$ is $p$-torsion-free and $p$-complete. Here we use regularity of $(p, E)$ on $\mathbb{A}_{\mathrm{inf}}$.

Using the completed period-ring base change of Lemma 2.18, the resulting exact sequence is

$$0 \longrightarrow \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\operatorname{Fil}^h\mathscr{M} \longrightarrow \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\mathscr{M} \longrightarrow \mathbb{A}_{\mathrm{inf}}\widehat{\otimes}_{\mathfrak{S}}Q \longrightarrow 0.$$

Its rightmost term is $p$-torsion-free, so reduction modulo $p^n$ is exact on the left for every $n$. Taking inverse limits shows that the leftmost term is $p$-complete, since the middle and rightmost terms are $p$-complete. Finally, the fixed submodules defining $T^h_{\mathfrak{S}}(\mathfrak{M})$ and $T_{\mathrm{cris}}(\mathscr{M})$ are $p$-complete: their defining maps $\varphi_h - 1$ have $p$-torsion-free targets, so these kernels are $p$-saturated, and limits of compatible elements remain in the kernels.

Consider the short exact sequences

$$0 \to \mathfrak{M} \xrightarrow{p} \mathfrak{M} \to \mathfrak{M}/p \to 0 \quad \text{and} \quad 0 \to \mathscr{M} \xrightarrow{p} \mathscr{M} \to \mathscr{M}/p \to 0.$$

The second is the Breuil-module sequence associated to the first by Lemma 4.5. Corollary 6.7 gives a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & T^h_{\mathfrak{S}}(\mathfrak{M}) & \xrightarrow{\times p} & T^h_{\mathfrak{S}}(\mathfrak{M}) & \longrightarrow & T^h_{\mathfrak{S}}(\mathfrak{M}/p) & \longrightarrow & 0 \\
 & & \downarrow\rho & & \downarrow\rho & & \downarrow\rho & & \\
0 & \longrightarrow & T_{\mathrm{cris}}(\mathscr{M}) & \xrightarrow{\times p} & T_{\mathrm{cris}}(\mathscr{M}) & \longrightarrow & T_{\mathrm{cris}}(\mathscr{M}/p) & \longrightarrow & 0.
\end{array}$$

The right vertical map is an isomorphism by Proposition 6.9. Exactness of the rows therefore shows that the middle vertical map induces an isomorphism

$$T^h_{\mathfrak{S}}(\mathfrak{M})/p \xrightarrow{\sim} T_{\mathrm{cris}}(\mathscr{M})/p.$$

Both modules are $p$-adically complete and separated by the first paragraph, and exactness of the rows shows that they are $p$-torsion-free. It follows that the middle vertical map is an isomorphism: injectivity follows by repeatedly dividing an element of its kernel by $p$ and using $p$-adic separatedness, while surjectivity follows by successively lifting a preimage modulo $p^n$ and using $p$-adic completeness. Thus

$$\rho : T^h_{\mathfrak{S}}(\mathfrak{M}) \xrightarrow{\sim} T_{\mathrm{cris}}(\mathscr{M}),$$

as required. □

6.5. **Extending the Galois actions.** So far, the $T_{\mathfrak{S}}$ and $T_{\mathrm{cris}}$ functors are defined on $\mathrm{Mod}^{\varphi,h}(\mathfrak{S})$ and $\mathrm{Mod}^{\varphi,h}(S)$. The resulting $\mathbb{Z}_p$-modules $T_{\mathfrak{S}}(\mathfrak{M})$ and $T_{\mathrm{cris}}(\mathscr{M})$ have $G_{R_\infty}$-actions. We extend the Galois actions to the categories $\mathrm{DD}^h_{\mathfrak{S}}$ and $\mathrm{Mod}^{\varphi,h,\nabla}(S)$ following [Hok24, §2.9.2, §2.9.4]. We state our results for the small affine case, but the same results and proofs hold for the CDVR case as well.

Fix an effective prismatic $F$-crystal of height at most $h$, $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}})$. Let $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon)$ denote the associated Breuil–Kisin module with descent datum. Recall our notation from 2.3. For each $g \in G_R$, let $\iota : \mathfrak{S} \to \mathbb{A}_{\mathrm{inf}}$ be the inclusion map from 2.3, and let $\iota_g = g \circ \iota$ with the usual $G_R$-action on $\mathbb{A}_{\mathrm{inf}}(\overline{R})$. In particular, $\iota_g(T_i) = g([T_i^\flat])$. Set $M := \mathbb{A}_{\mathrm{inf}} \otimes_{\iota,\mathfrak{S}} \mathfrak{M}$ and $M_g := \mathbb{A}_{\mathrm{inf}} \otimes_{\iota_g,\mathfrak{S}} \mathfrak{M}$ and equip them with the semilinear Frobenii $\varphi_{\mathbb{A}_{\mathrm{inf}}} \otimes \varphi_{\mathfrak{M}}$. There is a $g$-linear isomorphism

$$\tau_g : M \to M_g, \quad a \otimes m \mapsto g(a) \otimes m.$$

Note that this is well-defined:

$$g(a\iota(s)) \otimes m = g(a)\iota_g(s) \otimes m = g(a) \otimes sm.$$

The two maps $\iota_1, \iota_g : \mathfrak{S} \to \mathbb{A}_{\mathrm{inf}}$ induce morphisms of prisms, and thus determine a map $\kappa_g : \mathfrak{S}^{(1)} \to \mathbb{A}_{\mathrm{inf}}$. Pulling back the descent datum $\varepsilon$ along $\kappa_g$ yields an $\mathbb{A}_{\mathrm{inf}}$-linear isomorphism $\lambda_g : M \to M_g$. We define the $G_R$-action on $M$ by $g \cdot x := \lambda_g^{-1}(\tau_g(x))$ for $x \in M$.

**Lemma 6.10.** *The maps $r_g := \lambda_g^{-1}\tau_g$ define a continuous semilinear $G_R$-action on $M$ which commutes with Frobenius.*

*Proof.* Given any pair of maps $a, b : \mathfrak{S} \to \mathbb{A}_{\mathrm{inf}}$ inducing a morphism of log prisms, write $\kappa_{a,b} : \mathfrak{S}^{(1)} \to \mathbb{A}_{\mathrm{inf}}$ the induced map and $\lambda_{a,b} : M_a \to M_b$ the pullback of the descent datum along $\kappa_{a,b}$. The cocycle condition implies $\lambda_{a,a} = \mathrm{id}$ and $\lambda_{b,c}\lambda_{a,b} = \lambda_{a,c}$, and for any $g \in G_R$, we have

$$\tau_g \circ \lambda_{a,b}^{-1} = \lambda_{a_g,b_g}^{-1} \circ \tau_g.$$

The two occurrences of $\tau_g$ are the evident $g$-linear maps between the corresponding base changes, and $a_g := g \circ a$ and $b_g := g \circ b$. Consequently,

$$\begin{aligned} r_g \circ r_{g'} &= \lambda_{\iota,\iota_g}^{-1}\tau_g\lambda_{\iota,\iota_{g'}}^{-1}\tau_{g'} \\ &= \lambda_{\iota,\iota_g}^{-1}\lambda_{\iota_g,\iota_{gg'}}^{-1}\tau_{gg'} = \lambda_{\iota,\iota_{gg'}}^{-1}\tau_{gg'} = r_{gg'}. \end{aligned}$$

The identity $r_1 = \mathrm{id}$ follows in the same way. Continuity follows from continuity of the action on $\mathbb{A}_{\mathrm{inf}}$ and of the maps obtained by pulling back the descent datum. Finally, Frobenius compatibility of $\varepsilon$, together with the fact that Frobenius and the $G_R$-action on $\mathbb{A}_{\mathrm{inf}}$ commute, shows that every $r_g$ commutes with Frobenius. □

We now construct the action on the Breuil realization intrinsically. Let $(\mathscr{M}, \mathrm{Fil}^h \mathscr{M}, \varphi_h, \nabla)$ be an object of $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S)$. The map $\iota : \mathfrak{S} \to \mathbb{A}_{\mathrm{inf}}$ extends uniquely across the divided-power completion to a map $\iota : S \to \mathbb{A}_{\mathrm{cris}}$, which we use to form the completed tensor products below. In the small affine case, write

$$\nabla(m) = \sum_{j=1}^{d} N_j(m) \otimes d\log T_j.$$

(The CDVR case is identical, with the coordinates $u, X_1, \ldots, X_b$.) For $g \in G_R$, put

$$\ell_j(g) := \log\left(\frac{g([T_j^\flat])}{[T_j^\flat]}\right) \in \mathrm{Fil}^1 \mathbb{A}_{\mathrm{cris}}.$$

The ratio is the Teichmüller lift of the compatible system of $p$-power roots of unity $g(T_j^\flat)/T_j^\flat$; in particular it is a unit even when $T_j$ is not, and its logarithm converges in the divided-power ideal of $\mathbb{A}_{\mathrm{cris}}$. We have the classical identities

$$\ell_j(g'g) = \ell_j(g') + g'(\ell_j(g)) \quad \text{and} \quad \varphi(\ell_j(g)) = p\ell_j(g).$$

For $J = (j_1, \dots, j_d) \in \mathbb{N}^d$, write $|J| := \sum_{k=1}^d j_k$, $J! = \prod_{k=1}^d j_k!$, $N^J := \prod_{k=1}^d N_k^{j_k}$, and $\gamma_J(\ell(g)) := \prod_{k=1}^d \gamma_{j_k}(\ell_k(g))$. Define

$$g(a \otimes m) := \sum_{J \in \mathbb{N}^d} g(a)\gamma_J(\ell(g)) \otimes N^J(m). \tag{6.10}$$

**Lemma 6.11.** *The series* (6.10) *converges and defines a continuous semilinear $G_R$-action on $\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\mathscr{M}$. It preserves the submodule*

$$\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M} \subset \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S\mathscr{M}$$

*and commutes with $\varphi_h$. Consequently, it induces a $G_R$-action on $T_{\mathrm{cris}}(\mathscr{M})$ extending its natural $G_{R_\infty}$-action.*

*Proof.* Quasi-nilpotence gives convergence of the Taylor series: modulo any power of $p$, only finitely many multi-indices contribute for a fixed $m$. The logarithmic Leibniz rule shows that Taylor transport carries $1 \otimes sm$ to $\iota_g(s)$ times the transport of $1 \otimes m$. Thus the formula respects the tensor relations and is $g$-semilinear. Integrability gives $[N_j, N_k] = 0$. We compute

$$\begin{aligned} g'\big(g(a \otimes m)\big) &= \sum_{I,J} (g'g)(a) g'\big(\gamma_I(\ell(g))\big)\gamma_J(\ell(g')) \otimes N^J N^I(m) \\ &= \sum_{I,J} (g'g)(a)\gamma_I(g'\ell(g))\gamma_J(\ell(g')) \otimes N^{I+J}(m) \\ &= \sum_{K} (g'g)(a)\gamma_K(g'\ell(g) + \ell(g')) \otimes N^K(m) \\ &= \sum_{K} (g'g)(a)\gamma_K(\ell(g'g)) \otimes N^K(m) = (g'g)(a \otimes m). \end{aligned}$$

The third equality follows from the divided-power binomial identity and the commutativity of the $N_j$. Quasi-nilpotence permits the indicated rearrangement of the Taylor series. Since $\ell(1) = 0$, the identity element acts trivially. The same convergence estimate on a finite set of generators, together with continuity of the coefficient action and of $g \mapsto \ell(g)$, gives continuity of the $G_R$-action above.

We show that $G_R$ fixes $\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M}$. Write $\ell_j(g) = Eb_j(g)$ with $b_j(g) \in \mathbb{A}_{\mathrm{cris}}$ and put $b(g)^J = \prod_j b_j(g)^{j_j}$. The divided-power identities give

$$\gamma_J(\ell(g)) = b(g)^J \frac{|J|!}{J!}\gamma_{|J|}(E).$$

For $|J| \geq h$, the scalar $\gamma_{|J|}(E)$ belongs to $\operatorname{Fil}^h S$, so for $m \in \operatorname{Fil}^h \mathscr{M}$ the $J$-term of the Taylor series of $g(a \otimes m)$ reads

$$g(a)\frac{|J|!}{J!}b(g)^J \otimes \gamma_{|J|}(E)N^J(m)$$

in $\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M}$, because $(\operatorname{Fil}^h S)\mathscr{M} \subset \operatorname{Fil}^h \mathscr{M}$. For $k = |J| < h$, we show that $E^k N^J(\operatorname{Fil}^h \mathscr{M}) \subset \operatorname{Fil}^h \mathscr{M}$ by Griffiths transversality and induction. Let $D_j$ denote the derivation of $S$ underlying $N_j$. If $E^k m \in \operatorname{Fil}^h \mathscr{M}$, then

$$E^{k+1}N_j(m) = EN_j(E^k m) - kD_j(E)E^k m \in \operatorname{Fil}^h \mathscr{M}.$$

By Griffiths transversality $EN_j(\operatorname{Fil}^h \mathscr{M}) \subset \operatorname{Fil}^h \mathscr{M}$. Since $k < h \leq p-2$, $J!$ is a unit in $S$, and the $J$-term above is now

$$g(a)\frac{b(g)^J}{J!} \otimes E^k N^J(m) \in \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M}$$

as needed.

For Frobenius compatibility, put $\varphi_{\mathscr{M}} = c_1^{-h}\varphi_h(E^h -) : \mathscr{M} \to \mathscr{M}$. The horizontality condition of Definition 5.1 implies

$$N_j\varphi_{\mathscr{M}} = p\varphi_0 N_j.$$

Hence the linearization of $\varphi_{\mathscr{M}}$ is horizontal for the Frobenius-pullback connection and therefore respects its HPD-stratification, by Proposition 2.20. Pulling back the HPD-stratification $(\iota_g, \iota) : S^{(1)} \to \mathbb{A}_{\mathrm{cris}}$ shows that the ambient Frobenius $\varphi_{\mathbb{A}_{\mathrm{cris}}} \otimes \varphi_0$ commutes with Taylor transport: both coefficient maps commute with Frobenius, and its action on the displacement is $\ell_j \mapsto p\ell_j$. On $\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \operatorname{Fil}^h \mathscr{M}$ we have

$$\varphi_{\mathbb{A}_{\mathrm{cris}}} \otimes \varphi_{\mathscr{M}} = p^h \varphi_h,$$

since $\varphi_{\mathscr{M}}(m) = c_1^{-h}\varphi_h(E^h m) = p^h \varphi_h(m)$ for $m \in \operatorname{Fil}^h \mathscr{M}$. As $\mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_S \mathscr{M}$ is $p$-torsion-free, we may cancel $p^h$ and obtain $\varphi_h g = g\varphi_h$. Consequently $T_{\mathrm{cris}}(\mathscr{M})$ is $G_R$-stable. For $g \in G_{R_\infty}$ all the $\ell_j(g)$ vanish, so the action restricts to the previously defined coefficient-wise action. □

**Lemma 6.12.** *Suppose that $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h, \nabla)$ is the Breuil realization of a prismatic $F$-crystal, and let $\varepsilon$ be the corresponding Breuil–Kisin descent datum. Then the intrinsic action* (6.10) *agrees with the action obtained by evaluating the crystalline descent datum induced by $\varepsilon$ at the pair of maps $\iota_g, \iota : S \to \mathbb{A}_{\mathrm{cris}}$.*

*Proof.* The HPD-stratification of $\mathscr{M}$ is the base change of $\varepsilon$ from the Breuil–Kisin log prism to the Breuil log prism. Under the equivalence of Proposition 2.20, its logarithmic Taylor expansion is (2.6). Evaluating at $(\iota_g, \iota)$ sends $\ell_{T_j}$ to $\ell_j(g)$. The resulting series is exactly (6.10). Hence the two actions agree. □

Thus the $G_R$-representation attached to a relative Breuil module is defined purely from the Breuil data $(\mathscr{M}, \operatorname{Fil}^h \mathscr{M}, \varphi_h, \nabla)$; the preceding lemma identifies it with crystalline descent when the module comes from a prismatic $F$-crystal.

**Proposition 6.13.** *Let $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \operatorname{Vect}^{\varphi,h}((X, M_X)_{\mathbb{\Delta}})$. There is a natural $G_R$-equivariant isomorphism*

$$T_{\mathrm{cris}}\big(\mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}})\big) \xrightarrow{\sim} T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})(h).$$

*Proof.* Let $(\mathfrak{M}, \varphi_{\mathfrak{M}}, \varepsilon)$ be the Breuil–Kisin module with descent datum obtained by evaluating $\mathcal{E}_{\mathbb{\Delta}}$ on the Breuil–Kisin log prism, and set $\mathscr{M} := \mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}})$. Theorem 6.8 and Lemma 6.1 give natural isomorphisms

$$T_{\mathrm{cris}}(\mathscr{M}) \xleftarrow[\rho]{\sim} T^h_{\mathfrak{S}}(\mathfrak{M}) \xrightarrow[\vartheta]{\sim} T_{\mathfrak{S}}(\mathfrak{M})(h) \xrightarrow{\sim} T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})(h)$$

of $G_{R_\infty}$-representations. Here $\rho(x) = \mathfrak{c}^h x$, $\vartheta(x) = x/\beta^h$, and $t = \mathfrak{c}\beta$. We prove directly that the resulting composite is equivariant for the full group $G_R$. There is no $G_R$-equivariance for the middle term.

For $\iota_\bullet \in \{\iota_1, \iota_g\}$, the unscaled base-change maps

$$\theta_\bullet : \mathbb{A}_{\mathrm{inf}}\widehat{\otimes}_{\iota_\bullet, \mathfrak{S}_R} \mathfrak{M}^* \longrightarrow \mathbb{A}_{\mathrm{cris}}\widehat{\otimes}_{\iota_\bullet, S_R} \mathscr{M}$$

and their restrictions to the $\operatorname{Fil}^h$-submodules are compatible with the descent maps attached to $\varepsilon$. Indeed, the stratification on $\mathscr{M}$ is obtained from $\varepsilon$ by base change from the Breuil–Kisin prism to the Breuil prism, and Lemma 6.12 identifies the resulting action with the intrinsic Taylor action above. Thus these maps identify the ordinary Frobenius-compatible ambient $\mathbb{B}_{\mathrm{cris}}(\overline{R})$-realizations together with their full $G_R$-actions.

In this common rational realization, the composite $\Xi := \vartheta \circ \rho^{-1}$ is multiplication by $\frac{1}{\mathfrak{c}^h \beta^h} = \frac{1}{t^h}$. It lands in the integral lattice $T_{\mathfrak{S}}(\mathfrak{M})(h)$ because $\rho$ and $\vartheta$ are integral isomorphisms. Identify $T_{\mathfrak{S}}(\mathfrak{M})$ with $T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$ using ordinary Frobenius and the descent datum $\varepsilon$; this identification is $G_R$-equivariant. For $g \in G_R$, using $g(t) = \chi_{\mathrm{cyc}}(g)t$ and the action on the Tate twist, we obtain

$$g \cdot \Xi(y) = \chi_{\mathrm{cyc}}(g)^h \frac{g(y)}{g(t)^h} = \frac{g(y)}{t^h} = \Xi(g \cdot y).$$

Consequently the composite above is $G_R$-equivariant, giving the required natural isomorphism

$$T_{\mathrm{cris}}\big(\mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}})\big) \xrightarrow{\sim} T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})(h)$$

as desired. □

**Lemma 6.14.** *Let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$, and set $\mathscr{M}_j := \mathscr{M} \otimes_{S_R} S_{L_j}$. There is a natural $G_{L_j}$-equivariant isomorphism*

$$T_{\mathrm{cris}}(\mathscr{M})|_{G_{L_j}} \xrightarrow{\sim} T_{\mathrm{cris}}(\mathscr{M}_j).$$

*Proof.* Choose an extension $\overline{\iota}_j^\sharp : \overline{R} \to \mathcal{O}_{\overline{L_j}}$ of the ring map $\iota_j^\sharp : R \to \mathcal{O}_{L_j}$. It induces compatible maps of the $\mathbb{A}_{\mathrm{inf}}$ and $\mathbb{A}_{\mathrm{cris}}$ period rings and a homomorphism $G_{L_j} \to G_R$. Together with the base-change map $\mathrm{Fil}^h \mathscr{M} \to \mathrm{Fil}^h \mathscr{M}_j$, the map on $\mathbb{A}_{\mathrm{cris}}$-period rings defines a natural map

$$b_j : T_{\mathrm{cris}}(\mathscr{M})|_{G_{L_j}} \longrightarrow T_{\mathrm{cris}}(\mathscr{M}_j).$$

It is $G_{L_j}$-equivariant: by Lemma 5.6, the HPD-stratification, and hence the Taylor formula defining the action on the target, is the base change of the one defining the action on the source.

It remains to prove that $b_j$ is an isomorphism. Let $\mathfrak{M} \in \mathrm{Mod}^{\varphi,h,\mathrm{proj}}(\mathfrak{S}_R)$ be the Breuil–Kisin module corresponding to the underlying Breuil module of $\mathscr{M}$, and set $\mathfrak{M}_j := \mathfrak{M} \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j}$. Naturality of the comparison morphism in Theorem 6.8 gives a commutative diagram

$$\begin{array}{ccc} T^h_{\mathfrak{S}_R}(\mathfrak{M}) & \longrightarrow & T^h_{\mathfrak{S}_{L_j}}(\mathfrak{M}_j) \\ \sim\downarrow & & \downarrow\sim \\ T_{\mathrm{cris}}(\mathscr{M}) & \xrightarrow{b_j} & T_{\mathrm{cris}}(\mathscr{M}_j). \end{array}$$

The top horizontal map is an isomorphism. Indeed, after twisting by $(-h)$, Lemma 6.1 identifies it with the natural base-change map

$$T_{\mathcal{O}_{\mathcal{E}_R}}\big(\mathfrak{M} \otimes_{\mathfrak{S}_R} \mathcal{O}_{\mathcal{E}_R}\big) \longrightarrow T_{\mathcal{O}_{\mathcal{E}_{L_j}}}\big(\mathfrak{M}_j \otimes_{\mathfrak{S}_{L_j}} \mathcal{O}_{\mathcal{E}_{L_j}}\big),$$

which is an isomorphism by (A.2). Since the vertical maps are isomorphisms, the commutative diagram shows that $b_j$ is an isomorphism. □

**Proposition 6.15.** *Let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$. Then $T_{\mathrm{cris}}(\mathscr{M})$ is a prismatic semistable $\mathbb{Z}_p$-representation with Hodge–Tate weights in $[0,h]$.*

*Proof.* Let $(\mathcal{E}, \varphi_{\mathcal{E}})$ be the Laurent $F$-crystal associated to $T_{\mathrm{cris}}(\mathscr{M})$ by Theorem 3.10. For each completed Shilov trait $\iota_j : \Delta_j \to X$, set

$$\mathscr{M}_j := S_{L_j} \otimes_{S_R} \mathscr{M}.$$

Lemma 6.14 gives a $G_{L_j}$-equivariant identification

$$T_{\mathrm{cris}}(\mathscr{M})|_{G_{L_j}} \simeq T_{\mathrm{cris}}(\mathscr{M}_j).$$

By Corollary B.8, the representation on the right is semistable with Hodge–Tate weights in $[0,h]$. Equivalently, $(\iota_j)^*_{\mathbb{\Delta}}(\mathcal{E}, \varphi_{\mathcal{E}})$ extends to an analytic prismatic $F$-crystal for every $j$. Theorem 3.14 now implies that $(\mathcal{E}, \varphi_{\mathcal{E}})$ extends to an analytic prismatic $F$-crystal, and therefore that $T_{\mathrm{cris}}(\mathscr{M})$ is prismatic semistable.

It remains to verify the global Hodge–Tate range. By Proposition 3.18, $D_{\mathrm{dR}}(T_{\mathrm{cris}}(\mathscr{M}))$ is a filtered vector bundle with locally free graded pieces, and its formation is compatible with Shilov base change. Hence, for every $i \in \mathbb{Z}$ and every $j$, we have

$$\mathrm{gr}^{-i} D_{\mathrm{dR}}(T_{\mathrm{cris}}(\mathscr{M})) \otimes_{R[1/p]} L_j \simeq \mathrm{gr}^{-i} D_{\mathrm{dR}}\big(T_{\mathrm{cris}}(\mathscr{M})|_{G_{L_j}}\big).$$

The right-hand side vanishes for $i \notin [0,h]$ by the local classification. Since $R[1/p] \to \prod_j L_j$ is injective and the graded pieces on the left are finite projective, they also vanish before base change. Thus the Hodge–Tate weights of $T_{\mathrm{cris}}(\mathscr{M})$ lie in $[0,h]$. □

## 7. Proof of the main theorem

7.1. **Construction of the quasi-inverse.** Let $(X, M_X)$ be a small affine or log formal spectrum of a CDVR, and fix an integer $h \geq 0$ so that $eh < p-1$.

**Lemma 7.1.** *Let $(\mathcal{E}_{\mathbb{\Delta}}, \varphi_{\mathcal{E}_{\mathbb{\Delta}}}) \in \mathrm{Vect}^{\mathrm{an},\varphi}((X, M_X)_{\mathbb{\Delta}})$, and suppose that $T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})$ has Hodge–Tate weights in $[0, h]$. Then $\mathcal{E}_{\mathbb{\Delta}}\{-h\}$ is a genuine effective prismatic $F$-crystal of height at most $h$.*

*Proof.* Put $\mathcal{F}_{\mathbb{\Delta}} := \mathcal{E}_{\mathbb{\Delta}}\{-h\}$, and let $(\mathfrak{N}, \varphi_{\mathfrak{N}}, \varepsilon_{\mathfrak{N}})$ be its evaluation on the Breuil–Kisin log prism, and for each $1 \leq j \leq m$, let $(\mathfrak{N}_j, \varphi_{\mathfrak{N}_j}, \varepsilon_{\mathfrak{N}_j})$ denote its base change along $\mathfrak{S}_R \to \mathfrak{S}_{L_j}$. Write $\iota_j : (\Delta_j, M_{\Delta_j}) \to (X, M_X)$ for the inclusion of each Shilov point, and set $\mathcal{F}_j := \iota_j^* \mathcal{F}_{\mathbb{\Delta}}$ By Lemma 3.3, $T_{\mathrm{et}}(\mathcal{F}_j) \simeq T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}})|_{G_{L_j}}(-h)$ whose Hodge–Tate weights lie in $[-h, 0]$. This is precisely the $G_{L_j}$-representation $T_{\mathfrak{S}_{L_j}}(\mathfrak{N}_j)$. The statement of [ČGM26, Prop. 2.12(3)] holds in the imperfect residue field case by the arguments of [Gao20, Thrm. 3.2.3, Lemma 4.2.9] (which give the contravariant form of *loc. cit.*). Hence each $\mathfrak{N}_j$ has height at most $h$, i.e.

$$E^h \mathfrak{N}_j \subseteq \Phi_{\mathfrak{N}_j}(\mathfrak{N}_j^*) \subseteq \mathfrak{N}_j. \tag{7.1}$$

This same argument solves the general CDVR case. Inverting $p$ in equation (2.2) and tensoring with $\mathfrak{N}[p^{-1}]$, we obtain (by [DLMS24a, Lem. 3.1])

$$\mathfrak{N}[p^{-1}] = (\mathfrak{N}[p^{-1}] \otimes_{\mathfrak{S}_R[p^{-1}]} \mathcal{O}_{\mathcal{E}_R}[p^{-1}]) \cap \prod_{j=1}^{m} \mathfrak{N}_j[p^{-1}]. \tag{7.2}$$

Now take $x \in \mathfrak{N}^*$, and put $y = \Phi_{\mathfrak{N}}(x)$. By definition of $\Phi_{\mathfrak{N}}$, $y \in \mathfrak{N}[E^{-1}]$, so in particular $y$ is in the first factor of the right-hand-side. By (7.1), $y$ also belongs to the second factor, hence $y \in \mathfrak{N}[p^{-1}] \cap \mathfrak{N}[E^{-1}] = \mathfrak{N}$ (Definition 3.4). Thus $\Phi_{\mathfrak{N}}(\mathfrak{N}^*) \subset \mathfrak{N}$, meaning $\mathcal{F}_{\mathbb{\Delta}}$ is effective. Now let $y \in \mathfrak{N}$, and set $z := \Phi_{\mathfrak{N}}^{-1}(E^h y) \in \mathfrak{N}^*[E^{-1}]$; (7.1) implies $z_j \in \mathfrak{N}_j^*$ for each $1 \leq j \leq m$. Replacing $\mathfrak{N}[p^{-1}]$ with $\mathfrak{N}^*[p^{-1}]$ in (7.2), we find that $z \in \mathfrak{N}^*[p^{-1}]$. We have that

$$\mathfrak{N}^* = \mathfrak{N}^*[p^{-1}] \cap \mathfrak{N}^*[E^{-1}]$$

by faithful flatness of the Frobenius and because $\sqrt{(p, E)} = \sqrt{(p, \varphi(E))}$ since $\varphi(E) \equiv E^p \bmod p$. Hence $z \in \mathfrak{N}^*$, and thus $E^h y = \Phi_{\mathfrak{N}}(z) \in \Phi_{\mathfrak{N}}(\mathfrak{N}^*)$, proving the inclusion $E^h \mathfrak{N} \subset \Phi_{\mathfrak{N}}(\mathfrak{N}^*)$. Therefore $\mathcal{F}_{\mathbb{\Delta}}$ has height at most $h$ (Lemma 3.5), and thus by Proposition 3.6, $\mathcal{F}_{\mathbb{\Delta}} = \mathcal{E}_{\mathbb{\Delta}}\{-h\}$ is a genuine prismatic $F$-crystal of height at most $h$. □

Let $\mathbb{M}'_{\mathrm{Breuil}} : \mathrm{Rep}^{\mathrm{st},h}_{\mathbb{Z}_p}(G_R) \to \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$ be the functor defined as follows. For a prismatic semistable representation $T$ with Hodge–Tate weights in $[0, h]$, let $\mathcal{E}_{\mathbb{\Delta}}$ be the unique analytic prismatic $F$-crystal whose étale realization is $T$, and set

$$\mathbb{M}'_{\mathrm{Breuil}}(T) = \mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\}).$$

This is well-defined by the previous lemma. Similarly, we may define $\mathbb{M}'_{\mathrm{Breuil}}$ in the CDVR setting.

*Remark* 7.2. Let $(\Delta, M_\Delta)$ be the log formal spectrum of a CDVR, and write $\mathcal{O}_L := \Gamma(\Delta, \mathcal{O}_\Delta)$. By Proposition 6.13 and Corollary B.8, we have a triangle of equivalences

$$\begin{array}{ccc} & \mathrm{Vect}^{\varphi,h}((\Delta, M_\Delta)_{\mathbb{\Delta}}) & \\ {\scriptstyle \mathbb{M}_{\mathrm{Breuil}}} \swarrow & & \searrow {\scriptstyle T_{\mathrm{et}}(-)(h)} \\ \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L) & \xrightarrow{\quad T_{\mathrm{cris}} \quad} & \mathrm{Rep}^{\mathrm{st}}_{\mathbb{Z}_p,[0,h]}(G_L) \end{array}$$

In particular, $\mathbb{M}'_{\mathrm{Breuil}}$ is quasi-inverse to $T_{\mathrm{cris}}$.

7.2. **The counit comparison.**

**Proposition 7.3.** *Let $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$, and let $\mathcal{E}_{\mathbb{\Delta}}$ be the analytic prismatic $F$-crystal whose étale realization gives $T_{\mathrm{cris}}(\mathscr{M})$. Then*

$$\mathscr{M} \simeq \mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\}).$$

*Proof.* Set $\mathcal{F}_{\mathbb{\Delta}} := \mathcal{E}_{\mathbb{\Delta}}\{-h\}$. By Lemma 7.1, $\mathcal{F}_{\mathbb{\Delta}}$ is effective of height at most $h$, so we may set $\mathscr{N} := \mathbb{M}_{\mathrm{Breuil}}(\mathcal{F}_{\mathbb{\Delta}})$. Proposition 6.13 and Lemma 3.3 give a natural isomorphism

$$\theta : T_{\mathrm{cris}}(\mathscr{M}) \xrightarrow{\sim} T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}}) \xrightarrow{\sim} T_{\mathrm{et}}(\mathcal{F}_{\mathbb{\Delta}})(h) \xrightarrow{\sim} T_{\mathrm{cris}}(\mathscr{N}).$$

Forgetting the connections, let $\mathfrak{M}$ and $\mathfrak{N}$ be the Breuil–Kisin modules corresponding to $\mathscr{M}$ and $\mathscr{N}$ under Proposition 4.12. By construction of $\mathbb{M}_{\mathrm{Breuil}}$, the module $\mathfrak{N}$ is the evaluation of $\mathcal{F}_{\mathbb{\Delta}}$ on the Breuil–Kisin log prism. Put

$$\mathcal{M} := \mathfrak{M} \otimes_{\mathfrak{S}_R} \mathcal{O}_{\mathcal{E}_R}, \quad \text{and} \quad \mathcal{N} := \mathfrak{N} \otimes_{\mathfrak{S}_R} \mathcal{O}_{\mathcal{E}_R}.$$

Theorem 6.8 and Lemma 6.1 give natural identifications

$$T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M}) \simeq T_{\mathrm{cris}}(\mathscr{M})(-h)|_{G_{R_\infty}}, \quad \text{and} \quad T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{N}) \simeq T_{\mathrm{cris}}(\mathscr{N})(-h)|_{G_{R_\infty}}.$$

Hence $\theta(-h)|_{G_{R_\infty}}$ induces an isomorphism between the two étale realizations. By Theorem A.7, there is a unique isomorphism of étale $\varphi$-modules $\alpha : \mathcal{M} \xrightarrow{\sim} \mathcal{N}$ whose étale realization is $\theta(-h)|_{G_{R_\infty}}$.

For every Shilov point, set

$$\mathscr{M}_j := \mathscr{M} \otimes_{S_R} S_{L_j}, \quad \text{and} \quad \mathscr{N}_j := \mathscr{N} \otimes_{S_R} S_{L_j},$$

as well as

$$\mathfrak{M}_j := \mathfrak{M} \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j}, \quad \text{and} \quad \mathfrak{N}_j := \mathfrak{N} \otimes_{\mathfrak{S}_R} \mathfrak{S}_{L_j}.$$

Let $\theta_j : T_{\mathrm{cris}}(\mathscr{M}_j) \xrightarrow{\sim} T_{\mathrm{cris}}(\mathscr{N}_j)$ be the restrictions of $\theta$ along each $G_{L_j} \to G_R$, where we identify $T_{\mathrm{cris}}(\mathscr{M}_j) \simeq T_{\mathrm{cris}}(\mathscr{M})|_{G_{L_j}}$ and $T_{\mathrm{cris}}(\mathscr{N}_j) \simeq T_{\mathrm{cris}}(\mathscr{N})|_{G_{L_j}}$. Since $T_{\mathrm{cris}}$ is fully faithful over $S_{L_j}$ by Corollary B.8, there are unique isomorphisms $f_j : \mathscr{M}_j \xrightarrow{\sim} \mathscr{N}_j$ satisfying $T_{\mathrm{cris}}(f_j) = \theta_j$. Under Proposition 4.12, the underlying morphism of Breuil modules corresponds to an isomorphism $\beta_j : \mathfrak{M}_j \xrightarrow{\sim} \mathfrak{N}_j$. Let

$$\alpha_j : \mathcal{M} \otimes_{\mathcal{O}_{\mathcal{E}_R}} \mathcal{O}_{\mathcal{E}_{L_j}} \longrightarrow \mathcal{N} \otimes_{\mathcal{O}_{\mathcal{E}_R}} \mathcal{O}_{\mathcal{E}_{L_j}}$$

be the base change of $\alpha$. We claim that $\alpha_j = \beta_j \otimes_{\mathfrak{S}_{L_j}} \mathcal{O}_{\mathcal{E}_{L_j}}$. Indeed, by functoriality of the base-change isomorphism (A.2), the étale realization of $\alpha_j$ is the restriction of $\theta(-h)$ along $G_{\widetilde{L}_{j,\infty}} \to G_{R_\infty}$. On the other hand, by the naturality of Theorem 6.8 and Lemma 6.1, the étale realization of $\beta_j \otimes \mathcal{O}_{\mathcal{E}_{L_j}}$ is $\theta_j(-h)|_{G_{\widetilde{L}_{j,\infty}}}$. The local étale $\varphi$-module realization is faithful, so the two morphisms of étale $\varphi$-modules are equal, proving the claim.

We now use purity to descend to the $\mathfrak{S}_R$-level. If $x \in \mathfrak{M}$, then $\alpha(x) \in \mathcal{N}$, while for every $j$ its image in the $j$-th local factor is $\alpha_j(x) = \beta_j(x) \in \mathfrak{N}_j$. Therefore $\alpha(x) \in \mathfrak{N}$ by Lemma 2.9 applied to $\mathfrak{N}$. Hence $\alpha(\mathfrak{M}) \subseteq \mathfrak{N}$. Applying the same argument to $\alpha^{-1}$ gives the reverse inclusion. Thus $\alpha$ restricts to an isomorphism of Breuil–Kisin modules $\beta : \mathfrak{M} \xrightarrow{\sim} \mathfrak{N}$. Applying the functor to Breuil modules, we obtain an isomorphism $f : \mathscr{M} \xrightarrow{\sim} \mathscr{N}$ (preserving the filtration and divided Frobenius). Its base change to $S_{L_j}$ is $f_j$: indeed, the corresponding maps of Breuil–Kisin modules become equal to $\alpha_j$ after extending scalars to $\mathcal{O}_{\mathcal{E}_{L_j}}$, and $\mathfrak{N}_j \to \mathfrak{N}_j \otimes_{\mathfrak{S}_{L_j}} \mathcal{O}_{\mathcal{E}_{L_j}}$ is injective. It remains to check that $f$ commutes with the connections. Consider the $S_R$-linear map

$$\nabla_{\mathscr{N}} \circ f - (f \otimes 1) \circ \nabla_{\mathscr{M}} : \mathscr{M} \longrightarrow \mathscr{N} \otimes_{S_R} \omega^1_{S_R}.$$

Under the natural isomorphism $S_{L_j} \otimes_{S_R} \omega^1_{S_R} \simeq \omega^1_{S_{L_j}}$ recorded in Subsection 5.3, its base change to $S_{L_j}$ is the corresponding difference for $f_j$, and hence vanishes. Its source and target are finite projective $S_R$-modules. Since $S_R \to \prod_{j=1}^m S_{L_j}$ is injective (Lemma 2.16), it already vanishes over $S_R$. Thus $f$ defines an isomorphism

$$\mathscr{M} \xrightarrow{\sim} \mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\})$$

in $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$ as needed. Every step in the construction is functorial in $\mathscr{M}$: the isomorphism $\theta$ is natural, $\alpha$ is its unique lift under the equivalence of Theorem A.7, and $\beta$ and $f$ are obtained from $\alpha$ by restriction and scalar extension. Thus the displayed isomorphisms form a natural transformation. □

*Proof of the Main Theorem.* Proposition 6.13 and Lemma 3.3 give a natural isomorphism

$$T_{\mathrm{cris}}\big(\mathbb{M}'_{\mathrm{Breuil}}(T)\big) \simeq T_{\mathrm{et}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\})(h) \simeq T.$$

Conversely, Propositions 6.15 and 7.3, respectively, associate to every $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_R)$ an analytic prismatic $F$-crystal $\mathcal{E}_{\mathbb{\Delta}}$ whose étale realization is $T_{\mathrm{cris}}(\mathscr{M})$ and give a natural isomorphism

$$\mathbb{M}'_{\mathrm{Breuil}}\big(T_{\mathrm{cris}}(\mathscr{M})\big) = \mathbb{M}_{\mathrm{Breuil}}(\mathcal{E}_{\mathbb{\Delta}}\{-h\}) \simeq \mathscr{M}.$$

Thus $T_{\mathrm{cris}}$ and $\mathbb{M}'_{\mathrm{Breuil}}$ are quasi-inverse equivalences. □

## Appendix A. Étale $\varphi$-Modules and Galois Representations

We adapt [Kim15, Prop. 7.3] to our setting of a semistable base ring.

A.1. **The perfectoid tower and tilting.** Fix a semistable base ring $R$ with framing $\square : R_0 \to R$. Also fix algebraic closures $\overline{K}$ of $K$ and $\overline{R}$ of $R$ as in § 2, and a compatible system of $p$-power roots $\pi^{1/p^n} \in \overline{K}$. Let $R^{\mathrm{nc}}_{\pi,\infty} := \bigcup_{n\geq 0} R[\pi^{1/p^n}] \subset \overline{R}$, let $\mathcal{O}_{K_\infty}$ be the $p$-adic completion of $\bigcup_n \mathcal{O}_K[\pi^{1/p^n}]$, and put $K_\infty := \mathcal{O}_{K_\infty}[1/p]$. After choosing the compatible roots of $\pi$, there is a canonical identification

$$\mathcal{O}^\flat_{K_\infty} \simeq \big(k[u^{1/p^\infty}]\big)^\wedge_{(u)}, \quad \pi^\flat \to u. \tag{A.1}$$

Define $E^+_R := \mathfrak{S}_R/p$, and $E_R := E^+_R[1/u]$, as their perfected variants $\widetilde{E}^+_{R_\infty} := \big((E^+_R)^{\mathrm{perf}}\big)^\wedge_{(u)}$ and $\widetilde{E}_{R_\infty} := \widetilde{E}^+_{R_\infty}[1/u]$. We have that $(\widetilde{E}_{R_\infty}, \widetilde{E}^+_{R_\infty})$ is a perfectoid $(K^\flat_\infty, \mathcal{O}^\flat_{K_\infty})$-algebra via (A.1). When $R = R_0$, this is clear from the explicit description of $\widetilde{E}^+_{R_\infty}$ as the $u$-adic completion

$$\varinjlim_{n\geq 0} k[T_1^{1/p^n}, \dots, T_m^{1/p^n}, T_{m+1}^{\pm 1/p^n}, \dots, T_d^{\pm 1/p^n}, u^{1/p^n}]/(T_1^{1/p^n}\cdots T_m^{1/p^n} - u^{1/p^n} \mid n \geq 0),$$

and the assertion for $R$ follows by the completed étale base change defining $\mathfrak{S}_R$. Choose compatible systems of $p$-power roots $T_i^{1/p^n} \in \overline{R}$ for $1 \leq i \leq d$ compatibly with the chosen roots of $\pi$ in the sense that $\prod_{i=1}^m T_i^{1/p^n} = \pi^{1/p^n}$ for every $n \geq 0$. Define

$$R^{\mathrm{nc}}_\infty := R^{\mathrm{nc}}_{\pi,\infty}[T_1^{1/p^\infty}, \dots, T_d^{1/p^\infty}] \subset \overline{R} \quad \text{and} \quad R_\infty := \widehat{R^{\mathrm{nc}}_\infty}.$$

The extension $R \to R^{\mathrm{nc}}_\infty$ is integral and becomes ind-étale after inverting $p$. If $\xi_K$ generates the kernel of $\theta : W(\mathcal{O}^\flat_{K_\infty}) \to \mathcal{O}_{K_\infty}$, then by [Sch12, Rem. 5.19], this completion identifies with the untilt

$$R_\infty \simeq W(\widetilde{E}^+_{R_\infty})/(\xi_K) \simeq W(\widetilde{E}^+_{R_\infty})\widehat{\otimes}_{W(\mathcal{O}^\flat_{K_\infty}),\theta}\mathcal{O}_{K_\infty}.$$

We therefore obtain an injection of affinoid perfectoid Huber pairs $(R_\infty[1/p], R_\infty) \hookrightarrow (\widehat{\overline{R}}[1/p], \widehat{\overline{R}})$ whose tilt is $(\widetilde{E}_{R_\infty}, \widetilde{E}^+_{R_\infty}) \hookrightarrow (\overline{R}^\flat[1/\pi^\flat], \overline{R}^\flat)$ sending $u \to \pi^\flat$. After choosing compatible geometric points this gives a continuous map $G_{R_\infty} \to G_R$, where $G_{R_\infty} := \pi_1^{\text{ét}}(\mathrm{Spec}\, R_\infty[1/p])$. We summarize this as follows.

**Proposition A.1.** *The pair $(R_\infty[1/p], R_\infty)$ is an affinoid perfectoid $K_\infty$-algebra. Its tilt is canonically*

$$(R_\infty[1/p], R_\infty)^\flat \cong (\widetilde{E}_{R_\infty}, \widetilde{E}^+_{R_\infty}).$$

*In particular, there is a tilting equivalence*

$$\mathrm{Fét}(R_\infty[1/p]) \simeq \mathrm{Fét}(\widetilde{E}_{R_\infty})$$

*and hence*

$$G_{\widetilde{E}_{R_\infty}} \simeq G_{R_\infty}.$$

*Proof.* Perfectoidness and the identification of the tilt were proved above. The equivalence of finite étale sites is [Sch12, Thrm. 5.25]. □

Recall that $\mathcal{O}_{\mathcal{E}_R} := \mathfrak{S}_R[E^{-1}]^\wedge_p$.

**Lemma A.2** (cf. [Kim15, Eq. (7.2.3)])**.** *The natural map*

$$\mathcal{O}_{\mathcal{E}_R}/p = (\mathfrak{S}_R/p)[1/u] \longrightarrow \widetilde{E}_{R_\infty}$$

*induces an equivalence on finite étale sites.*

*Proof.* Put $A := \mathfrak{S}_R/p$. The map

$$A[1/u] \longrightarrow (A[1/u])^{\mathrm{perf}} = A^{\mathrm{perf}}[1/u]$$

is a universal homeomorphism, and hence induces an equivalence on finite étale sites by topological invariance of the étale site. It remains to pass from $A^{\mathrm{perf}}[1/u]$ to its $u$-adic completion $\widetilde{E}_{R_\infty} = \big(A^{\mathrm{perf}}\big)^\wedge_{(u)}[1/u]$. Since $A$ is $u$-adically complete, it is henselian along $(u)$. Moreover, writing $A^{\mathrm{perf}} = \varinjlim_\varphi A$, the pair $(A^{\mathrm{perf}}, (u))$ is the filtered colimit, along Frobenius, of the henselian pairs $(A, (u^{p^n}))$. Indeed, each ideal $(u^{p^n})$ defines the same adic topology as $(u)$, and henselian pairs are preserved under filtered colimits. Thus $A^{\mathrm{perf}}$ is henselian along $(u)$. The Elkik–Gabber–Ramero approximation theorem [GR03, Prop. 5.4.54], applied to $(A^{\mathrm{perf}}, (u))$, therefore gives an equivalence

$$\mathrm{Fét}\big(A^{\mathrm{perf}}[1/u]\big) \xrightarrow{\sim} \mathrm{Fét}\big((A^{\mathrm{perf}})^\wedge_{(u)}[1/u]\big).$$

Composing the two equivalences proves the assertion. □

**Corollary A.3.** *There are natural equivalences*

$$\mathrm{Fét}(\mathcal{O}_{\mathcal{E}_R}) \simeq \mathrm{Fét}(\mathcal{O}_{\mathcal{E}_R}/p) \simeq \mathrm{Fét}(\widetilde{E}_{R_\infty}) \simeq \mathrm{Fét}(R_\infty[1/p]).$$

*Consequently, after choosing compatible separable closures, one has a canonical identification*

$$\pi_1^{\acute{e}t}(\mathrm{Spec}\,\mathcal{O}_{\mathcal{E}_R}) \cong G_{R_\infty} := \pi_1^{\acute{e}t}(\mathrm{Spec}\,R_\infty[1/p]).$$

*Proof.* The first isomorphism follows from [Aut, Tag `0D49`] as the pair $(\mathcal{O}_{\mathcal{E}_R}, (p))$ is henselian. The rest follows from Lemma A.2 and Proposition A.1. □

The inclusion $E_R \hookrightarrow \widetilde{E}_{R_\infty}$ admits a unique lift $\mathcal{O}_{\mathcal{E}_R} \hookrightarrow W(\widetilde{E}_{R_\infty})$ with compatible Frobenius lifts (cf. [Kim15, §7.2]). Also, the map $\mathfrak{S}_R \to W(\overline{R}^\flat) : u \mapsto [\pi^\flat]$ uniquely lifts to $\mathcal{O}_{\mathcal{E}_R} \to W(\overline{R}^\flat[(\pi^\flat)^{-1}])$. Let $\mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}$ be the filtered union of the finite étale $\mathcal{O}_{\mathcal{E}_R}$-algebras embedded in $W(\overline{R}^\flat[(\pi^\flat)^{-1}])$ using a compatible geometric base point. The embeddings are the unique Frobenius-compatible lifts of the corresponding residue-ring embeddings; see the construction in the proof of [Kim15, Lem. 7.2.7]. Write $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}$ for the $p$-adic completion of $\mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}$. The residue-ring inclusion makes this completion a $p$-saturated subring of $W(\overline{R}^\flat[(\pi^\flat)^{-1}])$. Define $\mathfrak{S}^{\mathrm{unr}}_R$ to be the integral closure of $\mathfrak{S}_R$ in $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}$, and let $\widehat{\mathfrak{S}}^{\mathrm{unr}}_R$ be its $p$-adic closure. These lie in $W(\overline{R}^\flat)$, as in [Kim15, §7.2]. We have a diagram

$$\begin{array}{ccc} \widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R} & \hookrightarrow & W\Big(\overline{R}^\flat\big[(\pi^\flat)^{-1}\big]\Big) \\ \circlearrowleft & & \circlearrowleft \\ \mathrm{Aut}(\mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}/\mathcal{O}_{\mathcal{E}_R}) & \xrightarrow{\sim} & G_{R_\infty} \end{array}$$

where the lower horizontal isomorphism is Corollary A.3. Moreover, the $G_{R_\infty}$-action on $\mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}$ fixes $\mathfrak{S}_R$ and, in turn, acts on $\mathfrak{S}^{\mathrm{unr}}_R \subset \widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}$.

**Lemma A.4.** *We have* $\mathcal{O}_{\mathcal{E}_R} = (\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R})^{G_{R_\infty}}$ *and* $\mathfrak{S}_R = (\widehat{\mathfrak{S}}^{\mathrm{unr}}_R)^{G_{R_\infty}}$*. The same holds modulo* $p^n$*.*

*Proof.* [Kim15, Lem. 7.2.6]. □

**Lemma A.5.** *There is a unique $G_{R_\infty}$-equivariant endomorphism $\varphi : \widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R} \to \widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}$ lifting the $p$-power map $\varphi : \mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}/(p) \to \mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}/(p)$ and extending $\varphi : \mathcal{O}_{\mathcal{E}_R} \to \mathcal{O}_{\mathcal{E}_R}$. Moreover, this map restricts to $\varphi : \widehat{\mathfrak{S}}^{\mathrm{unr}}_R \to \widehat{\mathfrak{S}}^{\mathrm{unr}}_R$, and the inclusion $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R} \hookrightarrow W(\overline{R}^\flat[(\pi^\flat)^{-1}])$ is $\varphi$-equivariant.*

*Proof.* [Kim15, Lem. 7.2.7]. □

## A.2. Étale $\varphi$-modules and Galois representations.

**Definition A.6.** Let $\mathrm{Mod}^{\varphi,\text{ét}}(\mathcal{O}_{\mathcal{E}_R})$ denote the category of pairs $(\mathcal{M}, \varphi_{\mathcal{M}})$ where $\mathcal{M}$ is a finite $\mathcal{O}_{\mathcal{E}_R}$-module and $\varphi_{\mathcal{M}} : \mathcal{M} \longrightarrow \mathcal{M}$ is a $\varphi$-semilinear endomorphism whose linearization

$$\Phi_{\mathcal{M}} : \varphi^*\mathcal{M} := \mathcal{O}_{\mathcal{E}_R} \otimes_{\varphi, \mathcal{O}_{\mathcal{E}_R}} \mathcal{M} \longrightarrow \mathcal{M}, \quad f \otimes m \mapsto f\varphi_{\mathcal{M}}(m)$$

is an isomorphism. Let $\mathrm{Mod}^{\varphi,\text{ét},\mathrm{proj}}(\mathcal{O}_{\mathcal{E}_R})$ be the full subcategory where $\mathcal{M}$ is finite projective, and let $\mathrm{Mod}^{\varphi,\text{ét},\mathrm{tor}}(\mathcal{O}_{\mathcal{E}_R})$ be the full subcategory of $p$-power torsion objects.

For $(\mathcal{M}, \varphi_{\mathcal{M}})$ define

$$T(\mathcal{M}) := \left(\mathcal{M} \otimes_{\mathcal{O}_{\mathcal{E}_R}} \widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}\right)^{\varphi=1}.$$

For a continuous $\mathbb{Z}_p$-representation $T$ of $G_{R_\infty}$ define

$$\mathcal{M}(T) := \left(T \otimes_{\mathbb{Z}_p} \widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}\right)^{G_{R_\infty}}.$$

The Frobenius on $\mathcal{M}(T)$ is induced from Frobenius on $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}$.

**Theorem A.7** (Semistable relative $\varphi$-module correspondence)**.** *The functors*

$$T : \mathrm{Mod}^{\varphi,\text{ét}}(\mathcal{O}_{\mathcal{E}_R}) \longrightarrow \mathrm{Rep}_{\mathbb{Z}_p}(G_{R_\infty})$$

*and*

$$\mathcal{M} : \mathrm{Rep}_{\mathbb{Z}_p}(G_{R_\infty}) \longrightarrow \mathrm{Mod}^{\varphi,\text{ét}}(\mathcal{O}_{\mathcal{E}_R})$$

*are exact quasi-inverse tensor equivalences. They restrict to rank-preserving equivalences*

$$\mathrm{Mod}^{\varphi,\text{ét},\mathrm{proj}}(\mathcal{O}_{\mathcal{E}_R}) \simeq \mathrm{Rep}^{\mathrm{free}}_{\mathbb{Z}_p}(G_{R_\infty})$$

*and length-preserving equivalences between $p$-power torsion étale $\varphi$-modules and finite $\mathbb{Z}_p$-torsion representations of $G_{R_\infty}$.*

*Proof.* Put $E_R := \mathcal{O}_{\mathcal{E}_R}/p = (\mathfrak{S}_R/p)[1/u]$. The ring $\mathfrak{S}_{R_0}/p$ is the $(T_1 \cdots T_m)$-adic completion of $k[T_1, \ldots, T_m, T_{m+1}^{\pm 1}, \ldots, T_d^{\pm 1}]$. It is regular, since completion preserves regularity for this excellent Noetherian ring. Regularity is preserved by the étale extension to $\mathfrak{S}_R/p$ and by localization at $u$. Hence $E_R$ is regular, and in particular normal. By Corollary A.3, after choosing compatible geometric base points, we have

$$\pi_1^{\text{ét}}(\operatorname{Spec} E_R) \simeq G_{R_\infty}.$$

For objects annihilated by $p$, the required equivalence is the Artin–Schreier correspondence of [BS21, Prop. 3.6], applied to $E_R$, together with the preceding identification of fundamental groups. The general result follows by dévissage through the powers of $p$ and passage to inverse limits. More precisely, this is the argument of [Kim15, Prop. 7.3]; its proof uses only the characteristic-$p$ correspondence, the fixed-point identities of Lemma A.4, and the compatible Frobenius of Lemma A.5, all of which hold in the present setting. The same argument shows that the equivalence is exact and compatible with tensor products, and that it preserves ranks in the projective case and lengths in the $p$-power torsion case. □

A.3. **Evaluation of Laurent $F$-crystals.**

**Proposition A.8.** *Let $(\mathcal{E},\varphi_{\mathcal{E}})$ be a Laurent $F$-crystal on $(X,M_X)_{\mathbb{\Delta}}$, $T=\mathcal{E}(\mathbb{A}_{\mathrm{inf}})^{\varphi=1}$ the associated $G_R$-representation, and $(\mathcal{M},\varphi_{\mathcal{M}})$ the étale $\varphi$-module given by evaluation of $(\mathcal{E},\varphi_{\mathcal{E}})$ on the Breuil–Kisin log prism $(\mathfrak{S}_R,(E),M_{\operatorname{Spf}\mathfrak{S}_R})$. Then*

$$T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M})\simeq T|_{G_{R_\infty}},$$

*where the right-hand side denotes the restriction of the $G_R$-representation $T$ along the inclusion $G_{R_\infty}\to G_R$.*

*Proof.* The natural map $\mathfrak{S}_R\to W(\overline{R}^\flat):u\mapsto[\pi^\flat]$ naturally extends to $\mathcal{O}_{\mathcal{E}_R}\to W(\overline{R}^\flat[(\pi^\flat)^{-1}])$, and in turn to $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}\to W(\overline{R}^\flat[(\pi^\flat)^{-1}])$. By definition of the Laurent crystal, we have

$$\mathcal{E}\big(\mathbb{A}_{\mathrm{inf}}(\overline{R})\big)\simeq\mathcal{M}\otimes_{\mathcal{O}_{\mathcal{E}_R}}W(\overline{R}^\flat[(\pi^\flat)^{-1}]).$$

In particular, the associated $G_R$-representation is given by

$$T\simeq\Big(\mathcal{M}\otimes_{\mathcal{O}_{\mathcal{E}_R}}W(\overline{R}^\flat[(\pi^\flat)^{-1}])\Big)^{\varphi=1}.$$

Using Theorem A.7, we have

$$T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M})=(\mathcal{M}\otimes_{\mathcal{O}_{\mathcal{E}_R}}\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R})^{\varphi=1}\hookrightarrow(\mathcal{M}\otimes_{\mathcal{O}_{\mathcal{E}_R}}W(\overline{R}^\flat[(\pi^\flat)^{-1}]))^{\varphi=1}=T$$

which is an injection of finite free $\mathbb{Z}_p$-modules. We claim that this map is $p$-saturated, and thus we have equality. For this, note that $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}\to W(\overline{R}^\flat[(\pi^\flat)^{-1}])$ is $p$-saturated as the mod $p$ reduction is the inclusion of $\mathcal{O}^{\mathrm{unr}}_{\mathcal{E}_R}/p$ in $\overline{R}^\flat[(\pi^\flat)^{-1}]$. Since $\mathcal{M}$ is finite projective over $\mathcal{O}_{\mathcal{E}_R}$, the same holds for

$$\mathcal{M}\otimes_{\mathcal{O}_{\mathcal{E}_R}}\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}\hookrightarrow\mathcal{M}\otimes_{\mathcal{O}_{\mathcal{E}_R}}W(\overline{R}^\flat[(\pi^\flat)^{-1}]),$$

and in turn, $T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M})\to T$ is $p$-saturated. Finally,

$$\operatorname{rank}_{\mathbb{Z}_p}(T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M}))=\operatorname{rank}_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M})=\operatorname{rank}_{\mathbb{Z}_p}(T),$$

the first equality by Theorem A.7 and the second follows from [DLMS24b, Lem. 3.25]. Thus the cokernel of $T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M})\to T|_{G_{R_\infty}}$ is both finite and $p$-torsion-free, so it vanishes. □

A.4. **Base change to Shilov points.** We follow [Kim15, §7.4] almost *verbatim*. Let $R$ be a semistable base ring as above, and let $\mathfrak{p}_j$, $j=1,\dots,m$, be the minimal prime ideals of $R/p$ corresponding to the Shilov points of $\operatorname{Spf}R$. Recall that the morphisms $\iota_j:\Delta_j\to X$ induce ring maps $\iota_j^\sharp:R\to\mathcal{O}_{L_j}:=R^\wedge_{\mathfrak{p}_j}$. Choose extensions $\bar{\iota}_j^\sharp:\overline{R}\to\mathcal{O}_{\overline{L_j}}$. Thus $\bar{\iota}_j^\sharp$ induces a continuous group homomorphism $G_{L_j}\to G_R$ and a continuous ring homomorphism $\bar{\iota}_j^\flat:\overline{R}^\flat\to\mathcal{O}^\flat_{\overline{L_j}}$. Let $\mathfrak{S}_{L_j}$ and $\mathcal{O}_{\mathcal{E}_{L_j}}$ be as in 2.1. We define $\mathfrak{S}^{\mathrm{unr}}_{L_j}$ and $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_{L_j}}$ analogously to the semistable case (cf. [Moo24, §3]). Thus $\mathfrak{S}^{\mathrm{unr}}_R\subset W(\overline{R}^\flat)$ and $\mathfrak{S}^{\mathrm{unr}}_{L_j}\subset W(\mathcal{O}^\flat_{\overline{L_j}})$, and $W(\bar{\iota}_j^\flat):W(\overline{R}^\flat)\to W(\mathcal{O}^\flat_{\overline{L_j}})$ carries $\mathfrak{S}_R$ to $\mathfrak{S}_{L_j}$ and $\widehat{\mathfrak{S}}^{\mathrm{unr}}_R$ to $\widehat{\mathfrak{S}}^{\mathrm{unr}}_{L_j}$, and commutes with Frobenius and the $G_{L_j}$-actions (where $G_{L_j}$ acts on $\widehat{\mathfrak{S}}^{\mathrm{unr}}_R$ via restriction along $G_{L_j}\to G_R$). We also obtain a map $\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_R}\to\widehat{\mathcal{O}}^{\mathrm{unr}}_{\mathcal{E}_{L_j}}$ commuting with Frobenius and the Galois actions.

Choose the compatible $p$-power roots used in the construction of $R_\infty$ so that $\bar{\iota}_j^\sharp$ carries them to compatible roots in $\overline{L_j}$. For each $i=1,\dots,d$ and $j=1,\dots,m$, let $T_i\in\mathcal{O}_{L_j}$ denote the image of $T_i\in R$ under $\iota_j^\sharp$. Set

$$\widetilde{L}^{\mathrm{nc}}_{j,\infty}:=\bigcup_{n\geq0}L_j(\pi^{1/p^n},T_1^{1/p^n},\dots,T_d^{1/p^n}),\quad\text{and}\quad\widetilde{L}_{j,\infty}:=(\widetilde{L}^{\mathrm{nc}}_{j,\infty})^\wedge.$$

Then $\vec{\iota}_j^{\sharp}$ extends continuously to a map $R_\infty \longrightarrow \mathcal{O}_{\widetilde{L}_{j,\infty}}$ and in turn induces a continuous homomorphism $G_{\widetilde{L}_{j,\infty}} \longrightarrow G_{R_\infty}$. Given an étale $(\varphi, \mathcal{O}_{\mathcal{E}_R})$-module $\mathcal{M}$, we therefore obtain a natural $G_{\widetilde{L}_{j,\infty}}$-equivariant map

$$T_{\mathcal{O}_{\mathcal{E}_R}}(\mathcal{M}) \to T_{\mathcal{O}_{\mathcal{E}_{L_j}}}(\mathcal{O}_{\mathcal{E}_{L_j}} \otimes_{\mathcal{O}_{\mathcal{E}_R}} \mathcal{M}), \tag{A.2}$$

where $G_{\widetilde{L}_{j,\infty}}$ acts on the source via restriction along $G_{\widetilde{L}_{j,\infty}} \to G_{R_\infty}$. We claim that (A.2) is an isomorphism. It is injective, and its image is saturated: setting $\mathcal{E}_R := \mathcal{O}_{\mathcal{E}_R}[p^{-1}]$, $\mathcal{E}_{L_j} := \mathcal{O}_{\mathcal{E}_{L_j}}[p^{-1}]$, we have $\mathcal{E}_R \cap \mathcal{O}_{\mathcal{E}_{L_j}} = \mathcal{O}_{\mathcal{E}_R}$ inside $\mathcal{E}_{L_j}$. This is the saturation argument of [Kim15, §7.4]. Both sides of (A.2) are finite free $\mathbb{Z}_p$-modules of the same rank, using Theorem A.7 for the source and [Kim15, Prop. 7.3] for the target, so the map is an isomorphism.

## Appendix B. Intrinsic Strongly Divisible Modules over CDVRs

In this appendix, we restrict to the CDVR setup of [Moo24, §2]. In particular, if $k' := \mathcal{O}_{L_0}/(p)$, we assume that $k'$ is an extension of the fixed perfect field $k$ with a finite $p$-basis, endow $\mathcal{O}_{L_0}$ with the unique Frobenius-compatible $W(k)$-algebra structure lifting $k \to k'$, and assume that $\mathcal{O}_L = \mathcal{O}_{L_0} \otimes_{W(k)} \mathcal{O}_K$ (hence $L = L_0 \otimes_{K_0} K$). In particular, we take $\pi_L = \pi$ and $E_L(u) = E(u)$ in the notation of Subsection 2.1. These assumptions hold for the CDVRs $\mathcal{O}_{L_j} := R^{\wedge}_{(T_j)}$ considered in this paper. Let $S_L$ be the resulting Breuil ring. We compare the category $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$ of Definition 5.1 with the category of strongly divisible lattices defined in [Moo24, Def. 4.3]. In particular, we show that Breuil modules in CDVR case give rise to strongly divisible lattices inside the $S_L$-module attached from a weakly-admissible filtered $(\varphi, N, \nabla)$-module, establishing the analogue of [Liu08, Prop. 2.3.4] in the imperfect residue field case.

Fix $0 \le h \le p-2$. Following [Moo24, §2.1], set $R_{0,g}$ to be the $p$-adic completion of $\varinjlim_{\varphi} \mathcal{O}_{L_0}$, and let $k_g := R_{0,g}/(p)$. Then $k_g$ is a perfect extension of $k$, and there is a Frobenius-compatible identification $R_{0,g} \simeq W(k_g)$. Let $b_g : \mathcal{O}_{L_0} \longrightarrow W(k_g)$ denote the resulting Frobenius-compatible $W(k)$-algebra map, and let $S_g$ be the corresponding classical Breuil ring. The map $b_g$ extends to $S_L \to S_g$. By the second paragraph in the proof of [Moo24, Prop. 5.5], the maps $S_L/(p^n) \longrightarrow S_g/(p^n)$ are faithfully flat for every $n \ge 1$. The module of log differentials admits a decomposition

$$\omega^1_{S_L/\mathbb{Z}_p} \simeq S_L \, d\log u \oplus \left( S_L \otimes_{\mathcal{O}_{L_0}} \widehat{\Omega}^1_{\mathcal{O}_{L_0}/W(k)} \right).$$

We can split the connection on an object $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$ as

$$\nabla_{\mathscr{M}} = (-N_u \otimes d\log u, \nabla^0_{\mathscr{M}}),$$

where the sign is chosen so that the derivation on $S_L$ underlying $N_u$ sends $u$ to $-u$, in accordance with the convention of [Moo24]. Choose lifts $X_1, \dots, X_b$ of a $p$-basis of $k'$ with $\varphi(X_i) = X_i^p$, and let $N_{X_i}$ be the $d\log X_i$-component of $\nabla^0_{\mathscr{M}}$. Integrability implies that $N_u$ commutes with these components, while the horizontality condition in Definition 5.1 gives the usual compatibilities with Frobenius, namely

$$N_u\varphi = p\varphi N_u, \quad \text{and} \quad N_{X_i}\varphi = p\varphi N_{X_i},\ 1 \le i \le b.$$

The category $\mathrm{Mod}^{\varphi,N}_{/S_g}$ of what we call *abstract classical strongly divisible lattices* is defined in [Liu08, §2.3]. Its intrinsic characterization is due to Breuil: see [Bre02, Thrm. 2.2.3].

**Lemma B.1.** *Fix $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$. Set $\mathscr{M}_g := S_g \otimes_{S_L} \mathscr{M}$, and let $\mathrm{Fil}^h \mathscr{M}_g$ be the $p$-adic closure of $\mathrm{im}(S_g \otimes_{S_L} \mathrm{Fil}^h \mathscr{M} \to \mathscr{M}_g)$ inside $\mathscr{M}_g$, and equip the pair $(\mathscr{M}_g, \mathrm{Fil}^h \mathscr{M}_g)$ with the induced divided Frobenius and monodromy. We obtain an object of the category $\mathrm{Mod}^{\varphi,N}_{/S_g}$.*

*Proof.* The quotient $\mathscr{M}/\mathrm{Fil}^h \mathscr{M}$ is finite over $S_L/\mathrm{Fil}^h S_L \simeq \mathfrak{S}_L/(E^h)$, and in particular finite over $\mathcal{O}_{L_0}$. It is $p$-torsion-free by Remark 4.11, and therefore finite free over $\mathcal{O}_{L_0}$. Tensoring the quotient sequence modulo $p^n$ along $S_L/(p^n) \to S_g/(p^n)$ and taking inverse limits gives

$$\mathscr{M}_g/\mathrm{Fil}^h \mathscr{M}_g \simeq W(k_g) \otimes_{b_g, \mathcal{O}_{L_0}} (\mathscr{M}/\mathrm{Fil}^h \mathscr{M}).$$

In particular, $\mathscr{M}_g/\operatorname{Fil}^h \mathscr{M}_g$ is $p$-torsion-free. The divided Frobenius extends continuously to $\operatorname{Fil}^h \mathscr{M}_g$, and clearly $\varphi_h(\operatorname{Fil}^h \mathscr{M}_g)$ generates $\mathscr{M}_g$. The compatibility of $N_u$ with the filtration and divided Frobenius is exactly the $d\log u$-component of Griffiths transversality and horizontality. The base-changed coefficient connection is obtained by composing with the natural map

$$W(k_g)\widehat{\otimes}_{\mathcal{O}_{L_0},b_g}\widehat{\Omega}^1_{\mathcal{O}_{L_0}/W(k)} \longrightarrow \widehat{\Omega}^1_{W(k_g)/W(k)}.$$

The target vanishes: indeed, reduction modulo $p$ gives $\widehat{\Omega}^1_{W(k_g)/W(k)}/p \simeq \Omega^1_{k_g/k} = 0$, where the last equality follows because both $k_g$ and $k$ are perfect, hence $\widehat{\Omega}^1_{W(k_g)/W(k)} = 0$ by $p$-completeness. Thus the coefficient connection vanishes after applying $b_g$, whereas the operator $N_u$ remains. What we recover is precisely an object of the classical category $\operatorname{Mod}^{\varphi,N}_{/S_g}$ of [Liu08]. □

Let $I_0 := \ker(S_L \xrightarrow{u\mapsto 0} \mathcal{O}_{L_0})$, set $D := D(\mathscr{M}) := (\mathscr{M}/I_0\mathscr{M})[p^{-1}]$, $D_L := L \otimes_{L_0} D$, and write $q : \mathscr{M}[p^{-1}] \to D$ for the projection. The ideal $I_0$ is stable under Frobenius, monodromy, and the coefficient derivations, so $D$ inherits operators $\varphi_D$, $N_D$, and $\nabla_D$. Set $\mathscr{D}(D) := \mathscr{D}(D(\mathscr{M})) := S_L[p^{-1}] \otimes_{L_0} D$, with Frobenius, monodromy, and coefficient connection

$$\begin{aligned}\varphi(f \otimes d) &= \varphi(f) \otimes \varphi_D(d),\\ N_u(f \otimes d) &= -u\frac{df}{du} \otimes d + f \otimes N_D(d),\\ \nabla^0(f \otimes d) &= d^0 f \otimes d + f\nabla_D(d),\end{aligned}$$

where $d^0$ differentiates the $L_0$-coefficients and kills $u$. The full log connection is $-N_u \otimes d\log u + \nabla^0$. Let $q_\pi : S_L[p^{-1}] \to L$ denote evaluation at $u = \pi$.

Put $K_{g,0} := W(k_g)[p^{-1}]$ and $K_g := K_{g,0} \otimes_{K_0} K$. By [Liu08, Prop. 2.3.4(2)], which records the intrinsic characterization of strongly divisible modules due to Breuil, there is a weakly admissible filtered $(\varphi, N)$-module $D_g$ over $K_{g,0}$. Its rational Breuil realization is

$$\mathscr{D}(D_g) := S_g[p^{-1}] \otimes_{K_{g,0}} D_g,$$

with the tensor product Frobenius and monodromy above. The same result supplies an isomorphism

$$\mathscr{D}(D_g) \xrightarrow{\sim} \mathscr{M}_g[p^{-1}] \tag{B.1}$$

compatible with Frobenius, monodromy, and filtration; the weight-zero case is the classical case recalled in [Liu08, Rem. 2.3.2(1)]. Let $q_{\pi,g} : \mathscr{D}(D_g) \to (D_g)_{K_g}$ denote specialization at $u = \pi$. The filtration on $\mathscr{D}(D_g)$ is defined recursively by $\operatorname{Fil}^0 \mathscr{D}(D_g) = \mathscr{D}(D_g)$ and

$$\operatorname{Fil}^{i+1} \mathscr{D}(D_g) = \left\{x \in \mathscr{D}(D_g) \;\middle|\; N(x) \in \operatorname{Fil}^i \mathscr{D}(D_g),\ q_{\pi,g}(x) \in \operatorname{Fil}^{i+1}(D_g)_{K_g}\right\}.$$

**Lemma B.2.** *For every $i \geq 0$,*

$$q_{\pi,g}\big(\operatorname{Fil}^i \mathscr{D}(D_g)\big) = \operatorname{Fil}^i (D_g)_{K_g}. \tag{B.2}$$

*Proof.* Write $\mathscr{D}_g := \mathscr{D}(D_g)$ and $q := q_{\pi,g}$. The inclusion $q\big(\operatorname{Fil}^i \mathscr{D}_g\big) \subseteq \operatorname{Fil}^i (D_g)_{K_g}$ is tautological, so we show the reverse inclusion. Observe that the recursive definition of the filtration gives

$$x \in \operatorname{Fil}^i \mathscr{D}_g \quad\Longleftrightarrow\quad q\big(N^a(x)\big) \in \operatorname{Fil}^{i-a}(D_g)_{K_g} \quad \text{for } 0 \leq a \leq i. \tag{B.3}$$

Fix $y \in \operatorname{Fil}^i (D_g)_{K_g}$ and choose $x_0 \in \mathscr{D}_g$ such that $q(x_0) = y$: such a lift exists because $S_g[p^{-1}] \to K_g : u \mapsto \pi$ is surjective. For $i = 0, 1$, this lift already suffices. Set $c := q\big(N(E(u))\big) = -\pi E'(\pi) \neq 0$. We will successively modify $x_0$ by multiples of $E, E^2, \ldots, E^{i-1}$. For $z \in \mathscr{D}_g$ and integers $a, b \geq 0$, the Leibniz rule gives

$$q\big(N^a(E^b z)\big) = 0 \quad \text{if } a < b, \tag{B.4}$$

because every term in the iterated Leibniz expansion still contains a factor of $E$. If $a = b$, the only terms which survive specialization are those in which each of the $b$ factors of $E$ is differentiated exactly once, which simplifies to

$$q\big(N^b(E^b z)\big) = b!c^b q(z) \tag{B.5}$$

for any $z \in \mathscr{D}_g$. The scalar $b!c^b$ is invertible in $K_g$. Suppose inductively that, for some $1 \leq b < i$, we have constructed $x_{b-1} \in \mathscr{D}_g$ such that $q(x_{b-1}) = y$ and $q\big(N^a(x_{b-1})\big) = 0$ for each $1 \leq a < b$. Choose $z_b \in \mathscr{D}_g$ lifting $-\frac{q(N^b(x_{b-1}))}{b!c^b}$ (note that we cannot divide by $c$ in $\mathscr{D}_g$, as $c \in K_g$), and set $x_b := x_{b-1} + E^b z_b$. Since $q(E) = 0$, we still have $q(x_b) = y$. Equations (B.4) and (B.5) show that $q\big(N^a(x_b)\big) = 0$ for $1 \leq a \leq b$. Carrying out this construction for $b = 1, \dots, i-1$, we obtain an element $x := x_{i-1}$ such that $q(x) = y$ and $q(N^a(x)) = 0$ for all $1 \leq a < i$. Since zero belongs to every step of the filtration, criterion (B.3) implies that $x \in \mathrm{Fil}^i \mathscr{D}_g$. Hence every element of $\mathrm{Fil}^i(D_g)_{K_g}$ lifts to $\mathrm{Fil}^i \mathscr{D}_g$, proving the asserted surjectivity. □

**Lemma B.3.** *For $0 \leq i \leq h$, one has*

$$\mathrm{Fil}^i(D_g)_{K_g} = q_{\pi,g}\left(\left\{x \in \mathscr{D}(D_g) \;\middle|\; E(u)^{h-i}x \in \mathrm{Fil}^h \mathscr{D}(D_g)\right\}\right).$$

*Proof.* If $y \in \mathrm{Fil}^i(D_g)_{K_g}$, equation (B.2) allows us to choose a lift $x \in \mathrm{Fil}^i \mathscr{D}(D_g)$ of $y$. Set $j = h - i$. Since the Breuil filtration is multiplicative, we have $E(u)^j x \in \mathrm{Fil}^h \mathscr{D}(D_g)$. This proves one inclusion.

Conversely, suppose that $E(u)^j x \in \mathrm{Fil}^h \mathscr{D}(D_g)$. The recursive filtration gives $N^j(E(u)^j x) \in \mathrm{Fil}^i \mathscr{D}(D_g)$. Applying (B.5), we obtain

$$q_{\pi,g}\left(N^j\big(E(u)^j x\big)\right) = j!\big(-\pi E'(\pi)\big)^j q_{\pi,g}(x).$$

The left side belongs to $\mathrm{Fil}^i(D_g)_{K_g}$ by definition. Invertibility of $j!(-\pi E'(\pi))^j$ in $K_g$ therefore gives $q_{\pi,g}(x) \in \mathrm{Fil}^i(D_g)_{K_g}$, proving the reverse inclusion. □

The Breuil–Kisin equivalence of Proposition 4.12, applied to $\mathfrak{S}_L \to S_L$, writes the underlying Breuil module as $\mathscr{M} \simeq S_L \otimes_{\varphi,\mathfrak{S}_L} \mathfrak{M}$ for a finite free Breuil–Kisin module $\mathfrak{M}$ of height at most $h$; for $h > 0$, this is also [Gao20, Thm. 5.1.3(1)]. Applying [DLMS24a, Lem. 4.2] to $\mathfrak{M}$ gives a unique Frobenius-compatible section $s : D \to \mathscr{M}[p^{-1}]$ of $q$ and an isomorphism

$$\iota := 1 \otimes s : S_L[p^{-1}] \otimes_{L_0} D \xrightarrow{\sim} \mathscr{M}[p^{-1}] \tag{B.6}$$

of $S_L[p^{-1}]$-modules. This application only uses finite height and projectivity of $\mathfrak{M}$, not a prior realization by a Galois representation. Let $q_{\pi,\mathscr{M}} := (q_\pi \otimes 1)\iota^{-1}$ and, for $0 \leq i \leq h$, set $F^i\mathscr{M} := \{x \in \mathscr{M} : E(u)^{h-i}x \in \mathrm{Fil}^h \mathscr{M}\}$. We endow $D_L$ with the decreasing filtration

$$\mathrm{Fil}^i D_L := \begin{cases} D_L & i < 0, \\ q_{\pi,\mathscr{M}}((F^i\mathscr{M})[p^{-1}]) & 0 \leq i \leq h, \\ 0 & i > h. \end{cases}$$

Thus, just as for $\mathscr{D}(D_g)$, we can equip $\mathscr{D}(D)$ with the recursively defined filtration given by $\mathrm{Fil}^i \mathscr{D}(D) = \mathscr{D}(D)$ for $i \leq 0$, and

$$\mathrm{Fil}^{i+1} \mathscr{D}(D) = \big\{x \in \mathscr{D}(D) : N_u(x) \in \mathrm{Fil}^i \mathscr{D}(D),\ (q_\pi \otimes 1)(x) \in \mathrm{Fil}^{i+1} D_L\big\}$$

for any $i \geq 0$.

**Lemma B.4.** *The isomorphism*

$$\iota : \mathscr{D}(D(\mathscr{M})) \xrightarrow{\sim} \mathscr{M}[p^{-1}]$$

*provided by* (B.6) *commutes with Frobenius, monodromy, and connection, and satisfies*

$$\iota(\mathrm{Fil}^h \mathscr{D}(D(\mathscr{M}))) = (\mathrm{Fil}^h \mathscr{M})[p^{-1}]$$

*Moreover, the natural isomorphism*

$$K_{g,0} \otimes_{b_g,L_0} D(\mathscr{M}) \simeq D_g$$

*is compatible with Frobenius, monodromy, and the filtrations after extension to $K_g$.*

*Proof.* By Lemma B.1 and (B.1), we obtain the comparison

$$\mathscr{D}(D_g) \simeq S_g[p^{-1}] \otimes_{b_g, S_L[p^{-1}]} \mathscr{D}(D).$$

Reduction modulo $I_{0,g}$ identifies $K_{g,0} \otimes_{L_0, b_g} D$ with $D_g$; uniqueness of the section identifies the base change of $\iota$ with the classical comparison.

We first prove that $s$ is horizontal. Let $N$ denote $N_u$ or $N_{X_i}$, and denote the induced operators on $\mathscr{M}[p^{-1}]$ and $D$ by $N_{\mathscr{M}}$ and $N_D$. The Leibniz terms cancel in $\delta_N := N_{\mathscr{M}} s - s N_D$, so this is an $L_0$-linear map. It vanishes modulo $I_0$ and satisfies

$$\delta_N \varphi_D = p \varphi_{\mathscr{M}} \delta_N. \tag{B.7}$$

Under $\iota$ and the injection $S_L[p^{-1}] \hookrightarrow L_0[[u]]$ (cf. [Gao20, Not. 3.2.4]), expand $\iota^{-1}\delta_N = \sum_{m \geq 1} u^m \delta_m$, with $\delta_m \in \mathrm{Hom}_{L_0}(D, D)$. Equation (B.7) becomes

$$\sum_{m \geq 1} u^m \delta_m \varphi_D = p \sum_{m \geq 1} u^{pm} \varphi_D \delta_m.$$

If $n$ is the least index with $\delta_n \neq 0$, the coefficient of $u^n$ on the right is zero. Thus $\delta_n \varphi_D = 0$. Since the linearized Frobenius on $D$ is an isomorphism, $\varphi_D(D)$ spans $D$, contradicting $\delta_n \neq 0$. Hence $s$ is horizontal. Integrability and quasi-nilpotence descend to the stable lattice $\mathscr{M}/I_0\mathscr{M}$, and nilpotence of monodromy follows by the faithful extension to $D_g$.

We now verify the filtration comparison using the modules $F^i\mathscr{M}$ defined above. Since $E^h\mathscr{M} \subseteq \mathrm{Fil}^h \mathscr{M}$, we have $F^0\mathscr{M} = \mathscr{M}$ and $\mathrm{Fil}^0 D_L = D_L$. Moreover, $F^{i+1}\mathscr{M} \subseteq F^i\mathscr{M}$, so their images under $q_{\pi,\mathscr{M}}$ form a decreasing filtration. Since

$$S_L / \mathrm{Fil}^h S_L \simeq \mathcal{O}_{L_0}[u]/(E^h),$$

both $\mathscr{M} / \mathrm{Fil}^h S_L\mathscr{M}$ and $\mathscr{M} / \mathrm{Fil}^h \mathscr{M}$ are finite $\mathcal{O}_{L_0}$-modules. For $0 \leq i \leq h$, we have

$$\frac{F^i\mathscr{M}}{\mathrm{Fil}^h S_L\mathscr{M}} = \ker\left(\frac{\mathscr{M}}{\mathrm{Fil}^h S_L\mathscr{M}} \xrightarrow{E^{h-i}} \frac{\mathscr{M}}{\mathrm{Fil}^h \mathscr{M}}\right).$$

Since $q_\pi(\mathrm{Fil}^h S_L) = 0$, the map $q_{\pi,\mathscr{M}}$ factors through $\mathscr{M} / \mathrm{Fil}^h S_L\mathscr{M}$, hence

$$\mathrm{Fil}^i D_L = \mathrm{im}\left(\left(\frac{F^i\mathscr{M}}{\mathrm{Fil}^h S_L\mathscr{M}}\right)[p^{-1}] \xrightarrow{q_{\pi,\mathscr{M}}} D_L\right).$$

Now $W(k_g)$ is flat over $\mathcal{O}_{L_0}$, so we have identifications

$$W(k_g) \otimes_{\mathcal{O}_{L_0}} \frac{\mathscr{M}}{\mathrm{Fil}^h S_L\mathscr{M}} \simeq \frac{\mathscr{M}_g}{\mathrm{Fil}^h S_g\mathscr{M}_g}, \quad \text{and} \quad W(k_g) \otimes_{\mathcal{O}_{L_0}} \frac{\mathscr{M}}{\mathrm{Fil}^h \mathscr{M}} \simeq \frac{\mathscr{M}_g}{\mathrm{Fil}^h \mathscr{M}_g}.$$

Hence base change identifies the displayed kernel with

$$\ker\left(\frac{\mathscr{M}_g}{\mathrm{Fil}^h S_g\mathscr{M}_g} \xrightarrow{E^{h-i}} \frac{\mathscr{M}_g}{\mathrm{Fil}^h \mathscr{M}_g}\right) = \frac{F^i\mathscr{M}_g}{\mathrm{Fil}^h S_g\mathscr{M}_g}.$$

and $q_{\pi,\mathscr{M}}$ becomes $q_{\pi,g}$. It follows that

$$K_g \otimes_L \mathrm{Fil}^i D_L = q_{\pi,g}((F^i\mathscr{M}_g)[p^{-1}]).$$

Lemma B.3 identifies the right-hand side with $\mathrm{Fil}^i(D_g)_{K_g}$, and therefore

$$K_g \otimes_L \mathrm{Fil}^i D_L = \mathrm{Fil}^i(D_g)_{K_g}, \quad \text{for } 0 \leq i \leq h.$$

We now follow the intersection argument of [DLMS24a, Lem. 4.15], as used in the semistable setting in [DLMS24b, Lem. 4.5]. View $\mathscr{D}(D)$ inside $\mathscr{D}(D_g)$ using the natural injection $S_L \hookrightarrow S_g$. The preceding equality gives $D_L \cap \mathrm{Fil}^i(D_g)_{K_g} = \mathrm{Fil}^i D_L$. Since monodromy and specialization commute with this inclusion, induction in the recursive definition gives

$$\mathscr{D}(D) \cap \mathrm{Fil}^i \mathscr{D}(D_g) = \mathrm{Fil}^i \mathscr{D}(D), \quad \text{for } i \geq 0.$$

On the other hand, the quotient identification in Lemma B.1 makes the map

$$(\mathscr{M}/\operatorname{Fil}^h \mathscr{M})[p^{-1}] \longrightarrow (\mathscr{M}_g/\operatorname{Fil}^h \mathscr{M}_g)[p^{-1}]$$

injective. Taking kernels of the quotient maps therefore gives

$$\mathscr{M}[p^{-1}] \cap (\operatorname{Fil}^h \mathscr{M}_g)[p^{-1}] = (\operatorname{Fil}^h \mathscr{M})[p^{-1}].$$

The classical comparison identifies the two filtrations over $S_g[p^{-1}]$, so these two intersection identities imply

$$\iota\big(\operatorname{Fil}^h \mathscr{D}(D)\big) = (\operatorname{Fil}^h \mathscr{M})[p^{-1}].$$

Finally, we check that the coefficient connection $\nabla_D$ lowers the filtration on $D_L$. As $E(u) \in W(k)[u]$, we have $N_{X_j}(E) = 0$. For $1 \le i \le h$ and $x \in F^i\mathscr{M}$, Griffiths transversality for $\mathscr{M}$ gives

$$E^{h-i+1} N_{X_j}(x) = E N_{X_j}(E^{h-i}x) \in \operatorname{Fil}^h \mathscr{M},$$

and hence $N_{X_j}(F^i\mathscr{M}) \subseteq F^{i-1}\mathscr{M}$. Horizontality of $\iota$ and the fact that the coefficient derivations commute with evaluation at $u = \pi$ show that $q_{\pi,\mathscr{M}}$ commutes with these operators. Applying $q_{\pi,\mathscr{M}}$ after inverting $p$ gives

$$\nabla_D(\operatorname{Fil}^i D_L) \subseteq \operatorname{Fil}^{i-1} D_L \otimes_L \widehat{\Omega}^1_{L/K}.$$

For the remaining indices this follows from $\operatorname{Fil}^i D_L = D_L$ for $i \le 0$ and $\operatorname{Fil}^i D_L = 0$ for $i > h$. The induction in the proof of [Moo24, Lem. 4.2] now gives Griffiths transversality for the full recursive filtration on $\mathscr{D}(D)$. □

**Lemma B.5.** *The filtered $(\varphi, N, \nabla)$-module $D(\mathscr{M})$ is weakly admissible.*

*Proof.* By Lemma B.4, $D_g$ is the filtered base change of $D(\mathscr{M})$. Extension from $L$ to $K_g$ preserves the dimensions of the graded pieces, and $b_g : L_0 \to K_{g,0}$ preserves the $p$-adic valuation. Hence Hodge and Newton numbers are unchanged, giving

$$t_H(D(\mathscr{M})) = t_H(D_g) = t_N(D_g) = t_N(D(\mathscr{M})).$$

For any $(\varphi, N)$-stable subspace $D' \subset D(\mathscr{M})$ with its induced filtration, set $D'_g := K_{g,0} \otimes_{b_g, L_0} D'$. Flatness identifies its induced filtration with the scalar extension of that on $D'$. The weak admissibility of $D_g$ therefore gives

$$t_H(D') = t_H(D'_g) \le t_N(D'_g) = t_N(D').$$

This proves weak admissibility of $D(\mathscr{M})$. □

**Theorem B.6.** *Let $0 \le h \le p-2$. Every $\mathscr{M} \in \operatorname{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$ admits a canonical weakly admissible filtered $(\varphi, N, \nabla)$-module $D(\mathscr{M})$ and an identification*

$$\mathscr{M}[p^{-1}] \simeq \mathscr{D}(D(\mathscr{M}))$$

*under which $\mathscr{M}$ is a strongly divisible lattice in the sense of* [Moo24, Def. 4.3]*. In particular, the functor from Moon's category of strongly divisible lattices of weight $h$ over $S_L$ to $\operatorname{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$ given by forgetting the ambient weakly admissible filtered $(\varphi, N, \nabla)$-module is an equivalence.*

*Proof.* Lemmas B.4 and B.5 show that every object $\mathscr{M}$ of $\operatorname{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$ is a strongly divisible lattice in $\mathscr{D}(D(\mathscr{M}))$ except for the equality $\operatorname{Fil}^h \mathscr{M} = \mathscr{M} \cap \operatorname{Fil}^h \mathscr{M}[p^{-1}]$ which follows as the quotient $\mathscr{M}/\operatorname{Fil}^h \mathscr{M}$ is $p$-torsion-free (Remark 4.11).

Conversely, let $\mathscr{M}$ be a strongly divisible lattice in $\mathscr{D}(D)$ in the sense of [Moo24, Def. 4.3]. Then equality $\operatorname{Fil}^h \mathscr{M} = \mathscr{M} \cap \operatorname{Fil}^h \mathscr{D}(D)$ implies $Q := \mathscr{M}/\operatorname{Fil}^h \mathscr{M}$ is $p$-torsion-free. Moreover, $Q$ is finite over $S_L/\operatorname{Fil}^h S_L \simeq \mathfrak{S}_L/(E^h)$, so Remark 4.11 gives $\operatorname{pd}_{\mathfrak{S}_L}(Q) \le 1$. The remaining filtration, monodromy, and connection conditions are inherited from $\mathscr{D}(D)$. The only non-formal point is the generation axiom for the divided Frobenius. It follows from weak admissibility by [Moo24, Lem. 4.4]; the proof there is the argument of [Bre99, Prop. 2.1.3]. Thus forgetting the ambient rational module does indeed produce an object of $\operatorname{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$.

The two constructions are quasi-inverse. Starting from an object $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$, the preceding lemmas recover its rational module and filtration. Starting from a strongly divisible lattice $\mathscr{M} \subset \mathscr{D}(D)$, taking the quotient by $I_0$ recovers $D$, while Lemmas B.2 and B.3, applied over $L$, recover its filtration from $\mathrm{Fil}^h \mathscr{M}$. Finally, a morphism of objects in $\mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$ preserves each $F^i\mathscr{M}$, and uniqueness of the Frobenius-compatible sections makes $\iota$ functorial. The induced map on $D$ therefore preserves its filtration, Frobenius, monodromy, and connection. □

**Corollary B.7.** *For every $\mathscr{M} \in \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L)$, the representation $T_{\mathrm{cris}}(\mathscr{M})$ is a $\mathbb{Z}_p$-lattice in a semistable representation of $G_L$ with Hodge–Tate weights in $[0,h]$.*

*Proof.* By Theorem B.6, $\mathscr{M}$ is a strongly divisible lattice in Moon's sense. Proposition 4.9 of [Moo24] gives the assertion for Moon's contravariant functor. Applying it to the Cartier dual lattice and using [Moo24, §4.2, Lem. 4.15] gives the covariant realization used here. The duality pairing respects the connections, so this identification also respects the Taylor $G_L$-actions. □

**Corollary B.8.** *The functor*

$$T_{\mathrm{cris}} : \mathrm{Mod}^{\varphi,h,\nabla,\mathrm{proj}}(S_L) \xrightarrow{\sim} \mathrm{Rep}^{\mathrm{st}}_{\mathbb{Z}_p,[0,h]}(G_L)$$

*is an equivalence of categories.*

TODO: change some of the wording since this moved to the appendix.

*Proof.* By Theorem B.6, our category is Moon's category of strongly divisible lattices of weight $h$. To keep track of variance, write

$$T^{\vee}_{\mathrm{cris}}(\mathscr{P}) := \mathrm{Hom}_{S_L,\mathrm{Fil},\varphi_h}\big(\mathscr{P}, \mathbb{A}_{\mathrm{cris}}(\mathcal{O}_{\overline{L}})\big)$$

for Moon's contravariant realization. For a strongly divisible lattice $\mathscr{P}$, let $\mathscr{P}^\vee$ be its Cartier dual, with

$$\mathrm{Fil}^h \mathscr{P}^\vee = \{f \in \mathrm{Hom}_{S_L}(\mathscr{P}, S_L) \mid f(\mathrm{Fil}^h \mathscr{P}) \subseteq \mathrm{Fil}^h S_L\},$$

and with the dual divided Frobenius, monodromy, and connection. By [Moo24, Lem. 4.12], the assignment $\mathscr{P} \mapsto \mathscr{P}^\vee$ is an involutive anti-equivalence. Applying [Moo24, Lem. 4.15] to $\mathscr{P}^\vee$, and using $(\mathscr{P}^\vee)^\vee \simeq \mathscr{P}$, the evaluation pairing gives a natural isomorphism

$$T^{\vee}_{\mathrm{cris}}(\mathscr{P}^\vee) \xrightarrow{\sim} \Big(\mathbb{A}_{\mathrm{cris}}(\mathcal{O}_{\overline{L}})\widehat{\otimes}_{S_L} \mathrm{Fil}^h \mathscr{P}\Big)^{\varphi_h=1} = T_{\mathrm{cris}}(\mathscr{P}).$$

This identification is $G_L$-equivariant. Indeed, the connection and monodromy on $\mathscr{P}^\vee$ are defined by requiring the evaluation pairing $\mathscr{P} \otimes \mathscr{P}^\vee \to S_L$ to be horizontal, and the $G_L$-actions on both sides are obtained from the corresponding Taylor formulas. Hence evaluation intertwines the two actions.

Moon's functor $T^{\vee}_{\mathrm{cris}}$ is an anti-equivalence onto the category of lattices in semistable representations with Hodge–Tate weights in $[0,h]$ by [Moo24, Thrm. 4.22]. Its composite with the Cartier-dual anti-equivalence is therefore a covariant equivalence, and the displayed isomorphism identifies that composite with our functor $T_{\mathrm{cris}}$. □

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